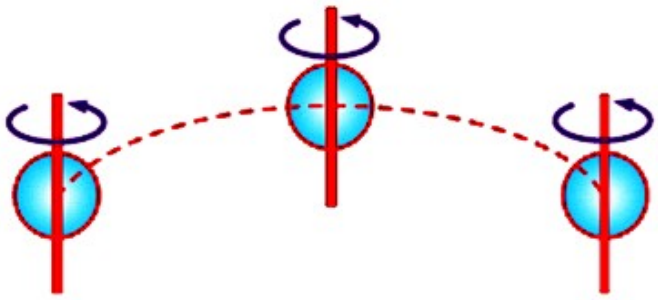
Centre of Mass, Impulse and Collision

DISCLAIMER

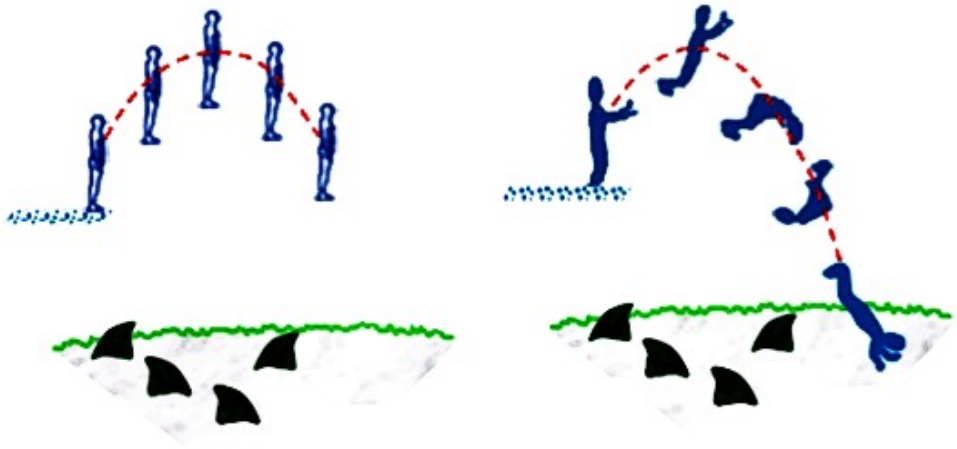
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Centre of Mass

Centre of mass is defined as a special point at which the whole mass of a body can be assumed to be concentrated. Studying the motion of this special point simplifies the study of motion of system consisting of large number of particles.
Consider a cricket ball spinning in air. Ball is spinning and rising at the same time. The path of any point on the surface of ball will be a complicated path. But, there is a special point which still moves on a fairly simple parabolic path which is the centre of the ball, and it is centre of mass of the ball as well.



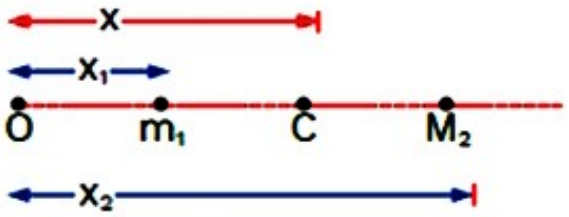
Another such example can be a swimmer diving in a pool.



We can balance a book by placing our finger exactly under a special point, that point is the centre of mass of the book.



Consider two masses m_1 and m_2 lie in a straight line at positions x_1 and x_2 from the origin (O) as shown below,



The position of centre of mass (C) from the origin is given by

$$x = \frac{m_1x_1 + m_2x_2}{m_1 + m_2}$$

If the two particles have same masses i.e. $m_1 = m_2 = m$ (say) then,

$$x = \frac{x_1 + x_2}{2}$$

i.e. the midpoint of the line joining m_1 and m_2. Thus, for two particles of equal masses the centre of mass lies exactly midway between the straight line joining them.

Now, suppose we have a system of n particles of masses m_1, m_2, m_3...m_n respectively along a straight line at distance x_1, x_2, x_3...x_n from the origin respectively. Then the centre of mass of the system is given by

$$X = \frac{m_1x_1 + m_2x_2 + ...m_nx_n}{m_1 + m_2 + ...m_n} = \frac{\sum_{i=1}^{n} m_ix_i}{\sum_{x=1}^{n} m_i}$$

Now, if there are n particles distributed in space, then each particle will have a

(x, y, z) coordinate and in that case the coordinates (x, y, z) of the centre of mass of the system is given by

$$x = \frac{\sum_{i=1}^{n} m_ix_i}{M}, y = \frac{\sum_{i=1}^{n} m_iy_i}{M} \text{ and } z = \frac{\sum_{i=1}^{n} m_iz_i}{M}$$

Q. **Two masses 1 kg and 2 kg are lying in xy plane at (–1, 2) and (2, 4) respectively. What are the coordinates of the centre of mass?**

Sol. Let the coordinates of the centre of mass be (x, y)

$$x = \frac{m_1x_1 + m_2x_2}{m_1 + m_2} = \frac{1\times(-1) + 2\times 2}{3} = \frac{-1+4}{3} = 1$$

$$y = \frac{m_1y_1 + m_2y_2}{m_1 + m_2} = \frac{1\times +2\times 2\times 4}{3} = \frac{2+8}{3} = \frac{10}{3}$$

Therefore, the coordinates of centre of mass are $\left(1, \frac{10}{3}\right)$.

Q. **The vertices of a triangle ABC are A (1, 2), B(4, 6) and C(–3, –2). Three particles of masses 1 kg, 2 kg and m kg are placed the vertices of the triangle. If the coordinates of the centre of mass are $\left(\frac{3}{5}, 2\right)$, calculate the mass m.**

Sol. $$x_{cm} = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3} = \frac{3}{5}$$

$$\Rightarrow \quad \frac{1+8-3m}{3+m} = \frac{3}{5}$$

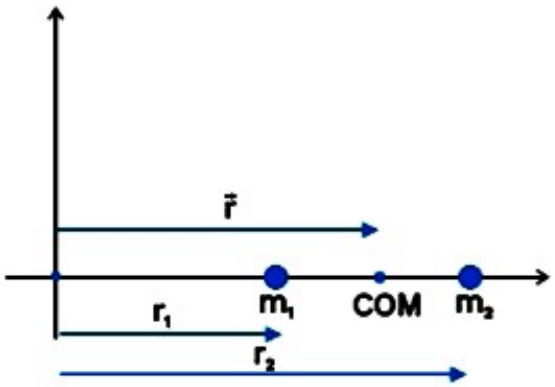
$$\Rightarrow \quad \frac{9-3m}{3+m} = \frac{3}{5}$$

$$\Rightarrow \quad 45 - 15\ m = 9 + 3m$$

$$\Rightarrow \quad 36 = 18\ m$$

$$\Rightarrow \quad m = 2\ kg$$

Shifting the location of COM by shifting the position of individual particle of the system



Position of centre of mass:

$$\vec{r} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1 + m_2}$$

$$\bar{x}_{COM} = \frac{m_1\,\bar{x}_1 + m_2\,\bar{x}_2}{m_1 + m_2} \qquad \ldots (1)$$

$$\bar{y}_{COM} = \frac{m_1\,\bar{y}_1 + m_2\,\bar{y}_2}{m_1 + m_2} \qquad \ldots (2)$$

$$\bar{z}_{COM} = \frac{m_1\,\bar{z}_1 + m_2\,\bar{z}_2}{m_1 + m_2} \qquad \ldots (3)$$

Now on differentiating (1), (2) and (3) wrt t, we get,

$$\frac{d\,\bar{x}_{COM}}{dt} = \frac{m_1\dfrac{d\bar{x}_1}{dt} + m_2\dfrac{d\bar{x}_2}{dt}}{m_1 + m_2}$$

$$d\bar{x}_{COM} = \frac{m_1\,d\bar{x}_2 + m_2\,d\bar{x}_2}{m_1 + m_2}$$

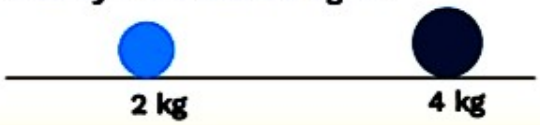
$$\Delta\bar{x}_{COM} = \frac{m_1\left(\Delta\bar{x}_1\right) + m_2\left(\Delta\bar{x}_2\right)}{m_1 + m_2}$$

Similarly, for $\bar{y}_{COM}$ and $\bar{z}_{COM}$,

$$\Delta\bar{y}_{COM} = \frac{m_1(\Delta\bar{y}_1) + m_2(\Delta\bar{y}_2)}{m_1 + m_2}$$

$$\Delta\bar{z}_{COM} = \frac{m_1(\Delta\vec{z}_1) + m_2(\Delta\vec{z}_2)}{m_1 + m_2}$$

Q. **In the figure given below, if the 4 kg block is shifted by 4 meter towards right, then by what distance the 2 kg block should be shifted so that the centre of mass is shifted by 1 m towards right?**



Sol. Using the relation:

$$\Delta\bar{x}_{COM} = \frac{m_1\Delta\bar{x}_1 + m_2\Delta\bar{x}_2}{(m_1 + m_2)}$$

Where $\Delta\bar{x}_{COM}$ = displacement of centre of mass in x direction.

$\Delta\bar{x}_1$ = Displacement of mass m_1.

$\Delta\bar{x}_2$ = Displacement of mass m_2.

Step 1: Assign direction. Let right direction be +ve.

Now put all the values with proper sign in the formula.

$$+1 = \frac{2(\Delta\bar{x}_1) + 4(+4)}{2 + 4}$$

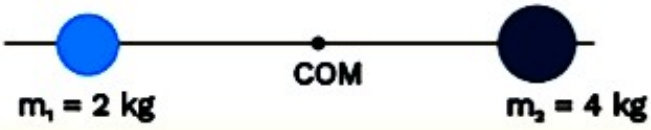
Positive sign is used since COM is shifted towards right.

$$6 = 2(\Delta\bar{x}_1) + 16$$

$$\Delta x_1 = -5 \text{ m}$$

Negative sign indicates that the 2 kg block should be shifted towards left by 5 m.

Q. **Given that the 4 kg block is shifted towards right by 4m, then find the displacement of 2 kg block in order to keep the centre of mass at the same initial position.**



Sol.

$$\Delta x_{COM} = \frac{m_1(\Delta x_1) + m_2(\Delta x_2)}{m_1 + m_2}$$

$$0 = \frac{2(\Delta \vec{x}_1) + 4(4)}{2 + 4}$$

$$0 = 2(\Delta \vec{x}_1) + 16$$

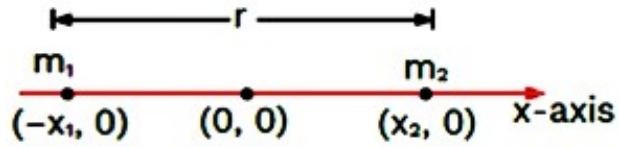
$$\Delta \vec{x}_1 = -8\,m$$

So, 2 kg block should be shifted by 8 m towards left.

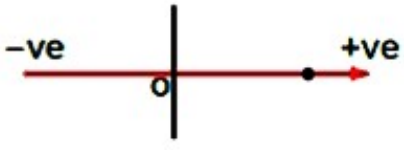
Q. **Two particles having masses m_1 and m_2 are placed at a separation r between them. Let m_2 be displaced away from m_1 by a distance x. By what distance, m_1 should be displaced in order to keep the position of centre of mass un-changed?**

Sol. Let the masses be placed on x-axis with centre of mass at origin.
Using,

$$\overline{x_{cm}} = \frac{m_1\overline{x_1} + m_2\overline{x_2}}{m_1 + m_2}$$



Thus, $\Delta \vec{x}_{CM} = \frac{m_1\Delta\vec{x}_1 + m_2\Delta\vec{x}_2}{m_1 + m_2}$



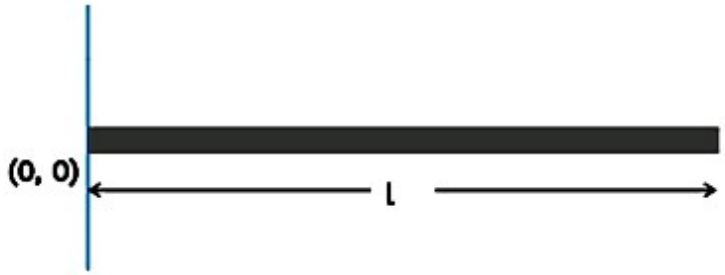
$$\Rightarrow \quad 0 = \frac{m_1\Delta\vec{x}_1 + m_2 x\,\hat{i}}{m_1 + m_2}$$

$\Delta \vec{x}_1 = \frac{-m_2 x}{m_1}\hat{i}$. Thus, m_1 should be displaced by $\frac{m_2 x}{m_1}$ away from origin.

CENTRE OF MASS FOR CONTINUOUS MASS DISTRIBUTION

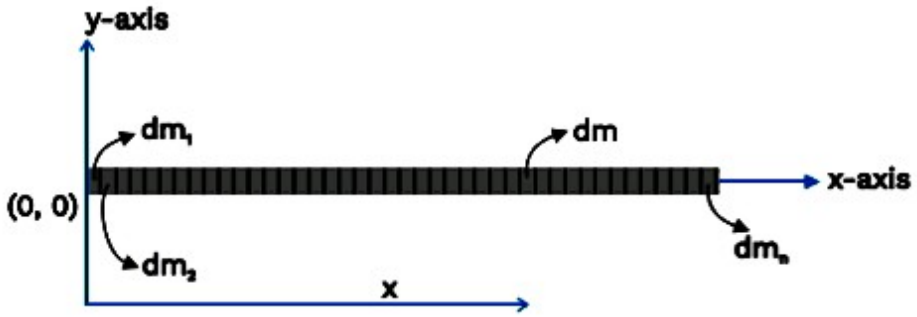
For a continuous distribution, the centre of mass is calculated by dividing the continuous distribution into discrete symmetrical parts. The centre of mass of a symmetric part is at its geometric centre.

Consider a rod of mass m and length l as shown in the figure. The mass m of the rod is continuously and uniformly distributed along the total length of the rod.



Formula derivation by using the formula for discrete mass system:

The rod can be assumed to be made up of large number of elementary particles of masses dm_1 , dm_2 , dm_3 ,.....dm_n.



Now we can use the formula of centre of mass of discrete mass system,

$$x_{COM} = \frac{(dm_1)(x_1) + (dm_2)(x_2) + (dm_3)(x_3) + ... + (dm_n)(x_n)}{dm_1 + dm_2 + dm_3 + ... + dm_n}$$

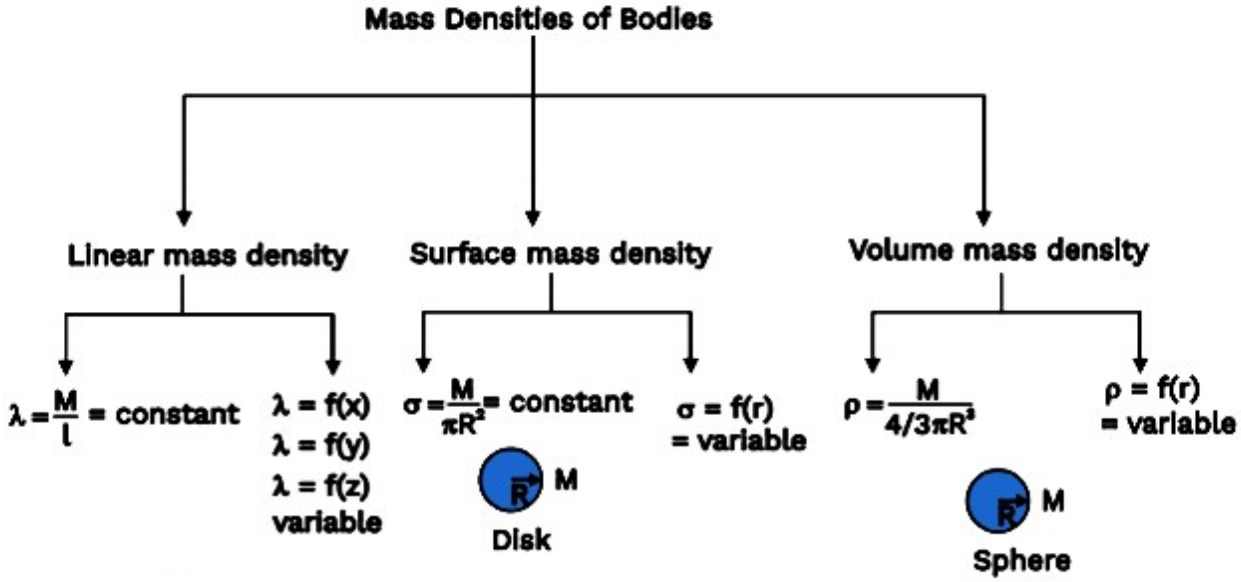
$$x_{COM} = \frac{\sum_{i=1}^{n} dm_i x_i}{\sum_{i=1}^{n} dm_i}$$

Since $n \to \infty$, so $\sum_{i=1}^{n} dm_i x_i = \int x.dm$ and $\sum_{i=1}^{n} dm_i = \int dm$.

So, we can write that,

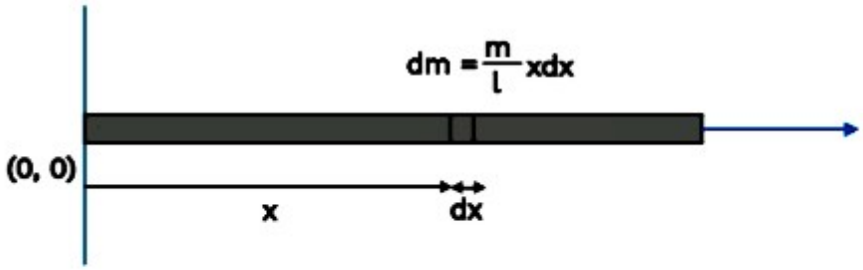
$$\boxed{x_{COM} = \frac{\int x dm}{\int dm}}$$ (where x is the location of centre of mass of a general dm mass element).

The mass distribution in a continuous body can be of three types. It can be linear, areal or volumetric mass distribution. Below figure describes the possible cases of mass distribution in a body.



Calculation of elemental mass of some standard bodies:

1. **Straight Rod:**

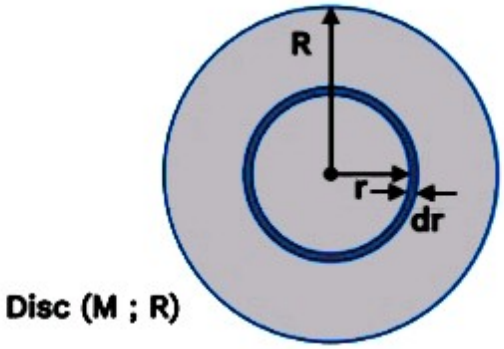


Elemental mass, $dm = \lambda dx = \left(\frac{m}{\ell}\right)(dx)$

If λ is not constant, then $dm = \lambda . dx$, where $\lambda = f(x)$.

2. **Circular Disc:**

Elemental mass in a circular disc is a circular ring of mass dm, radius r and very small thickness dr as shown in the figure.

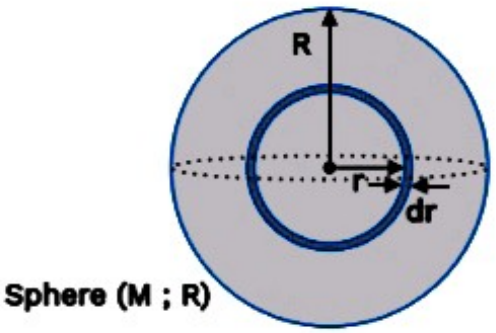


Mass of the elemental ring, $dm = \left(\frac{M}{\pi R^2}\right)(2\pi r\, dr) = \sigma(2\pi r\, dr)$, where σ = surface mass density.

If σ is not constant, then $dm = \sigma(2\pi r\, dr)$, where $\sigma = f(r)$.

3. Sphere:

Elemental mass in a sphere is a spherical shell of mass dm, radius r and thickness dr as shown in the figure.



Sphere (M ; R)

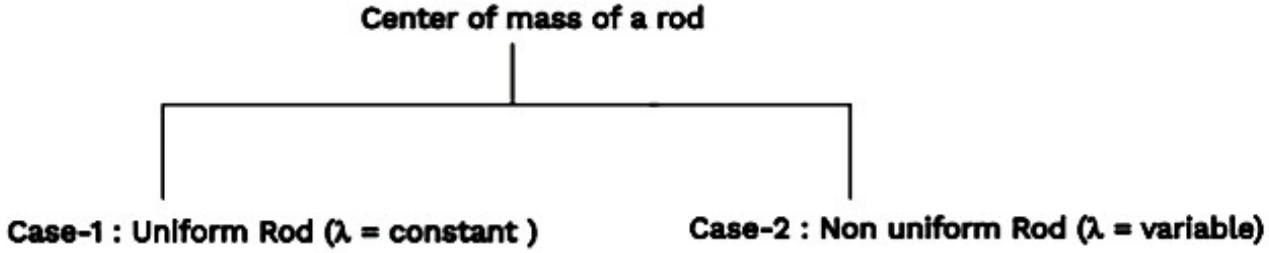
Mass of the elemental shell, $dm = \rho(4\pi r^2\, dr) = \left(\frac{M}{\frac{4}{3}\pi R^3}\right)(4\pi r^2\, dr)$

If ρ is not constant, then $dm = \rho(4\pi r^2\, dr)$ where $\rho = f(r)$.

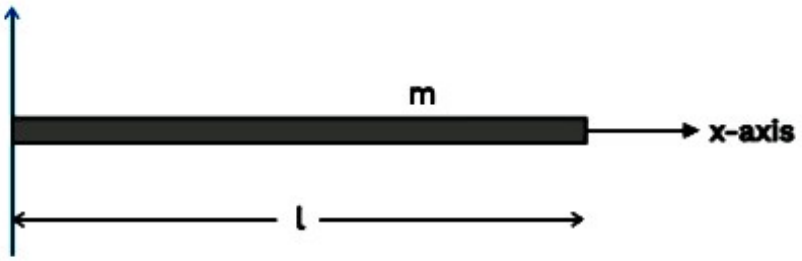
Calculation of centre of mass of some standard bodies like rod, half ring, half disc, hollow hemisphere, solid hemisphere, hollow cone and solid cone is described in this section.

1. Centre of mass of a rod

There are two cases possible for mass distribution in a rod.

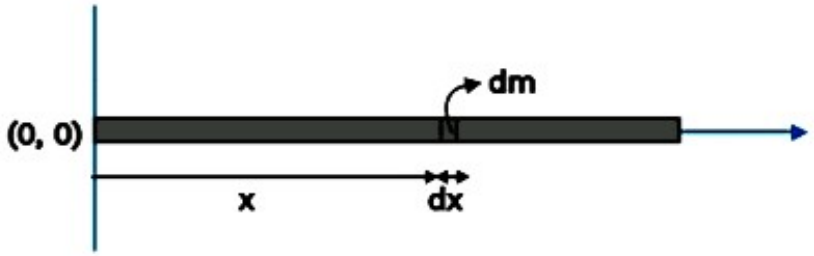


Case-1: Centre of mass of a uniform rod.



We know the formula of centre of mass of distributed mass system.
Since the rod is along the x-axis, so its centre of mass must exist on x-axis.

$x_{CM} = \frac{\int x.dm}{\int dm}$ where x = Location of centre of mass of dm (mass element) mass.

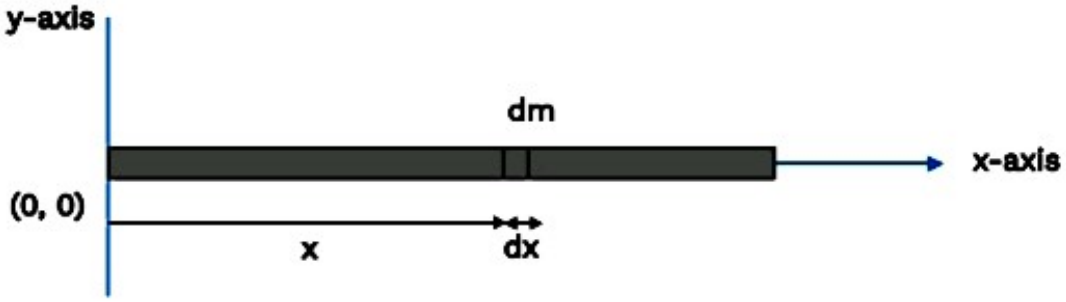


$$x_{CM} = \frac{\int_0^{\ell} x\left(\frac{m}{\ell}dx\right)}{\int_0^{\ell}\frac{m}{\ell}dx} = \frac{\frac{m}{\ell}\int_0^{\ell} xdx}{\frac{m}{\ell}\int_0^{\ell} dx} = \frac{\left[\frac{x^2}{2}\right]_0^{\ell}}{[x]_0^{\ell}} = \frac{\ell^2}{2\ell} = \frac{\ell}{2}$$

Case-2: Centre of mass of a non-uniform rod

Suppose the linear mass density of the rod is given as $\lambda = k x^2$, where k is a constant.

Now use following steps to find the centre of mass.



Step-1

Write the formula of centre of mass.

$x_{CM} = \frac{\int x.dm}{\int dm}$ (x= Location of centre of mass of dm mass)

Consider an elemental mass dm of length dx at distance x from the origin.

Since $dx \to 0$, so the location of centre of mass of dm mass is x.

Step-2

Find the mass dm.

dm = λ. dx ... (where, λ is the linear mass density)

In this case, λ is a variable. So, $dm = (k x^2)\,dx$

Step-3

Put limit on integration and calculate the location of centre of mass as,

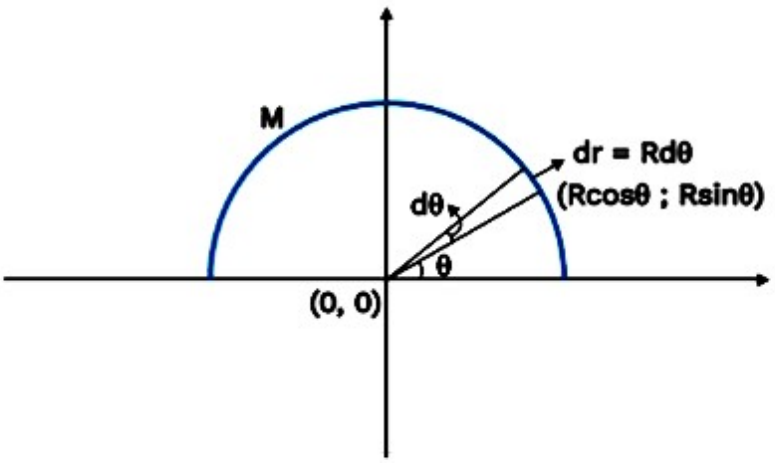
$$x_{CM} = \frac{\int_0^{\ell} x\left[k\,x^2 dx\right]}{\int_0^{\ell}\left(kx^2\right)dx} = \frac{k\int_0^{\ell} x^3 dx}{k\int_0^{\ell} x^2 dx} = \frac{\left[\frac{x^4}{4}\right]_0^{\ell}}{\left[\frac{x^3}{3}\right]_0^{\ell}}$$

$$= \frac{\ell^4}{4\times\frac{\ell^3}{3}} = \frac{3}{4}\ell$$

$x_{CM} = \frac{3\ell}{4}$, which is the location of centre of mass.

2. Centre of mass of a half Ring

Consider a small elemental mass on the half ring as shown in the figure.



If R is the geometrical radius and M is mass of the half ring, then linear mass density λ will be,

$$\lambda = \frac{M}{\pi R}$$

The centre of mass coordinates of the elemental mass are Rcosθ and Rsinθ.

$$x_{CM} = \frac{\int (x)\,dm}{\int dm} \qquad y_{CM} = \frac{\int y dm}{\int dm}$$

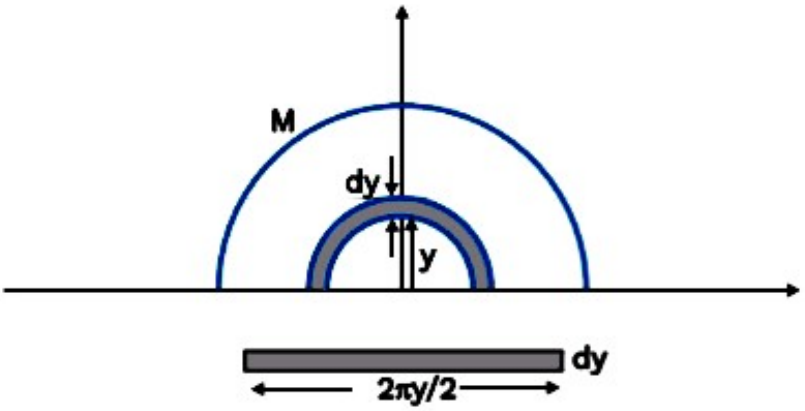
$$x_{CM} = \frac{\int (R\cos\theta)\left[\lambda\,dr\right]}{\int \lambda\,dr} \qquad y_{CM} = \frac{\int (R\sin\theta)\left[\lambda\,dr\right]}{\int \lambda\,dr}$$

$$= \frac{\int_0^{\pi} R\cos\theta\left[\frac{M}{\pi R}.Rd\theta\right]}{M} \qquad = \frac{\int_0^{\pi} R\sin\theta\left[\frac{M}{\pi R}.Rd\theta\right]}{M}$$

$$x_{CM} = \left(\frac{R}{\pi}\right)\int_0^{\pi}\cos\theta\, d\theta = \left(\frac{R}{\pi}\right)\left[(\sin\theta)\right]_0^{\pi} \qquad y_{CM} = \left(\frac{R}{\pi}\right)\int_0^{\pi}\sin\theta\, d\theta = \left(\frac{R}{\pi}\right)\left[(-\cos\theta)\right]_0^{\pi}$$

$$x_{CM} = \left(\frac{R}{\pi}\right)(\sin\pi - \sin 0) = 0 \qquad y_{CM} = \frac{R}{\pi}(-\cos\pi + \cos 0) = \frac{2R}{\pi}$$

3. Centre of mass of uniform half disc



Elemental half ring is as shown. Its mass can be calculated as,

$$dm = \frac{M}{\left(\frac{\pi R^2}{2}\right)}\left[\frac{2\pi y}{2}dy\right]$$

a. $x_{CM} = 0$ ... (By symmetry)

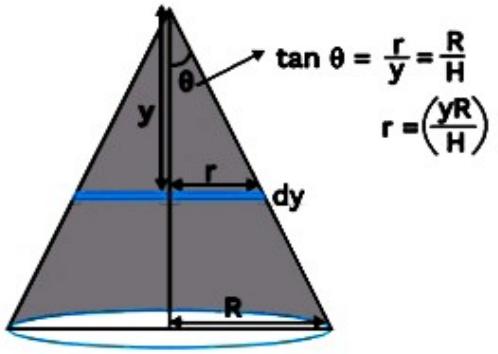
b. $y_{CM} = \frac{\int y'dm}{\int dm}$, where y' is the location of centre of mass of the elemental half ring.

Here, $y' = \frac{2y}{\pi}$

So, $y_{CM} = \frac{\int_0^R\left(\frac{2y}{\pi}\right).\frac{M}{\left(\frac{\pi R^2}{2}\right)}.\left[\frac{2\pi y}{2}dy\right]}{\int dm} = \frac{\frac{4M}{\pi R^2}\int_0^R y^2\, dy}{M} = \frac{4R}{3\pi}$

4. Centre of mass of solid cone

Consider a solid cone of base radius R and vertical height H having mass M as shown.



Volume mass density, $\rho = \dfrac{M}{\frac{1}{3}\pi R^2 H} = \dfrac{3M}{\pi R^2 H}$

Consider an elemental disc of thickness dy as shown in figure.

Mass of the elemental disc can be calculated as,

$$dm = \int \rho . dV = \int \frac{3M}{\pi R^2 H}\left(\pi r^2 dy\right)$$

Now,

$$y_{CM} = \frac{\int y . dm}{\int dm}$$

$$= \frac{\int_0^H y\left\{\frac{3Mr^2 dy}{R^2 H}\right\}}{\int dm} = \frac{\int_0^H y\left\{\frac{3Mr^2 dy}{R^2 H}\right\}}{M} = \int_0^H y\left\{\frac{3\, r^2 dy}{R^2 H}\right\}$$

Put $r = \dfrac{yR}{H}$, then,

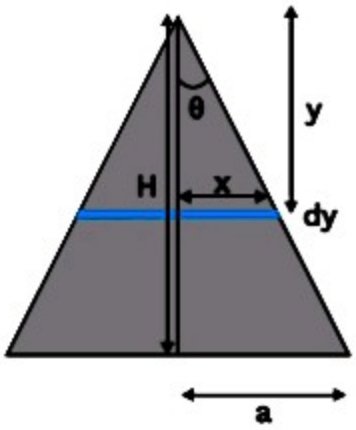
$$y_{CM} = \int_0^H y\left\{\frac{3\, y^2\, dy}{H^3}\right\} = \frac{3}{H^3} \times \left[\frac{y^4}{4}\right]_0^H = \frac{3H}{4}$$

This is the distance of centre of mass from the top of the cone.

Distance of centre of mass of solid cone from base = H/4.

5. Centre of mass of triangular plate

Consider a triangular plate of mass M, base length 2a and height H as shown in the figure.



From the figure, $\tan\theta = \frac{x}{y} = \frac{a}{H} \quad \Rightarrow x = \frac{a\,y}{H}$

Consider an elemental strip of thickness dy as shown. Mass of this elemental strip can be calculated as,

$$dm = \left\{\frac{M}{\frac{1}{2}\times 2a\times H}\right\}(2x.dy) = \frac{2M}{aH}x.dy$$

Centre of mass can be found as,

$$y_{CM} = \frac{\int y\,dm}{\int dm}$$

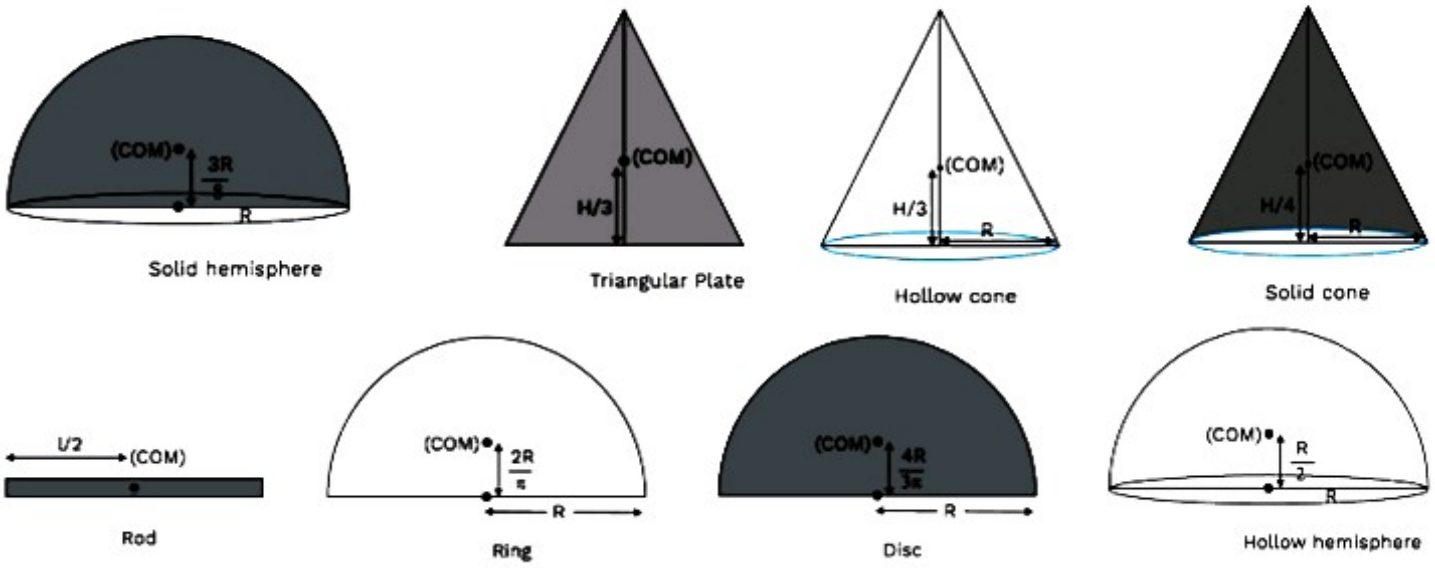
$$= \frac{\int y.\frac{2M}{aH}x\,dy}{\int dm} = \frac{\int y.\frac{2M}{aH}\left(\frac{a\,y}{H}\right)dy}{\int dm} = \frac{\int y.\frac{2M}{aH}\left(\frac{a\,y}{H}\right)dy}{M}$$

$$y_{CM} = \frac{2}{H^2}\int_0^H y^2\,dy = \frac{2}{3}H$$

This is the distance of centre of mass from the top corner of the triangle.

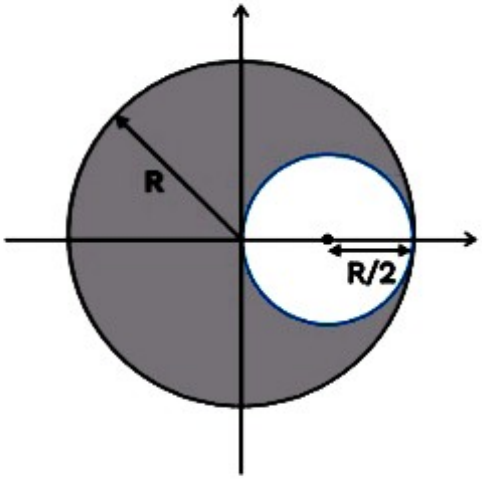
Then, the distance of centre of mass from bottom side = H/3.

Centre of mass of some standard bodies



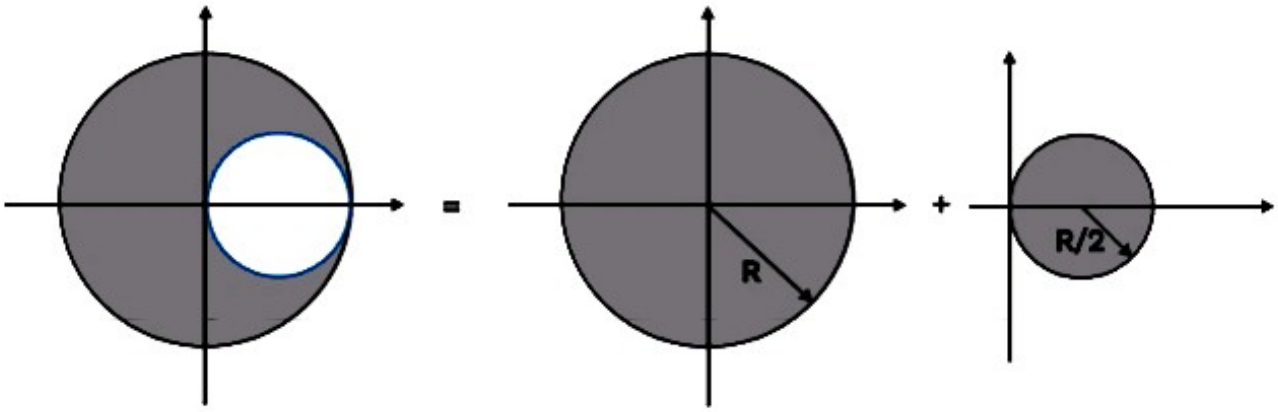
Centre of mass in case of cavities

Suppose there is a uniform circular lamina of radius R. Now, a circular portion of radius R/2 is removed from the lamina as shown in figure.



Now, to find the position of centre of mass of the remaining body, we use the concept of negative mass.

It can be seen that this body is a combination of two bodies as shown. The smaller circle of radius R/2 is to be taken as a negative mass which when subtracted from the complete circle, gives the desired lamina.



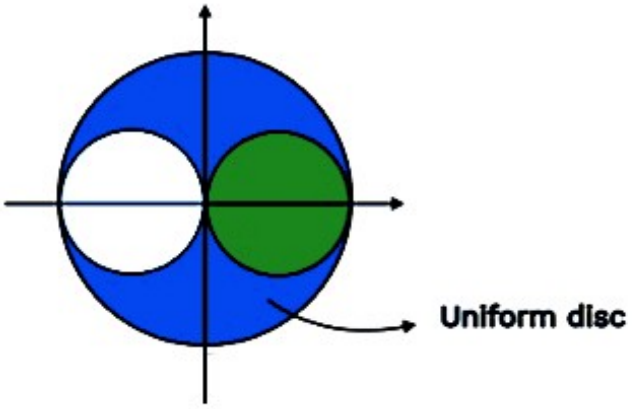
For the complete circular lamina, the centre of mass lies at centre (0, 0).
Now use the formula for discrete bodies considering the smaller disc to be negative mass.

$$x_{CM} = \frac{m_1x_1 + m_2x_2}{m_1 + m_2}$$

$$x_{CM} = \frac{\sigma\left(\pi R^2\right)\times(0) + \left(\frac{-\sigma\pi R^2}{4}\right)\times\left(+\frac{R}{2}\right)}{\sigma\left(\pi R^2\right) + \left(-\frac{\sigma\pi R^2}{4}\right)} = \frac{0 - \frac{\sigma\pi R^2 \times R}{8}}{\sigma\pi R^2\left[1-\frac{1}{4}\right]}$$

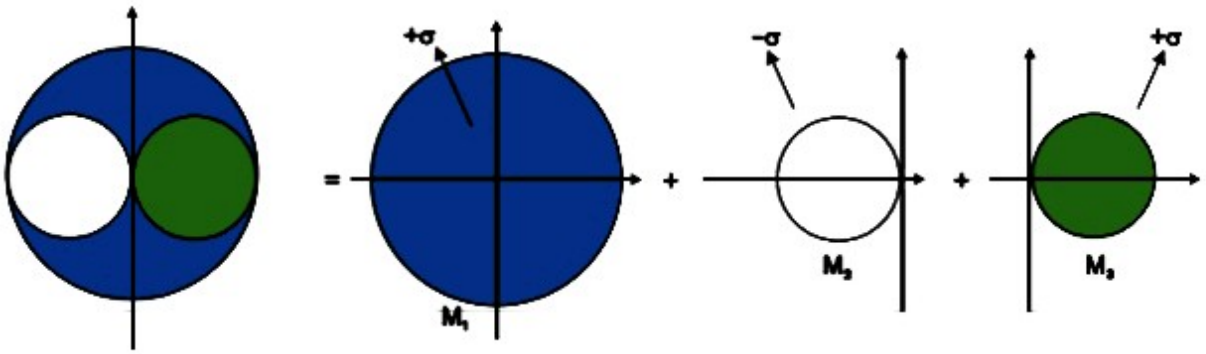
$$x_{CM} = \frac{-R}{8\left[\frac{3}{4}\right]} = -\frac{R}{6}, \text{ and } y_{CM} = 0 \text{ by symmetry.}$$

Q. **Figure shows a uniform disc of disc R, from which a hole of Radius $\frac{R}{2}$ has been cut-out from left of the centre and is placed on the right of the centre of the disc.**



Find the centre of mass of the resulting disc.

Sol.



Let surface mass density of disc be σ.

$M_1 = \sigma(\pi R^2)$; $x_1 = 0$ (location of COM of full disc)

$M_2 = \sigma(\pi)\left(\frac{R}{2}\right)^2$; $x_2 = \frac{R}{2}$ (location of COM of subtracted disc)

It is negative because this part is subtracted from the disc.

$M_3 = (+\sigma)(\pi)\left(\frac{R}{2}\right)^2$; $x_3 = \frac{R}{2}$ (location of COM of the added disc)

Now using the formula of centre of mass,

$$x_{CM} = \frac{M_1x_1 + M_2x_2 + M_3X_3}{M_1 + M_2 + M_3}$$

$$= \frac{\sigma\left(\pi R^2\right)(0) + \frac{(-\sigma)(\pi R^2)}{4}\left(-\frac{R}{2}\right) + \sigma\left(\frac{\pi R^2}{4}\right)\left(+\frac{R}{2}\right)}{\sigma(\pi R^2) + (-\sigma)\left(\frac{\pi R^2}{4}\right) + \sigma\left(\frac{\pi R^2}{4}\right)}$$

$$= 0 + \frac{\sigma\pi R^2\left(\frac{R}{8}\right) + \sigma\pi R^2 \times \left(\frac{R}{8}\right)}{\sigma\pi R^2}$$

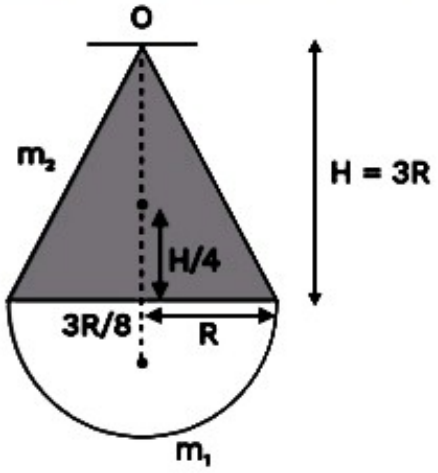
$$= \frac{\sigma\pi R^2\left[\frac{R}{8} + \frac{R}{8}\right]}{\sigma\pi R^2} = \frac{R}{4}$$

$$x_{CM} = \frac{R}{4}$$

And, $y_{CM} = 0$ due to symmetry.

Q. **Find the centre of mass of a uniformly dense object which consists of a solid hemisphere joined at the base of a cone. Height of the cone is H = 3R and the base radius is R.**

Sol. This is combination of two bodies. We can use the position of the centre of mass of the cone and the hemisphere as the position for their whole masses.



$$y_{CM} = \frac{m_2\left(\frac{H}{4}\right) + m_1\left(\frac{-3R}{8}\right)}{(m_1 + m_2)}$$

$$y_{CM} = \frac{m_2\left(\frac{3R}{4}\right) + m_1\left(\frac{-3R}{8}\right)}{m_1 + m_2}$$

It is given that mass density is uniform. So,

$$m_1 = \frac{2}{3}\pi R^3 \rho;$$

$$m_2 = \frac{1}{3}\pi R^2 H\rho = \left(\frac{1}{3}\pi R^2 \times 3R\right)\rho = \pi R^3 \rho$$

Now,

$$y_{CM} = \frac{\pi R^3\rho\left(\frac{3R}{4}\right) - \frac{2}{3}\pi R^3\rho\left(\frac{3R}{8}\right)}{\pi R^3\rho + \frac{2}{3}\pi R^3\rho} = \frac{3R - R}{4 \times \frac{5}{3}} = \frac{3R}{10} = \frac{H}{10}$$

MOTION OF CENTRE OF MASS

Position vector of centre of mass, $\vec{r}_{CM} = \frac{M_1\vec{r}_1 + M_2\vec{r}_2 + \ldots\ldots\ldots + M_n\vec{r}_n}{M_1 + M_2 + \ldots\ldots + M_n}$

On differentiating both sides wrt time t,

$$\frac{d\vec{r}_{CM}}{dt} = \frac{M_1\frac{d\vec{r}_1}{dt} + M_2\frac{d\vec{r}_2}{dt} + \ldots\ldots\ldots + M_n\frac{d\vec{r}_n}{dt}}{M_1 + M_2 + \ldots\ldots + M_n}$$

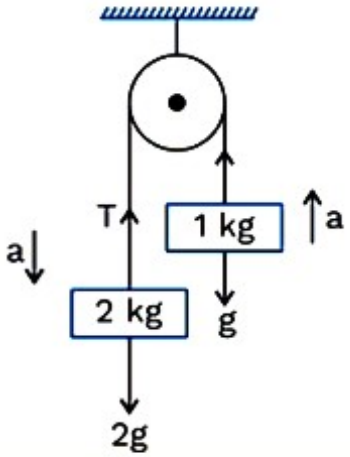
Velocity of centre of mass, $\vec{v}_{CM} = \dfrac{M_1\vec{v}_1 + M_2\vec{v}_2 + \ldots\ldots\ldots + M_n\vec{v}_n}{M_1 + M_2 + \ldots\ldots + M_n}$

On differentiating both sides wrt time t,

$$\frac{d\vec{v}_{CM}}{dt} = \frac{M_1\dfrac{d\vec{v}_1}{dt} + M_2\dfrac{d\vec{v}_2}{dt} + \ldots\ldots\ldots + M_n\dfrac{d\vec{v}_n}{dt}}{M_1 + M_2 + \ldots\ldots + M_n}$$

Acceleration of centre of mass, $\vec{a}_{CM} = \dfrac{M_1\vec{a}_1 + M_2\vec{a}_2 + \ldots\ldots\ldots + M_n\vec{a}_n}{M_1 + M_2 + \ldots\ldots + M_n}$

Q. **Calculate the acceleration of centre of mass in the following case.**



Sol. By Newton's 2nd law,

$$2g - T = 2a$$
$$T - 1g = 1a$$
$$2g - 1g = 2a + a$$
$$g = 3a$$

Acceleration of the masses, $a = \dfrac{g}{3}$

Acceleration of centre of mass, $\vec{a}_{CM} = \dfrac{M_1\vec{a}_1 + M_2\vec{a}_2}{M_1 + M_2}$

For $M_1 = 2kg$, $a_1 = \dfrac{g}{3}\downarrow$

For $M_2 = 1kg$, $a_2 = \dfrac{g}{3}\uparrow$

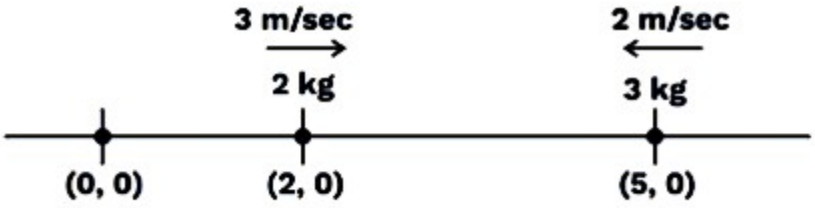
Taking downward direction as positive,

$$\vec{a}_{CM} = \frac{2\left(\frac{g}{3}\right) + 1\left(-\frac{g}{3}\right)}{2+1}$$

$$= \frac{\frac{2g}{3} - \frac{g}{3}}{3} = \frac{g}{9}$$

$\therefore$ $\boxed{\vec{a}_{cm} = \frac{g}{9}}$ in downward direction.

Q. **Calculate-**
a) Position of centre of mass
b) Velocity of centre of mass, for the given system.



Sol. Velocity of centre of mass, $\vec{v}_{CM} = \frac{M_1\vec{v}_1 + M_2\vec{v}_2}{M_1 + M_2}$

$$v_{CM} = \frac{2(3) + 3(-2)}{2+3} = 0$$

Position of centre of mass, $x_{CM} = \frac{M_1x_1 + M_2x_2}{M_1 + M_2} = \frac{2(2) + 3(5)}{2+3} = \frac{19}{5}\,m$

Since the velocity of centre of mass is zero, so the position of centre of mass will remain the same at every time.

Conceptual Points

$\vec{F}_{ext} = M\vec{a}_{CM}$

If $\vec{F}_{ext} = 0$,

$\Rightarrow \quad \vec{a}_{CM} = 0$

$\Downarrow$

$$\boxed{\vec{v}_{CM} = \text{constant}}$$

$\Rightarrow$ If initially both particles are at rest, i.e. $\vec{v}_{CM} = 0$, then the centre of mass remains at rest. Then,

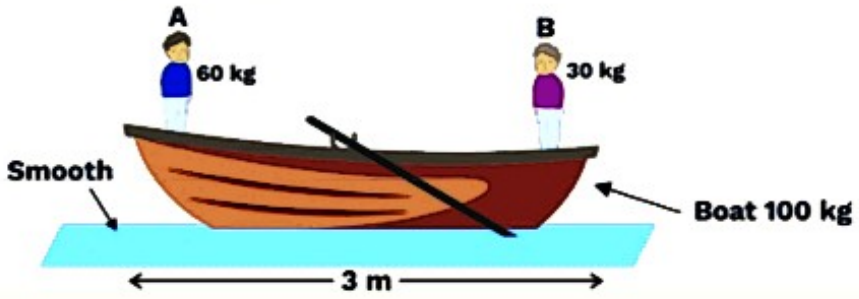
$$\Delta\vec{x}_{CM} = \frac{M_1\,\Delta\vec{x}_1 + M_2\,\Delta\vec{x}_2}{M_1 + M_2}$$

$$0 = \frac{M_1\,\Delta\vec{x}_1 + M_2\,\Delta\vec{x}_2}{M_1 + M_2}$$

$$\boxed{M_1\,\Delta\vec{x}_1 + M_2\,\Delta\vec{x}_2 = 0}$$

Above equation is applicable when external force acting on a system is zero but the individual bodies within the system are moving.

Q. **Initially the system of boat+2 men is at rest. If A goes towards B for a hand-shake, then find the displacement of boat.**



Sol. Initially the system is at rest, so displacement of centre of mass will be zero.

$$m_1 \Delta\vec{x}_1 + m_2 \Delta\vec{x}_2 + m_3 \Delta\vec{x}_3 = 0$$

Consider the displacement of the boat to be x towards left,

Then displacement of A = 3-x , towards right

And, displacement of B = x, towards left.

Then,

$$60(3-x) + 30(-x) + 100(-x) = 0$$

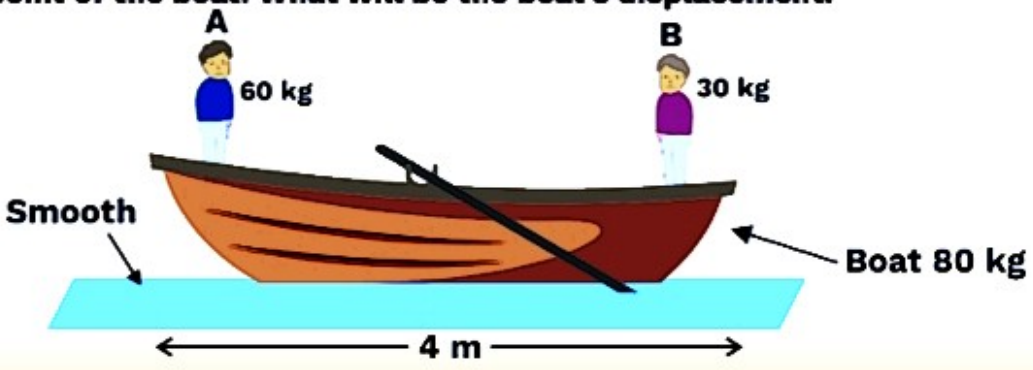
$$\Rightarrow \quad 180 - 60x - 30x - 100x = 0$$

$$\Rightarrow \quad 180 = 190x$$

$$\Rightarrow \quad \boxed{x = \frac{18}{19} m}$$

So, the boat is displaced by $\frac{18}{19}$ m towards left

Q. **Person A and person B are moving towards each other and they meet at the midpoint of the boat. What will be the boat's displacement.**



Sol. Let the displacement of boat be x towards left.

Then, displacement of A = (2 – x) towards right.

And, displacement of B = (2 + x) towards left.

Now,

$$M_1\vec{X}_1 + M_2\vec{X}_2 + M_3\vec{X}_3 = 0$$

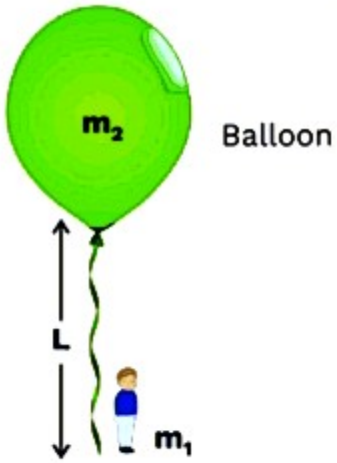
$$60(2-x) + \left[-30(2+x)\right] + (-80x) = 0$$

$$\Rightarrow \quad 120 - 60x - 60 - 30x - 80x = 0$$

$$\Rightarrow \quad 60 = 170x$$

$$\Rightarrow \quad \boxed{x = \frac{6}{17}\,m}$$

Q. **A balloon, a light rope and a man all are at rest in air as shown in the figure.**



If the man wants to reach the other end of the rope, then find by how much distance will the balloon descend?

Sol. If the balloon descends by x, then man has to move (L-x) to reach the other end of the rope.

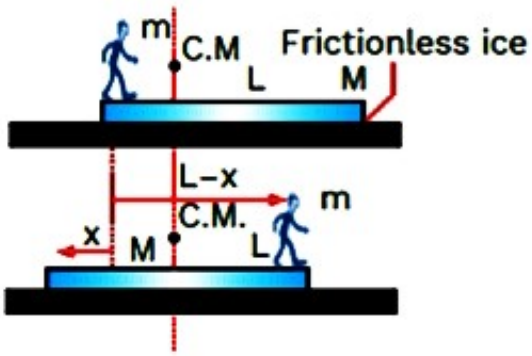
$$m_1 x_1 + m_2 x_2 = 0$$

$$m_1(L-x) + m_2(-x) = 0$$

$$\Rightarrow \quad \boxed{x = \frac{m_1 L}{m_1 + m_2}}.$$

Q. **A man of mass m stands at the left end of a uniform plank of length L and mass M, which lies on a frictionless surface of ice. If the man walks to the other end of the plank, then by what distance does the plank slide on the ice?**

Sol. The net external force on the system (man + plank) is zero.
So, the centre of mass of the system remains at same position during their motion. The plank will shift towards left.



Let the displacement of plank relative to ground = x

The displacement of man relative to ground= –(L – x)

Moments of masses about C.M. = 0

$Mx - m(L - x) = 0$

$\Rightarrow \quad Mx = m(L - x)$

$\Rightarrow \quad x = \frac{mL}{m + M}$

LINEAR MOMENTUM OF A SYSTEM OF PARTICLES

The centre of mass of a system of particles moves as if all the mass of the system is at the centre of mass and all the external forces are applied at that point.

The linear momentum of a particle is defined as

$$\vec{P} = m\vec{v}$$

and by Newton's second law,

$$\vec{F} = \frac{d\vec{p}}{dt}$$

i.e. the rate of change of linear momentum of a particle is equal to the net force acting on the object.

Now, $M\vec{v}_{cm} = \sum_{i=1}^{n} m_i \vec{v}_i$

And, $\sum_{i=1}^{n} \vec{P}_i = \sum_{i=1}^{n} m_i \vec{v}_i$

So, $\vec{P} = M\vec{v}_{cm}$ ...(i)

Differentiating the above equation w.r.t. time, we get

$$\frac{d\vec{p}}{dt} = M\frac{d\vec{v}_{cm}}{dt} = M\vec{a}_{cm} = \vec{F}_{ext}$$

Now, if the net external force on the system is zero, the linear momentum of the system is conserved and the centre of mass will move with constant velocity.

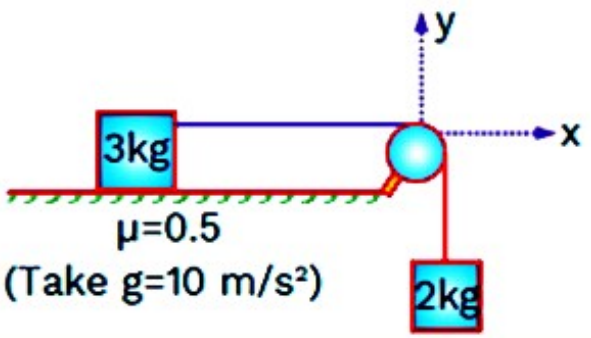
Q. **Two particles of mass 2kg and 5 kg are moving along the same line with speeds 3 m/s and 5 m/s respectively. Calculate the speed of the centre of mass of the system if both the particles are moving in the same direction.**

Sol. $$v_{cm} = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$$

Substituting the value, we get

$$v_{cm} = \frac{(2 \times 3) + (5 \times 5)}{3} = \frac{6 + 25}{3} = \frac{31}{3} m/s$$

Q. **Find the acceleration of centre of mass of the system shown.**



Sol. Weight of the 2 kg mass will be the driving force for the system.
Driving force on the system = 2 g = 20 N
Limiting friction = μ mg = 0.5 × 3 × 10 = 15 N
Magnitude of acceleration of both masses,

$$a = \frac{20N - 15N}{2+3} = \frac{5}{5} = 1 m/s^2$$

$$m_1 = 3kg, \quad \vec{a}_1 = 1\hat{i}\, m/s^2$$

$$m_2 = 2kg, \quad \vec{a}_2 = -1\hat{j}\, m/s^2$$

Acceleration of centre of mass, $\overline{a_{cm}} = \dfrac{m_1\overline{a_1} + m_2\overline{a_2}}{m_1 + m_2}$

$$= \frac{3\hat{i} - 2\hat{j}}{3+1} = \frac{3}{5}\hat{i} - \frac{2}{5}\hat{j}$$

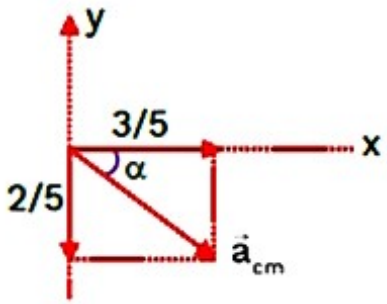
Magnitude of acceleration of centre of mass,

$$a_{cm} = \sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{2}{5}\right)^2}$$

$$= \frac{\sqrt{13}}{5} m/s^2$$

$$\tan\alpha = \frac{2/5}{3/5} = \frac{2}{3}$$

$\alpha = \tan^{-1}\left(\frac{2}{3}\right)$ below horizontal



Q. **Two particles A and B of masses 5 kg and 2 kg are projected simultaneously from same ground level as shown in figure. Find the maximum height reached by centre of mass.**
[Assume that only gravitational force of attraction by the earth is present]
[g = 10 m/s²]



Sol. Both particles are projected in the air, so both particles have acceleration g downwards.

Thus, $\vec{a}_{CM} = -g\hat{j}$

Now, if $\vec{a}_{CM}$ remains constant, then equation of motion can be applied to centre of mass. In this equation, maximum height attained by the centre of mass is asked, thus we need not to think about the horizontal movement of centre of mass.

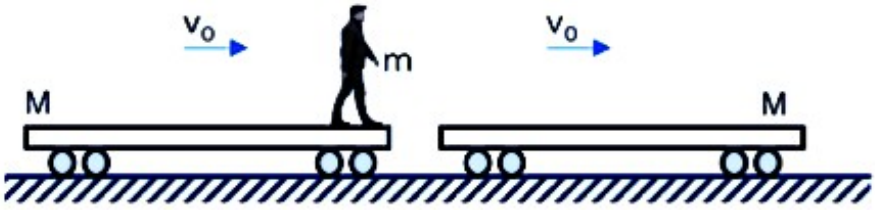
$$\vec{v}_{CMy} = \frac{5(10\sin 30°)\hat{j} + 2(17\sin 30°)\hat{j}}{5+2} = \frac{25+17}{7}\hat{j} = \frac{42}{7} = 6\,m/s\,\hat{j}$$

$$\vec{a}_{CMy} = -10\hat{j}$$

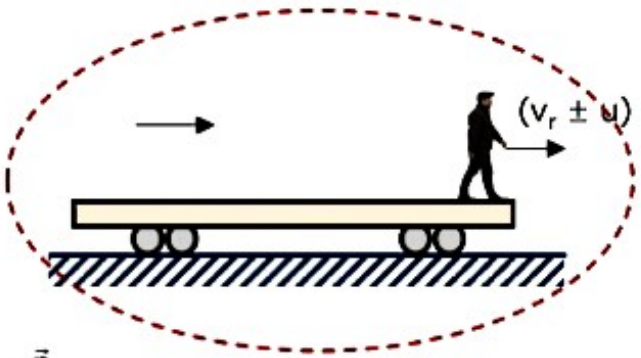
$$\vec{S}_{CM\,max} = \frac{(V_{CMy})^2}{2a_{CMy}} = \frac{36}{2\times 10} = 1.8\,m.$$

CART JUMPING PROBLEMS

Q. **Two identical buggies each of mass M are moving one after other on a frictionless surface with the some velocity. A person having mass m is on the rear buggy. At a certain moment, he jumps to the front buggy with some relative velocity to the buggy. What is the velocity with which the buggies will move after that.**



Sol. There is no external force acting on "rear buggy+ man" system in horizontal direction. So, linear momentum of the system remains conserved.
If velocity of rear buggy just after jumping is $\vec{V}_r$, then velocity of man just after jumping will be $(\vec{V}_r + \vec{u})$.



$\vec{p}_i = (M+m)\vec{V}_0$ $\vec{p}_f = M\vec{V}_r + m(\vec{V}_r + \vec{u})$ and $\vec{p}_i = \vec{p}_f$

$$\Rightarrow \quad (M+m)\vec{V}_0 = M\vec{V}_r + m(\vec{V}_r + \vec{u})$$

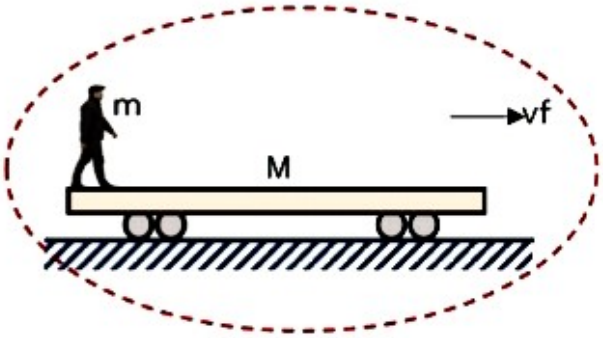
$$\vec{V}_f = \vec{V}_0 - \frac{m\vec{u}}{(M+m)}$$

Now considering front buggy and man as system, and again conserving linear momentum in horizontal direction

$$M_0\vec{V}_0 + m\left(\vec{u} + \vec{V}_f\right) = (m+M)\vec{V}_f$$

$$M\vec{V}_0 + m\left(\vec{u} + \vec{V}_0 - \frac{m\vec{u}}{(M+m)}\right) = (m+M)\vec{V}_f$$

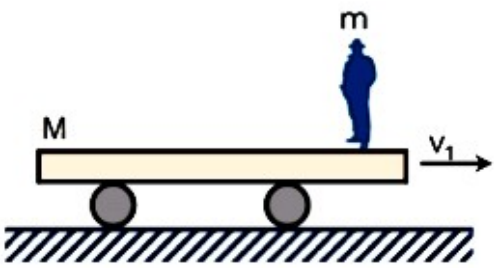
$$\Rightarrow \quad \vec{V}_f = \vec{V}_0 + \frac{Mm\vec{u}}{(M+m)^2}$$

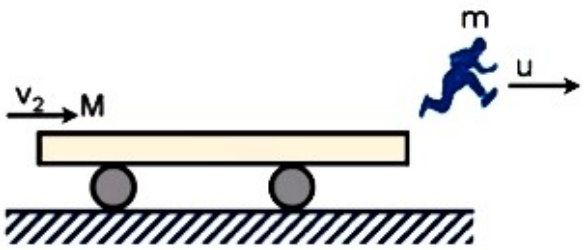


Front buggy

Q. **A car having mass M with a boy having mass m is moving with a velocity v_1. The boy jumps with a velocity u wrt the car in the direction of motion of car. Find the final velocities of the boy and that of the car after the jump.**

Sol.





According to momentum conservation,

$$(M + m)v_1 = Mv_2 + m(u + v_2)$$

Velocity of car after jump is $v_2 = \frac{(M+m)v_1 - mu}{M+m}$

Velocity of boy after jump is $u + v_2 = \frac{(M+m)v_1 + Mu}{M+m}$

LAW OF CONSERVATION OF LINEAR MOMENTUM

We know that,

$$\vec{F}_{ext} = (m_1 + m_2)\vec{a}_{CM}$$

Suppose that the sum of all the external forces is zero.

In this case, the acceleration of centre of mass becomes zero. The motion of centre of mass is governed by Newton's first law, and it moves in a straight line.

The velocity of centre of mass is given as,

$$\vec{v}_{CM} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2}$$

As $\vec{a}_{CM} = 0$, $\vec{v}_{CM}$ = constant. Therefore, it can be said that

$$\frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2} = \text{constant}$$

or, $m_1\vec{v}_1 + m_2\vec{v}_2$ = constant.

Let $\vec{p}_1$ and $\vec{p}_2$ be the momenta of two particles, then

$$\vec{p}_1 = m_1\vec{v}_1 ; \quad \vec{p}_2 = m_2\vec{v}_2$$

$$\vec{p}_1 + \vec{p}_2 = \text{constant}$$

The above result is expressed in the form of law of conservation of linear momentum. It can be stated as follows.

"If the resultant external force acting on a system is zero, the total linear momentum of the system remains constant".

Recoil of a Gun - A gun having a mass M, fires a bullet of mass m.

Initially, both M and m are at rest, therefore total linear momentum is

$$\vec{p}_B + \vec{p}_G = 0 \qquad \text{...(i)}$$

After firing,



Final total linear momentum is

$$M\vec{V} + m\vec{v} = \vec{p}'_G + \vec{p}'_B \qquad \text{...(ii)}$$

As $\vec{F}_{ext} = \vec{0}$, so applying conservation of linear momentum, we get

$$\vec{p}'_G + \vec{p}'_B = 0 \quad \text{or} \quad M\vec{V} + m\vec{v} = \vec{0}$$

$$\Rightarrow \quad \vec{V} = -\frac{m\vec{v}}{M} \qquad \ldots(iii)$$

Negative sign indicates that the gun recoils back or kicks back.

Also,

$$\frac{|p'_B|}{|p'_G|} = 1 \Rightarrow p'_B = p'_G = p \text{ (say)}$$

The ratio of kinetic energies of the bullet and gun is given by

$$\frac{K_B}{K_G} = \frac{\frac{p_B^2}{2m}}{\frac{p_G^2}{2M}} = \frac{M}{m}$$

The kinetic energy released $= \frac{1}{2}(mv^2 + MV^2) = \frac{p^2}{2\left(\frac{M.m}{M+m}\right)}$

Q. **A gun of mass M (without bullet) fires a bullet of mass m with muzzle velocity u. Find the recoil speed of the gun.**

Sol. "Muzzle velocity" means velocity of bullet with respect to gun after firing the bullet. Momentum of the system is conserved in ground frame of reference. Let the gun recoils with a speed v, then

$$\vec{v}_{bullet/ground} = \vec{v}_{bullet/gun} + \vec{v}_{gun/ground}$$

$$= u\hat{i} - v\hat{i}$$

$$= (u - v)\hat{i}$$

Applying conservation of linear momentum is horizontal direction,

$$0 = m(u - v) - Mv$$

$$v = \frac{mu}{m + M}$$

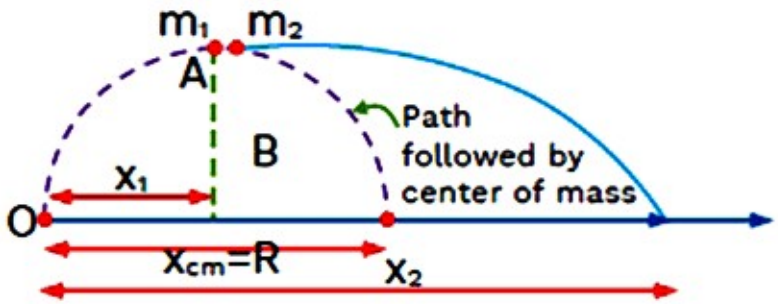
Q. **A body is projected from the ground for horizontal range R. At the highest point of its path, it breaks apart into two identical parts A and B.**

(i) **If A comes to rest, find the horizontal distance of the point, from the point of projection, where B lands on ground.**

(ii) **If A retraces its path, then find the horizontal distance other part B will travel from the point of projection.**

Sol. The motion of centre of mass is unaffected by the internal forces.

So X_{cm} = R (Range)



(i) $m_1 = m, m_2 = m$.

As A comes to rest, it falls vertically downward.

$\therefore \quad x_1 = \frac{R}{2}$

Now, $x_{cm} = \frac{m_1x_1 + m_2x_2}{m_1 + m_2}$

So, $R = \frac{m\frac{R}{2} + mx_2}{2m}$ $\quad (\because x_{cm} = R)$

Then, $x_2 = \frac{3R}{2} = 1.5\ R$

(ii) When A retraces its path, $x_1 = 0$

So, $R = \frac{m \times (0) + mx_2}{2m}$, then, $x_2 = 2R$

Bomb Explosion case

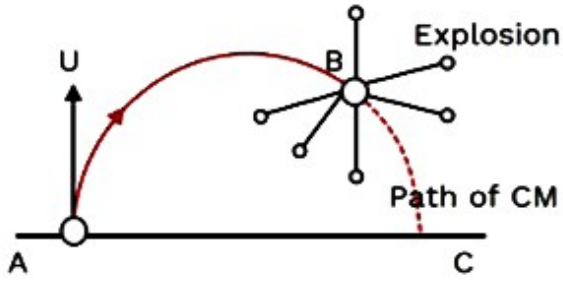
Important Points :

(1) Sum of internal forces on particles of a system is zero.

(2) Centre of mass is not accelerated by internal forces only.

(3) Action point of external force does not matter while producing acceleration in center of mass.

(4) If a projectile explodes in air in different parts, the path of the center of mass remains unchanged. This is because during explosion no external force (except gravity) acts on the center of mass. The situation is as shown in Figure.



(5) Path of center of mass is ABC, even though the different parts travel in different directions after explosion,

(6) For a two particle system, the ratio of velocities, displacements and accelerations of the particles is equal to the inverse ratio of their masses.

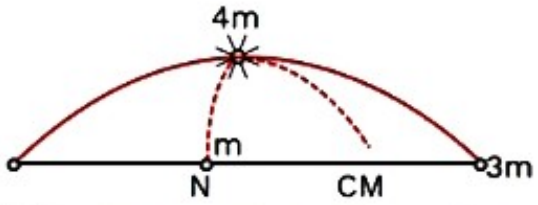
(7) If $\vec{F}_{ext} = 0$, the CM moves with constant velocity. Hence, the CM (but not necessarily other points of the system which may be accelerating) can be treated as an inertial frame (point).

(8) The net effect of internal forces is zero on the CM whereas it is not zero (in general) for other points of the system.

Q. **A projectile is fired at a speed of 100 m/s at an angle of 37° above the horizontal. At the highest point, the projectile breaks into two parts of mass ratio 1:3, the smaller coming to rest. Find the distance from the launching point to the point where the heavier piece lands.**

Sol. Internal forces do not affect the motion of the center of mass, the center of mass hits the ground at a position where the original projectile would have landed. The range of the original projectile is

$$x_{CM} = \frac{2u^2 \sin\theta\cos\theta}{g} = \frac{2\times 10^4 \times \frac{3}{5}\times\frac{4}{5}}{10}\text{m} = 960\text{m}$$

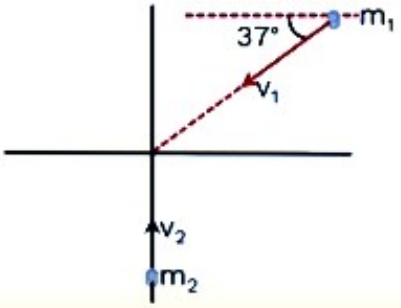


The center of mass will hit the ground at this position. As the smaller block comes to rest after breaking, it falls down vertically and hits the ground at half of the range, i.e., at x = 480 m. If heavier block hits the ground at x_2, then

$$x_{CM} = \frac{m_1x_1 + m_2x_2}{m_1 + m_2} \Rightarrow 960 = \frac{(m)(480) + (3m)(x_2)}{(m + 3m)}$$

$\therefore \quad x_2 = 1120$ m

Two balls with masses m_1 = 3 kg and m_2 = 5 m/s in the directions shown in figure. They collide at the origin.
(a) Find the velocity of the CM 3 s before the collision.
(b) Find the position of the CM 2 s after the collision.



(a) The given time is of no consequence since v_{CM} is fixed for all times.

$$v_{CM}(x) = \frac{m_1v_{1x} + m_2v_{2x}}{m_1 + m_2}$$

$$= \frac{(3)(-5\cos 37^\circ) + (5)(0)}{8kg} = -1.5m/s$$

$$v_{CM}(y) = \frac{m_1v_{1y} + m_2v_{2y}}{m_1 + m_2}$$

$$= \frac{(3)(-5\sin 37^\circ) + (5 \times 5)}{8kg} = +2m/s$$

$\therefore \quad \vec{v}_{CM} = -1.5\hat{i} + 2\hat{j}m/s$

(b) Since the collision occurs at the origin $(\vec{r}_1 = 0)$, the position of the center of mass 2 s later is $\vec{r}_{CM} = \vec{r}_1 + \vec{v}_{CM}t = -3\hat{i} + 4\hat{j}m$.

The centre of mass frame (C-Frame)

To describe the motion of the constituent particles of a system, find the velocity of centre of mass and then the velocities of the constituent particles with respect to centre of mass (called C-frame).

For simplicity, consider a two-particle system. The masses of the particles are m_1 and m_2 and their velocities in a stationary frame of reference are $\overline{v_1}$ and $\overline{v_2}$.

Then, velocity of centre of mass, $$\overline{v_{cm}} = \frac{m_1\overline{v_1} + m_2\overline{v_2}}{m_1 + m_2}$$

$$\overline{v_{1/cm}} = \overline{v_1} - \overline{v_{cm}} = \frac{m_2(\overline{v_1} - \overline{v_2})}{m_1 + m_2}$$

$$\overline{v}_{2/cm} = \overline{v_2} - \overline{v_{cm}} = \frac{m_1(\overline{v_2} - \overline{v_1})}{m_1 + m_2}$$

Now, total momentum of a system in C-frame=

$$m_1\overline{v_{1/cm}} + m_2\overline{v_{2/cm}} = \frac{m_1m_2(\overline{v_1} - \overline{v_2})}{m_1 + m_2} + \frac{m_2m_1(\overline{v_2} - \overline{v_1})}{m_1 + m_2} = 0$$

$$K_{1/cm} = \frac{1}{2}m_1v_{1/cm}^2$$

$$K_{2/cm} = \frac{1}{2}m_2v_{2/cm}^2$$

Total kinetic energy of system in C-frame

$$= K = \frac{1}{2}\frac{m_1m_2^2}{(m_1 + m_2)^2}(\overline{v_1} - \overline{v_2})^2 + \frac{1}{2}\frac{m_2m_1^2}{(m_1 + m_2)^2}(\overline{v_2} - \overline{v_1})^2$$

$$K = \frac{1}{2}\left(\frac{m_1m_2}{(m_1 + m_2)}\right)(\overline{v_1} - \overline{v_2})^2$$

Consider two particles of mass m_1 and m_2 having velocity $\bar{v}_1$ and $\bar{v}_2$ respectively in ground frame. Let velocity of centre of mass equals $\bar{v}_{CM}$ in ground frame, then

$$\bar{v}_{1/CM} = \bar{v}_1 - \bar{v}_{CM}$$

$$\bar{v}_{2/CM} = \bar{v}_2 - \bar{v}_{CM}$$

Kinetic energy in C-frame

$$= \frac{1}{2}m_1(v_{1/cm})^2 + \frac{1}{2}m_2(v_{2/cm})^2$$

$$= \frac{1}{2}m_1[v_1^2 + v_{Cm}^2 - 2\vec{v}_1.\vec{v}_{CM}] + \frac{1}{2}m_2[v_2^2 + v_{CM}^2 - 2\vec{v}_2.\vec{v}_{CM}]$$

$$= \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 + \frac{1}{2}(m_1 + m_2)v_{CM}^2 - (m_1\vec{v}_1 + m_2\vec{v}_2).\vec{v}_{CM}$$

$$= \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 - \frac{1}{2}(m_1 + m_2)v_{CM}^2$$

$$\Rightarrow \quad K.E._{\text{inC-frame}} = K.E._{\text{inground-frame}} - \frac{1}{2}(m_1 + m_2)v_{CM}^2$$

Here $m_1 + m_2 = m$, $v_{cm} = v$

$$\Rightarrow \quad KE_{\text{ingroundframe}} = K.E._{\text{inC-frame}} + \frac{1}{2}mv^2$$

Important Point

The kinetic energy of a system of particles (K) is equal to the kinetic energy of the system relative to centre of mass (K' plus kinetic energy of the centre of mass)

K_{total} = KE of system w.r.t centre of mass + KE of centre of mass

or $K_{total} = K' + \frac{1}{2}Mv_{cm}^2$

M = total mass of system of particles

Work Energy Theorem Relative to Centre of Mass

Work done by total forces (may be internal or external) is equal to change in kinetic energy of the system with respect to centre of mass.

$$W_{total} = \Delta K_{cm}$$

$$W_{internal} + W_{external} = \Delta K_{cm}$$

$$W_{internal} + W_{external} = K_{cm,\,final} - K_{cm,\,initial}$$

$$W_{internal} + W_{external} = \frac{1}{2}\mu(v_{rel,f}^2 - v_{rel,i}^2)$$

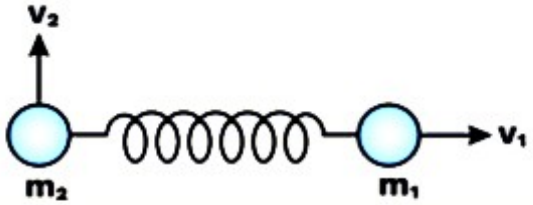
Note :

- If no external force acts on system of two particles system then $W_{ext} = 0$.

  We can write $W_{internal} = \frac{1}{2}\mu(v_{rel,f}^2 - v_{rel,i}^2)$
- The total work done by an internal force pair does not depend on the frame of reference.
- Kinetic energy of a system is minimum relative to CM.

- The work done by pseudo (internal) force in CM frame is zero for a translating system.
- The work done by gravity is zero relative to CM (for the bodies having size negligible compared to the radius of the earth).
- Kinetic energy of a system is (generally) greater than KE of the system about the CM of the system. However, both are equal in CM frame.
- For a two particle system $K_{int} = K_{system} = \frac{1}{2}\mu v_{rel}^2$ relative to CM, where μ = reduced mass of the system and v_{rel} = relative velocities between the particles.
- Internal forces as a whole cannot change the KE of the CM of the system whereas external force can change the KE of CM of the system.
- When the net work done by the internal forces is non-zero, the KE of the system about the CM changes due to the internal forces.

Q. **Two small discs of masses m_1 and m_2 are connected by a weightless spring resting on a smooth horizontal plane. The discs are set in motion with initial velocities v_1 and v_2, whose directions are mutually perpendicular and in the same horizontal plane. Find the total energy E of the system with reference to the frame fixed to the centre of mass.**



Sol. Let $\vec{v}_{cm}$ be the velocity of the centre of mass. Velocity of m_1 relative to the centre of mass is $(\vec{v}_1 - \vec{v}_{cm})$ and that of m_2 relative to the centre of mass is $(\vec{v}_2 - \vec{v}_1)$

Velocity of centre of mass $\vec{v}_{cm} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2}$

$$(\vec{v}_1 - \vec{v}_{cm}) = \frac{m_2}{m_1 + m_2}(\vec{v}_1 - \vec{v}_2)$$

Hence, $|\vec{v}_1 - \vec{v}_{cm}| = \frac{m_2}{m_1 + m_2}(v_1^2 + v_2^2)^{1/2}$

and $(\vec{v}_1 - \vec{v}_{cm}) = \frac{m_1}{(m_1 + m_2)}(\vec{v}_2 - \vec{v}_1)$

Hence, $|\vec{v}_2 - \vec{v}_{cm}| = \frac{m_1}{m_1 + m_2}(v_1^2 + v_2^2)^{1/2}$

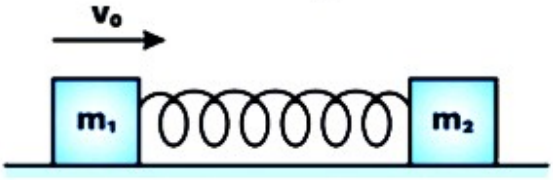
Total energy with respect to the centre of mass,

$$E = \frac{1}{2}m_1 |\vec{v}_1 - \vec{v}_{cm}|^2 + \frac{1}{2}m_1 |\vec{v}_2 - \vec{v}_{cm}|^2$$

$$= \frac{1}{2}\left(\frac{m_1 m_2}{m_1 + m_2}\right)(v_1^2 + v_2^2)$$

Where $\mu = \frac{m_1 m_2}{m_1 + m_2}$ is called reduced mass.

Q. **Two smooth blocks of masses m_1 and m_2 attached with an ideal spring of stiffness k and kept on a horizontal surface. If m_1 is projected with a horizontal velocity v_o, find the maximum compression of the spring.**



Sol. **Method 1 (From ground frame)**

$p_i = m_1 v_0 + 0$ and $p_f = (m_1 + m_2)V$

As $p_i = p_f \Rightarrow m_1 v_0 = (m_1 + m_2)V$

or $V = \frac{m_1 v_0}{(m_1 + m_2)}$ ...(1)

Now using conservation of mechanical energy.

$\Delta K + \Delta U = 0$

$(K_f - K_i) + \Delta U_{spring} = 0$

$$\left[\frac{1}{2}(m_1 + m_2)V^2 - \frac{1}{2}m_1 v_0^2\right] + \frac{1}{2}kx^2 = 0 \quad ...(2)$$

From (1) and (2), we get

$$x = \left[\sqrt{\frac{m_1 m_2}{(m_1 + m_2)k}}\right] v_0$$

Method 2 : (Work-energy theorem from centre of mass frame)

The linear momentum of system from CM frame is zero and no external force is acting on the system. We can apply work-energy theorem from CM frame

$$W_{ext} + W_{int} = \Delta K_{cm} = (K_f - K_i)_{cm} \qquad ...(3)$$

Initial kinetic energy of system w.r.t. CM

$$(K_i)_{cm} = \frac{1}{2}\mu(v_{rel})^2_{initial} = \frac{1}{2}\left(\frac{m_1 m_2}{m_1 + m_2}\right)(v_0 - 0)^2$$

$$\Rightarrow \quad (K_i)_{cm} = \frac{1}{2}\left(\frac{m_1 m_2}{m_1 + m_2}\right)v_0^2$$

Final kinetic energy of system w.r.t. CM

$$(K_f)_{cm} = \frac{1}{2}\mu(v_{rel})^2_{final} = 0$$

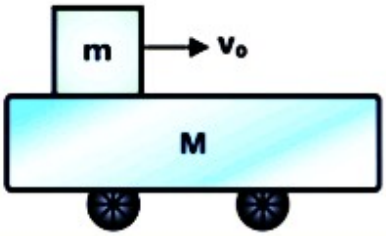
As both the blocks move with common velocity hence, $(v_{rel})_{final} = 0$ which gives $(K_f)_{cm} = 0$

Hence from (1)

$$0 + \left(-\frac{1}{2}kx^2\right) = 0 - \frac{1}{2}\left(\frac{m_1 m_2}{m_1 + m_2}\right)v_0^2$$

$$\Rightarrow \quad x = \left[\sqrt{\frac{m_1 m_2}{(m_1 + m_2)k}}\right] v_0$$

Q. A block of mass m is pushed with a velocity v_0 along the surface of a trolley car of mass M. If the horizontal ground is smooth and the coefficient of kinetic friction between the block and plank is μ, find the minimum distance of relative sliding between the block and plank.



Sol. Method 1 (Using ground frame method)

By conservation of linear momentum in horizontal direction.

$$mv_0 + 0 = (m+M)V \Rightarrow V = \frac{mv_0}{(m+M)} \qquad ...(1)$$

Now using WE theorem from ground frame

$$W_{ext} + W_{int} = \Delta K$$

$$W_{ext} = 0$$

$$W_{int} = W_{friction} = -fx = -\mu mgx$$

$$0 + (-\mu mgx) = \left(\frac{1}{2}(m+M)V^2 - \frac{1}{2}mv_0^2\right) \qquad ...(2)$$

From (1) and (2), relative distance of sliding

$$x = \frac{v_0^2}{2\left(1+\frac{m}{M}\right)\mu g}$$

Method 2 : (Work-energy theorem from centre of mass frame)

Considering the system from CM frame

$$W_{ext} + W_{int} = (\Delta K)_{cm} = (k_f - k_i)_{cm}$$

Initial kinetic energy of system w.r.t. CM

$$(K_i)_{cm} = \frac{1}{2}\mu(v_{rel})_{in}^2 = \frac{1}{2}\frac{mM}{(m+M)}(v_0 - 0)^2$$

$$\Rightarrow \quad (K_i)_{cm} = \frac{1}{2}\frac{mM}{(m+M)}v_0^2$$

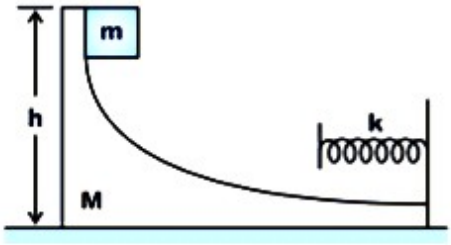
Final kinetic energy of system w.r.t. CM

$$(K_f)_{cm} = 0 \text{ or } |(v_{rel})_f| = 0$$

Hence, $0 + (-\mu mgx) = 0 - \frac{1}{2}\frac{mM}{(m+M)}v_0^2$

$$\Rightarrow \quad x = \frac{v_0^2}{2\left(1+\frac{m}{M}\right)\mu g}$$

Q. **A block of mass m is released from rest from a height h onto a smooth sledge of mass M fitted with an ideal spring of stiffness k.**



a) Find the velocity of the block and sledge just before the block touches the spring.

b) Find maximum compression of the spring.

Sol. **Method 1 :**

a) As no external force acts in horizontal direction, the linear momentum will be conserved in horizontal direction. Let the velocity of block and wedge be v and V respectively up to the case when the block reaches on horizontal surface on the wedge.

$$mv = MV \qquad ...(1)$$

Mechanical energy is also conserved, so,

$$\Delta K + \Delta U = 0$$

$$\left(\frac{1}{2}mv^2 + \frac{1}{2}MV^2 - 0\right) + (-mgh) = 0 \qquad ...(2)$$

$$\frac{1}{2}mv^2 + \frac{1}{2}M\left(\frac{m}{M}v\right)^2 = mgh$$

$$v = \sqrt{\frac{2Mgh}{M+m}} \quad \text{and} \quad V = \sqrt{\frac{2m^2gh}{M(M+m)}}$$

Method 2 : Applying work-energy theorem from centre of mass frame

$$W_{gravity} = (\Delta K)_{CM}$$

$$mgh = \frac{1}{2}\left(\frac{mM}{m+M}\right)(v+V)^2$$

Also, $mv = MV \Rightarrow v = \frac{MV}{m}$

$$\Rightarrow \quad mgh = \frac{1}{2}\left(\frac{mM}{m+M}\right)V^2\left(\frac{M}{m}+1\right)^2$$

$$\Rightarrow \quad V^2 = \sqrt{\frac{2m^2gh}{M(m+M)}}$$

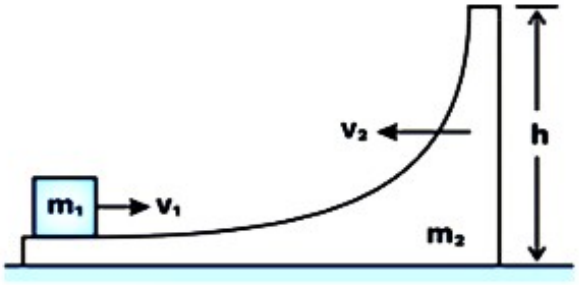
b) Initial linear momentum of the system is zero and there is no external force on the system, hence at the time of maximum compression of spring the linear momentum of the system should be zero. Hence final kinetic energy should be zero.

$$\Delta U + \Delta K = 0 \quad \Rightarrow \quad \Delta K = K_f - K_i = 0$$

Hence, $\Delta U_{gravity} + \Delta U_{spring} = 0$

$$-mgh + \frac{1}{2}kx^2 = 0 \quad \Rightarrow \quad x = \sqrt{\frac{2mgh}{k}}$$

Q. **A particle of mass m_1 is projected to the right with a speed v_1 onto a smooth wedge of mass m_2 which is simultaneously projected due to the left with a speed v_2. If the particle attains the highest point of the wedge, find h.**



Sol. Applying work – energy theorem

$$W_{total} = \Delta K_{blocks,\ cm} + \Delta K_{cm}$$

As no external forces acting, the velocity of centre of mass will be constant.

Hence, $\Delta K_{cm} = 0$

Hence, $W_{total} = \Delta K_{blocks,\ cm}$

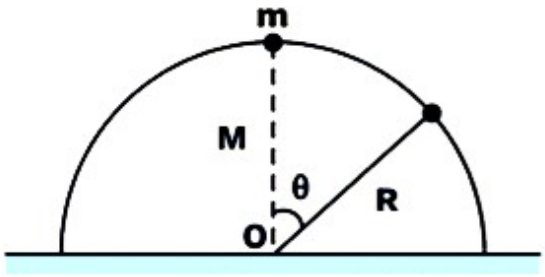
$$W_{gravity} = \left[\frac{1}{2}\mu v_{rel}^2\right]_{final} - \left[\frac{1}{2}\mu v_{rel}^2\right]_{initial}$$

$$-m_1gh = 0 - \frac{1}{2}\left(\frac{m_1m_2}{m_1+m_2}\right)(v_1+v_2)^2$$

$$-m_1gh = -\frac{1}{2}\left(\frac{m_1m_2}{m_1+m_2}\right)(v_1+v_2)^2$$

$$h = \frac{1}{2}\,\frac{m_2(v_1+v_2)^2}{g(m_1+m_2)}$$

Q. **A bead of mass m kept at the top of a smooth hemispherical wedge of mass M and radius R, is gently pushed towards right. As a result, the wedge slides due left. Find the**

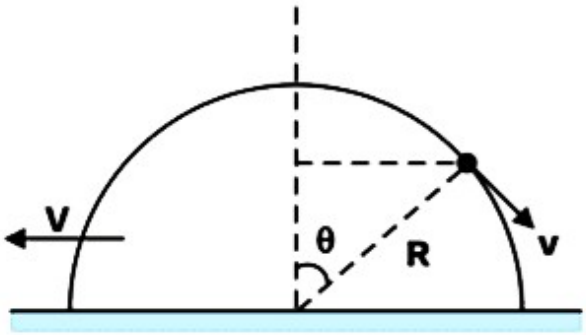


a) speed of the wedge.

b) magnitude of velocity of the bead relative to the wedge.

Sol. Applying work-energy theorem

$W_{gravity} = \Delta K' + \Delta K_c$ ...(1)



$\Delta K'$ = change in KE w.r.t. centre of mass frame and ΔK_c = change in kinetic energy of centre of mass w.r.t. ground.

Let v is the velocities of ball with respect to wedge and V is the velocity of wedge.

Linear momentum will be conserved in the horizontal direction.

Hence, $MV = m(v\cos\theta - V)$

$$\Rightarrow \quad V = \frac{mv\cos\theta}{(m+M)} \qquad ...(2)$$

The velocity of centre of mass will be zero in horizontal direction but the velocity of centre of mass will change in vertical direction.

$$(v_{cm})_y = \frac{mv\sin\theta}{(m+M)} \qquad ...(3)$$

Now applying WE theorem

$$-mgR(1-\cos\theta) = \frac{1}{2}\mu v^2 + \frac{1}{2}(m+M)v_{cm,y}^2$$

Here $\mu = \frac{mM}{(m+M)}$ and $v_{cm,y} = \frac{mv\sin\theta}{(m+M)}$

Which gives $v = \sqrt{\frac{2(M+m)gR(1-\cos\theta)}{(M+M\sin^2\theta)}}$

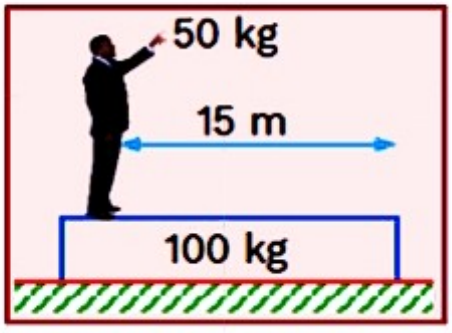
and from Eq. (ii) $V = \sqrt{\frac{2m^2gR(1-\cos\theta)\cos^2\theta}{(M+m)(M+m\sin^2\theta)}}$.

CONSERVATION OF LINEAR MOMENTUM AND MECHANICAL ENERGY

Linear momentum of a system is conserved when total external force acting on the system is zero.

Mechanical energy of a system is conserved when work done by external forces and dissipative force is zero.

Q. **A man of mass 50 kg is standing on a horizontal platform of mass 100 kg resting on a horizontal frictionless floor. If the man walks to the other end of the platform, then find**



(A) **Displacement of man w.r.t. floor**

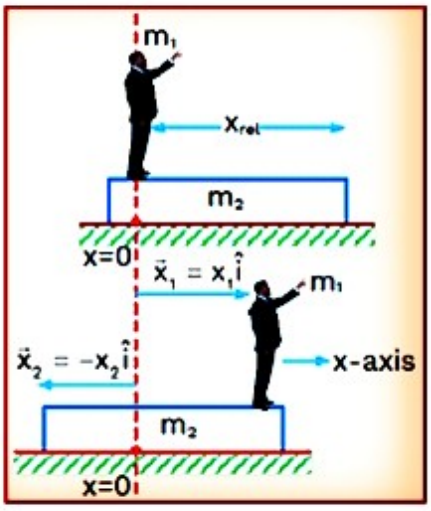
(B) **Displacement of platform w.r.t. floor**

(C) **If the velocity of the man at an instant is 1.5 m/s w.r.t. the platform, then find**

(i) Velocity of man w.r.t. floor

(ii) Velocity of platform w.r.t. floor

Sol. Initially both are at rest, so, $\vec{V}_{CM} = \vec{0}$



Along horizontal,

$$\vec{F}_{ext} = \vec{0}$$

$$\vec{V}_{CM} = \vec{0} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2} \quad \text{....(for all instants)}$$

$$\Rightarrow \quad \Delta x_{CM} = 0$$

$$\text{Now, } \Delta\vec{x}_{CM} = \frac{m_1\vec{x}_1 + m_2\vec{x}_2}{m_1 + m_2}$$

$$\text{or} \quad m_1\vec{x}_1 + m_2\vec{x}_2 = 0$$

$\vec{x}_1$ and $\vec{x}_2$ are actual displacements of man and platform respectively,

$$m_1x_1 - m_2x_2 = 0 \qquad \text{...(i)}$$

$$x_1 + x_2 = x_{rel} \qquad \text{...(ii)}$$

Solving (i) and (ii),

$$x_1 = \frac{m_2x_{rel}}{m_1 + m_2} = 10 \text{ m}$$

$$x_2 = \frac{m_1x_{rel}}{m_1 + m_2} = 5 \text{ m} \qquad \text{...(iii)}$$

Differentiating equation (iii)

$$\frac{dx_1}{dt} = \frac{m_2}{m_1 + m_2}\frac{dx_{rel}}{dt}$$

Actual velocity of man,

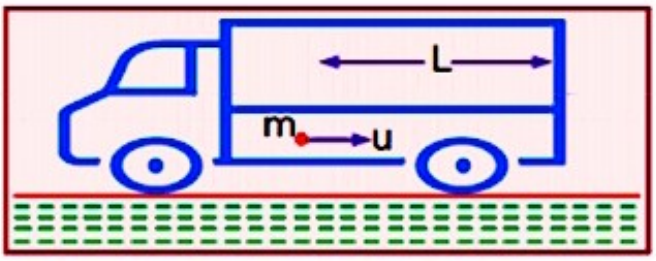
$$v_1 = \frac{m_2v_{rel}}{m_1 + m_2} = 1 \text{ m/s}$$

$$\frac{dx_2}{dt} = \frac{m_1}{m_1 + m_2}\frac{dx_{rel}}{dt}$$

Actual velocity of platform,

$$v_2 = \frac{m_1v_{rel}}{m_1 + m_2} = 0.5 \text{ m/s}$$

Q. **A cannon is bolted to the floor of a truck, which is free to move without friction along a straight track.**
The combined mass of cannon and the carriage is M.



The cannon fires a cannon ball, of mass m horizontally with velocity u.
The cannon ball travels the length of the truck, a distance L, and then becomes embedded in the truck's end wall.
What is the recoil speed of the truck right after the cannon is fired?
What is the velocity of the truck after the cannon ball strikes the far wall?
What net distance, and in what direction, does the truck moves as a result of the firing of the cannon?

Sol. As per conservation of linear momentum, the net horizontal momentum of the system is the same before and after the cannon is fired.
The momentum before the cannon is fired is zero, since nothing is initially moving. Hence, we can also set the momentum after the cannon is fired to zero, then,

$$0 = Mv + mu$$

Where v is recoil velocity of the truck.

$$\text{So, } v = -\frac{m}{M}u$$

-ve sign indicates that the recoil velocity of the truck is in the opposite direction of motion of the cannon ball.
Suppose, that after the cannon ball strikes the far wall of the truck, both the cannon ball and the truck move with common velocity v_c.
Then, by conservation of momentum, we can write,

$$Mv + mu = (M + m)v_c$$

However, $Mv + mu = 0$.
So, $v_c = 0$.
Thus, the truck is brought to complete halt when the cannon ball strikes its far wall.

In the frame of reference of the truck, the cannon ball moves with velocity u – v after the cannon is fired.
Hence, the time of flight of the cannon ball is

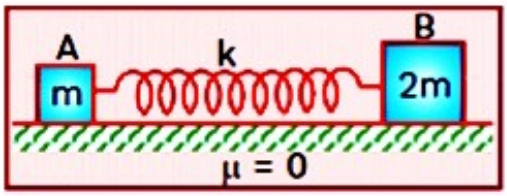
$$t = \frac{L}{u - v}$$

The displacement of the truck in this interval is

$$d = vt$$

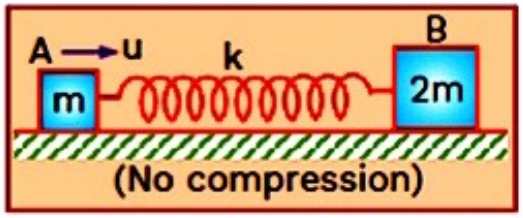
Thus, the truck moves by vt in the opposite direction to the direction of motion of cannon ball.

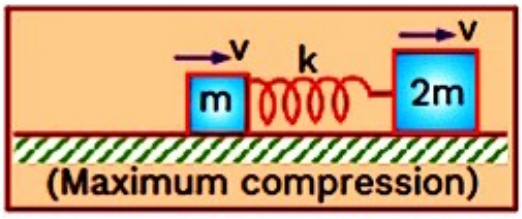
Q. **Two blocks A and B having masses m and 2m respectively are connected by a massless relaxed spring having spring constant k.**



A is given a velocity u towards right at this instant. Find their velocities at the instant of maximum compression of the spring.

Sol. As long as there is a relative velocity of approach between A and B, the compression of spring continues.
At maximum compression of spring, relative velocity becomes zero i.e. they have same velocity.





Spring force is an internal force, so external force on system is zero.
By conservation of linear momentum,

$$mu + 2m \times 0 = (m + 2m)v$$

$$v = \frac{u}{3}$$

By conservation of mechanical energy,

$$\frac{1}{2}mu^2 + \frac{1}{2}(2m)0^2 + \frac{1}{2}k0^2 = \frac{1}{2}(m+2m)\left(\frac{u}{3}\right)^2 + \frac{1}{2}k\left(x_{max}\right)^2$$

$$\Rightarrow \quad \frac{mu^2}{2} - \frac{mu^2}{6} = \frac{1}{2}kx^2$$

$$\Rightarrow \quad \frac{1}{3}mu^2 = \frac{1}{2}kx^2$$

$$\Rightarrow \quad x^2 = \frac{2mu^2}{3k}$$

$$\Rightarrow \quad x = \sqrt{\frac{2m}{3k}}u$$

Q. **A bullet of mass m travelling horizontally with speed u strikes a free wooden block of mass M perpendicularly. The penetration is assumed uniform and the bullet comes to rest after penetrating a distance 'a' into the block. Show that final common velocity of the system is $\frac{mu}{m+M}$ and that total loss of kinetic energy during the penetration is $\frac{mMu^2}{2(m+M)}$. Also find the resistive force acting on the bullet assuming it to be constant.**

Sol. By conservation of linear momentum,

$$mu = (M + m)v$$

$$\therefore \quad v = \frac{mu}{m+M}$$



$$\text{Energy loss} = \frac{1}{2}mu^2 - \frac{1}{2}(M+m)v^2 = \frac{1}{2}mu^2 - \frac{1}{2}(M+m)\left(\frac{mu}{m+M}\right)^2 = \frac{mMu^2}{2(m+M)}$$

For a constant resistive force (F),

$$\text{Energy loss} = (\text{Resistive force})(\text{displacement}) = Fa = \frac{mMu^2}{2(m+M)}$$

$$\therefore \quad F = \frac{mMu^2}{2a(m+M)}$$

Q. **A light spring of spring constant k is kept compressed between two blocks of masses m and M on a smooth horizontal surface. When released, the blocks acquire velocities in opposite directions. The spring loses contact with the blocks when it acquires natural length. If the spring was initially compressed through a distance x, find the final speeds of the two blocks.**

Sol. Consider the two blocks plus the spring to be the system. No external force acts on this system in horizontal direction. Hence, the linear momentum will remain constant. Suppose, the block of mass M moves with a speed V and the other block with a speed v after losing contact with the spring. From conservation of linear momentum in horizontal direction, we have

$$MV - mv = 0 \text{ or } V = \frac{mv}{M} \qquad \text{(i)}$$

Initially, the mechanical energy of the system $= \frac{1}{2}kx^2$

Finally, the mechanical energy of the system $= \frac{1}{2}mv^2 + \frac{1}{2}MV^2$

As there is no friction, mechanical energy will remain conserved. Therefore,

$$\frac{1}{2}mv^2 + \frac{1}{2}MV^2 = \frac{1}{2}kx^2 \qquad \text{(ii)}$$

Solving equation (i) and (ii), we get

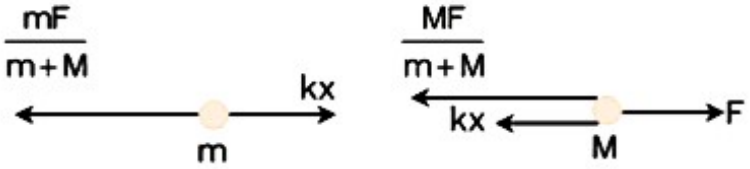
$$v = \left[\frac{kM}{m(M+m)}\right]^{\frac{1}{2}} x$$

and $$V = \left[\frac{kM}{M(M+m)}\right]^{\frac{1}{2}} x$$

Q. **A block of mass m is connected to another block of mass M by a massless spring of spring constant k. The blocks are kept on a smooth horizontal plane and are at rest. The spring is unstretched when a constant force F starts acting on the block of mass M to pull it. Find the maximum extension of the spring.**



Sol. We solve the situation in the reference frame of center of mass. As only F is the external force acting on the system, due to this force, the acceleration of the center of mass is F/(M + m). Thus, with respect to center of mass, there is a pseudo force on the two masses in the opposite direction, the free body diagram of m and M with respect to center of mass (taking center of mass at rest) is shown in figure.



Taking center of mass at rest, if m moves maximum by a distance x_1 and M moves maximum by a distance x_2, then the work done by external forces (including pseudo force) will be

$$W = \frac{mF}{m+M}x_1 + \left(F - \frac{MF}{m+M}\right)x_2 = \frac{mF}{m+M}(x_1 + x_2)$$

This work is stored in the form of potential energy of the spring as U = (1/2) $k(x_1 + x_2)^2$.

Thus, on equating we get the maximum extension in the spring, as after this instant the spring starts contracting.

$$\frac{1}{2}k(x_1 + x_2)^2 = \frac{mF}{m+M}(x_1 + x_2)$$

$$\Rightarrow \qquad x_{max} = x_1 + x_2 = \frac{2mF}{k(m+M)}$$

Impulse

When a force acts on a body for some time, it produces a change in momentum. This is called impulse.

Impulse of a force acting on a body is the change in momentum of the body as per impulse momentum theorem.

By Newton's Second Law.

$$\vec{F} = \frac{d\vec{p}}{dt}$$

$$\Rightarrow \quad d\vec{p} = \vec{F}dt\,.$$

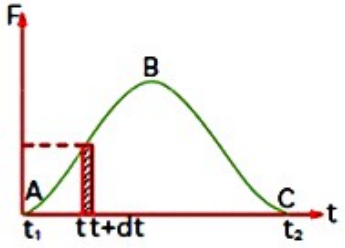
Integrating,

$$\int d\vec{p} = \int \vec{F}dt$$

$$\Rightarrow \quad \Delta\vec{p} = \int \vec{F}dt$$

This is written as impulse, $I = \int \vec{F}.dt = \Delta\vec{p}$

Impulse of variable force is equal to area enclosed by the force-time curve and the time axis.

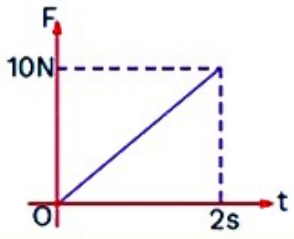


$$\text{Area} = \int_{t_1}^{t_2} F dt$$

= Area ABCA

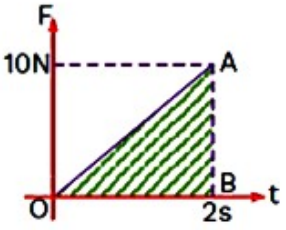
= Change in momentum

Q. **A variable force acts on a body of mass 2 kg initially at rest as shown in figure. Calculate the impulse of the force in first 2 seconds and speed of the body after 2 seconds.**



Sol. Impulse of a force equals area under force time curve.

Thus,



Area of $\Delta OAB = \frac{1}{2} \times 2 \times 10 = 10$ Ns

Thus, impulse equals 10 Ns.

Now, impulse is equal to change in momentum.

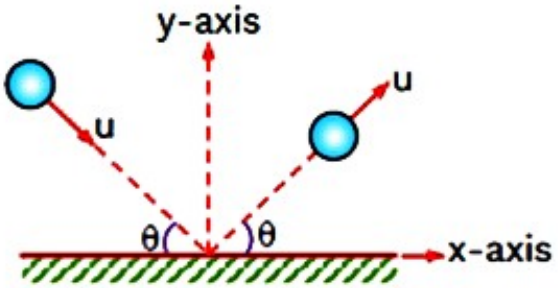
$\Rightarrow \quad 10 = mv - 0$

$$v = \frac{10}{2} = 5 \text{ m/s}$$

Q. **A rubber ball of mass 0.4 kg hits the ground at an angle of 30° with the ground and rebounds at the same angle without any change in speed. If the speed of ball is 1 m/s and the period of contact between the ball and floor is 0.4s, find the average force exerted by the ball on the ground.**

Sol. The initial momentum is $m\vec{u} = m(u\cos\theta\hat{i} - u\sin\theta\hat{j})$

The final momentum is $m\vec{v} = m(u\cos\theta\hat{i} + u\sin\theta\hat{j})$



Change in momentum of the ball = $\frac{m(\vec{v} - \vec{u})}{\Delta t} = 2mu\sin\theta\hat{j}$

$\Rightarrow$ The average force on the ball by the ground is

$$\vec{F}_{av} = \frac{\Delta\vec{P}}{\Delta t} = \frac{2mu\sin\theta\hat{j}}{\Delta t}$$

$\therefore$ Force exerted by the ball on the ground is

$$\frac{-2mu\sin\theta}{\Delta t}\hat{j} \qquad \begin{pmatrix}\text{an equal and opposite force}\\ \text{by Newton's third law}\end{pmatrix}$$

Therefore, $F = \frac{2\times 0.4\times 1\times \sin 30^\circ}{0.4} = 1N$, along the normal.

Note: As the force is along y-axis, the y-component of velocity changes.

Q. **The hero of a stunt film fires 25 gram bullets from a machine gun, each at a speed of 500 m/s. If he fires 25 bullets in 5 sec, what average force does he exert against the machine gun during this period?**

Sol. $F = \frac{\Delta P}{\Delta t}$ ..... (i)

Change in momentum (ΔP) for one bullet $= P_f - P_i$

$= mV_f - mV_i$

$= \left(\frac{25}{1000}\right) \times (500) - \frac{25}{1000} \times (0)$

$|\Delta \vec{P}| = \frac{25}{2} = 12.5$ kg m/sec

Momentum change for 25 bullets = $25 \times (1.25)$ kg m/sec

Force $= \frac{\Delta P}{\Delta t} = \frac{25 \times 12.5}{5}$

$= 5 \times 12.5$

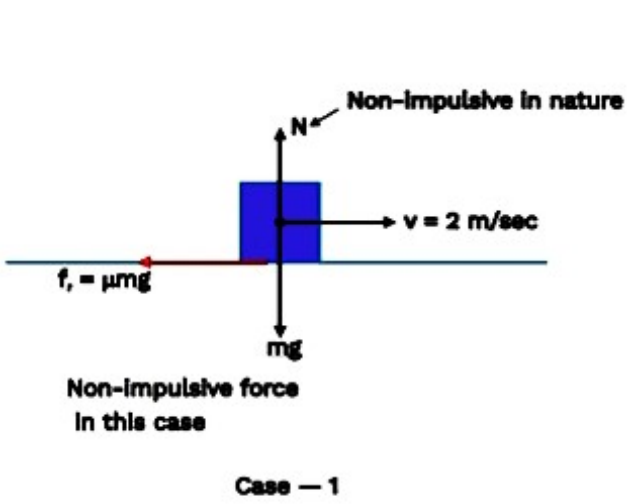
Force = 72.5 Newton

Impulsive force:

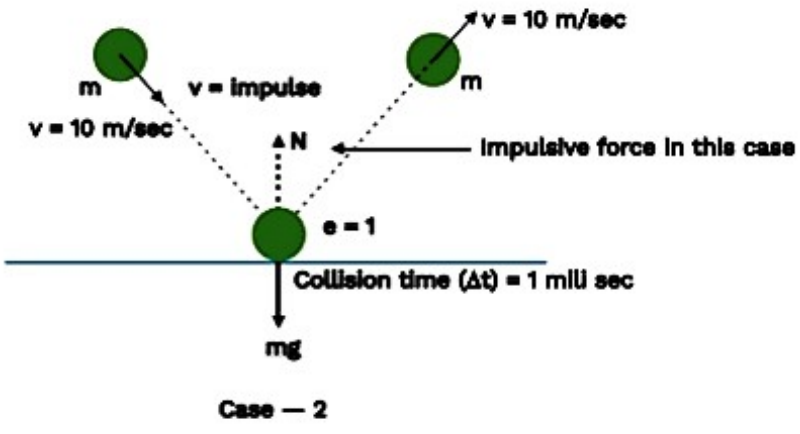
A force which has a very high magnitude and acts for a very short interval of time is called 'impulsive force'.

Important Point:

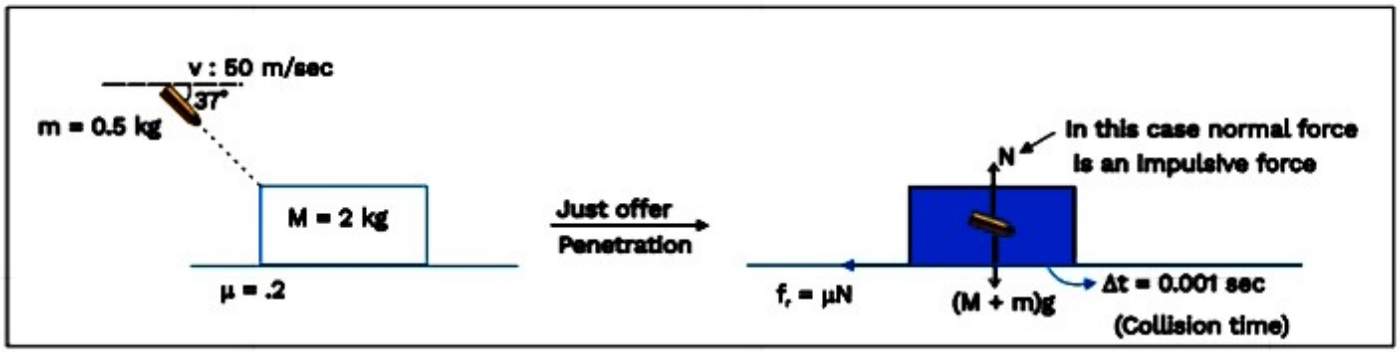
1. Impulsive force is a relative term, there is no clear boundary between an impulsive and a non – impulsive force.
2. Most of the times, colliding forces are impulsive in nature.
3. Gravitational and spring force are always non-impulsive.
4. Normal, Tension and friction are case dependent.



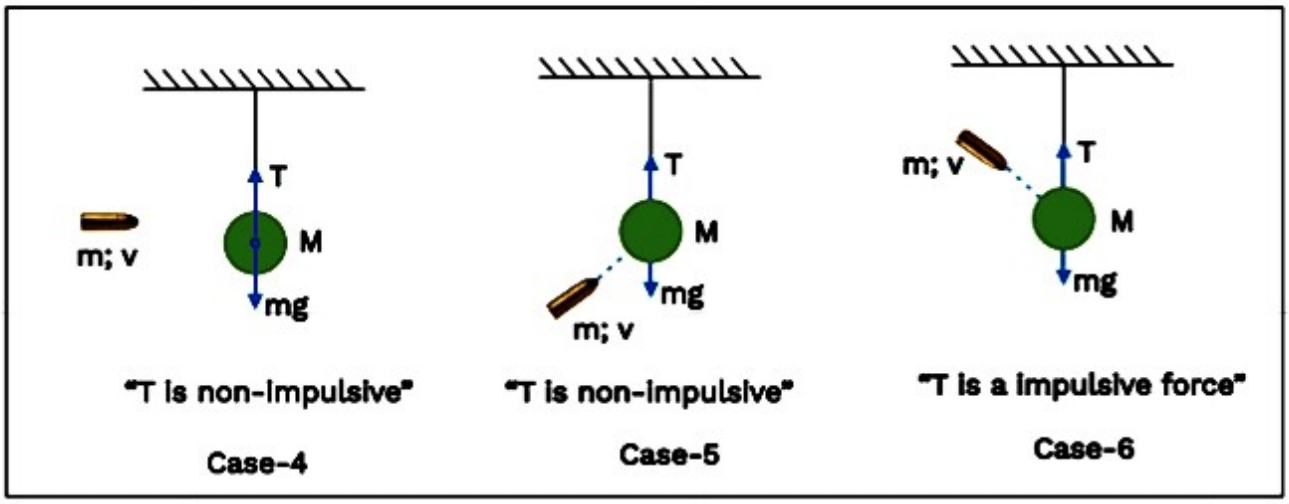
Case — 1



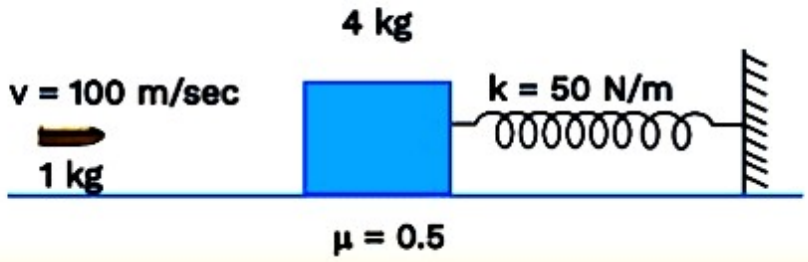
Case — 2



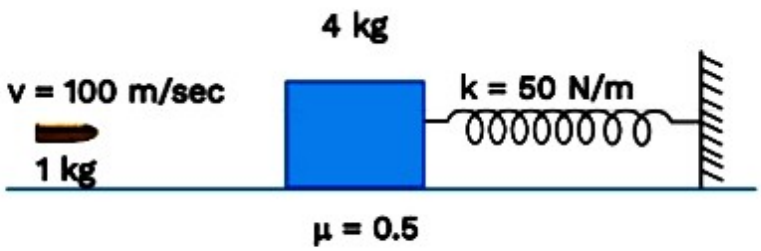
Case-3



Q. **Find the maximum compression in the spring when the bullet gets embedded in the block after hitting. Ignore heating effects due to collision.**



Sol.



Since spring force is non-impulsive force so we can conserve the momentum of block-bullet system.

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

$$1 \times 100 + 4 \times 0 = (1+4) \times V_{comm}$$

$$V_{comm} = 20 \text{ m/sec.}$$

Kinetic energy of system just after the penetration of bullet

$$= \frac{1}{2}(m_1 + m_2) \times V_{comm}^2$$

$$= \frac{1}{2}(1+4)(20)^2$$

$$= \frac{5}{2} \times 400$$

$$(KE)_{sys} = 1000\,J$$

Now, apply the energy conservation law.
Kinetic energy of system = | work done by frictional | + | work done by spring force|

$$1000 = \frac{1}{2}kx_{max}^2 + |-\mu mg\, x_{max}|$$

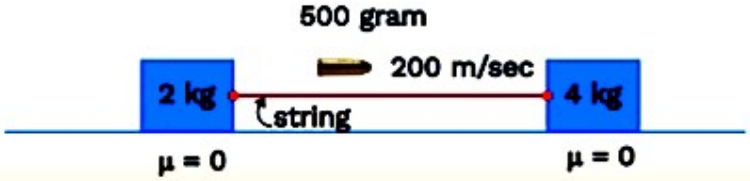
$$1000 = \frac{1}{2} \times 50 \times x_{max}^2 + |-0.5 \times 5 \times 10\, x_{max}|$$

$$40 = x^2 + x \Rightarrow x^2 + x - 40 = 0$$

$$x = \frac{-1 \pm \sqrt{1 - 4 \times 1 \times (-40)}}{2}$$

$x = 5.844$ m

Q. **A bullet of mass 500 gm with speed 20 m/sec penetrates the 4 kg block and gets embedded into it. Find the final speed of the system.**



Sol.



Concept:
In this case the tension force is a impulsive force so we can not apply the momentum conservation law. To solve this problem use impulse momentum theorem.

$$\overline{\Delta P} = \int \vec{F}.dt$$

$$\boxed{\vec{P}_f - \vec{P}_i = \int \vec{F}.dt}$$

Now write impulse-momentum theorem for each element of the system

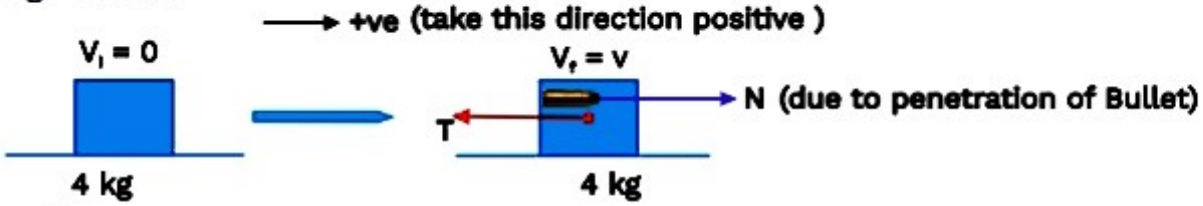
1. **For 2 kg Block:**



$$\vec{P}_f - \vec{P}_i = \int \vec{F}.dt$$

$$2 \times v - 2 \times 0 = \int T.dt \quad \text{..... (1)}$$

2. **For 4 kg – Block:**



$$\vec{P}_f - \vec{P}_i = \int \vec{F}.dt$$

$$4 \times v - 4 \times 0 = \int (N - T)dt \quad \text{..... (2)}$$

3. **For Bullet:**



m = 0.5 kg

$$\vec{P}_f - \vec{P}_i = \int \vec{F}.dt$$

$$0.5 \times v - 0.5 \times 20 = -\int N.dt \quad \text{..... (3)}$$

Now add Eq. 1 + Eq. 2 + Eq. 3

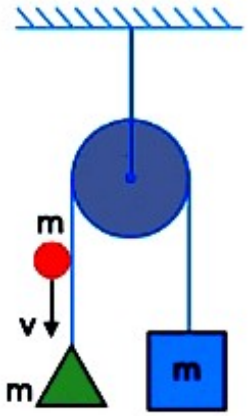
$$2 \times v - 0 + 4v - 0 + 0.5v - 0.5 \times 20 = \int T.dt + \int (N - T)dt + \int (-N).dt$$

$$\Rightarrow 2v + 4v + 0.5v - 10 = \int (T + N - T - N)\,dt$$

$$\Rightarrow 6.5v - 10 = 0$$

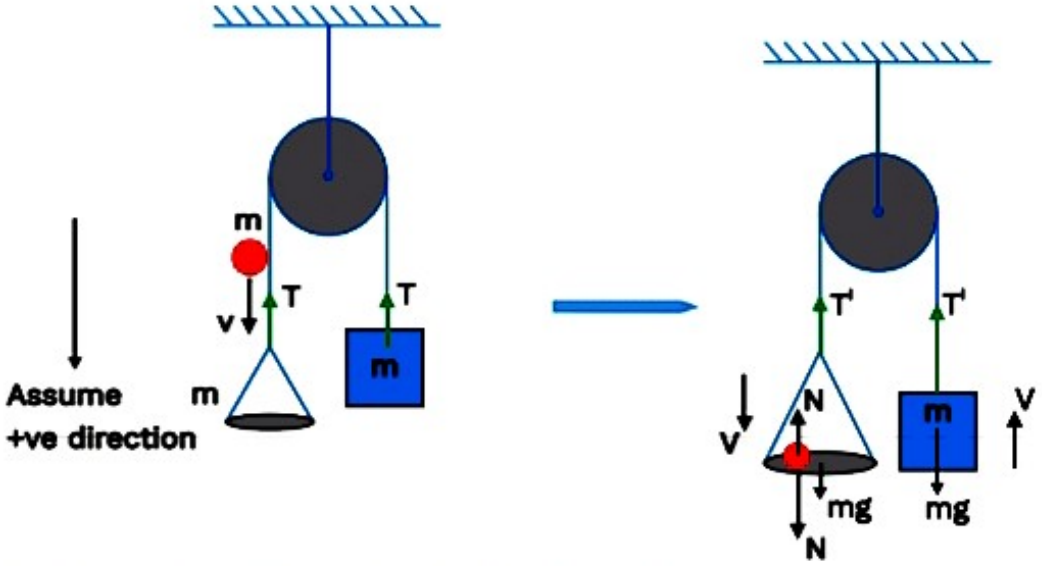
$$\Rightarrow v = \frac{10}{6.5} = 1.54\,m/sec.$$

Q. **A ball of mass 'm' and a pan of equal mass are connected by a string going over a smooth light pully. Initially the system is at rest then a particle of mass 'm' falls on the pan and sticks to it.**



It the particle strikes the pan with a speed v, find the speed with which the system moves just after the collision.

Sol.



Write impulse – momentum theorem for each element

The impulsive reaction comes due to collision between block and pan.

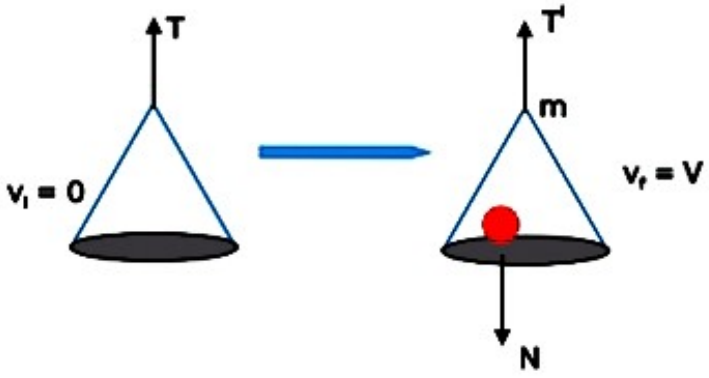
1. **For ball**



$$\vec{P}_f - \vec{P}_i = \int \vec{F}.dt$$

$$mV - m \times v = -\int N.dt \quad \text{..... (1)}$$

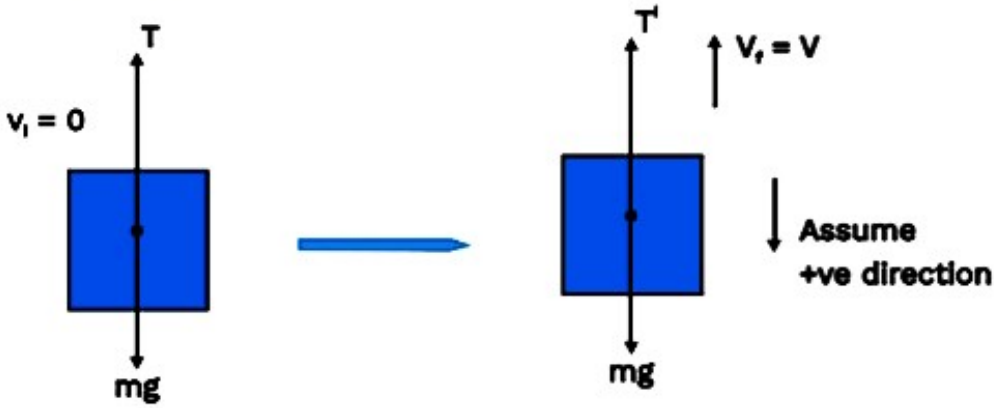
2. For Block



$$\vec{P}_f - \vec{P}_i = \int \vec{F}.dt$$

$$mV - m \times 0 = \int (N - T')dt \quad \text{..... (2)}$$

3. For block 'm':



$$\vec{P}_f - \vec{P}_i = \int \vec{F}.dt$$

$$-m \times V - m \times 0 = \int (-T')\, dt \quad \text{..... (3)}$$

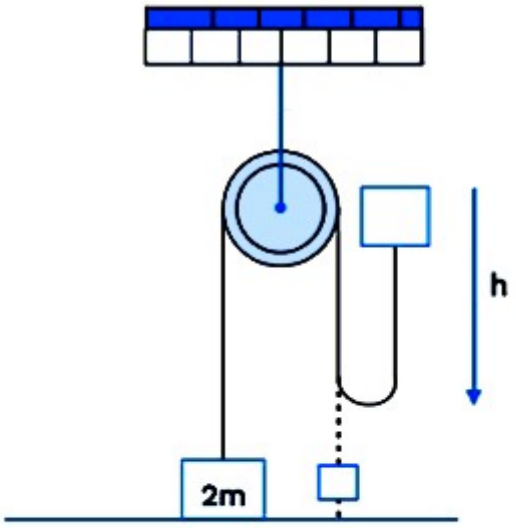
add — Eq. 1 + Eq. 2 — Eq. 3

$$mV - mv + mV + mV = -\int N.dt + \int (N - T')dt - \int (-T')dt$$

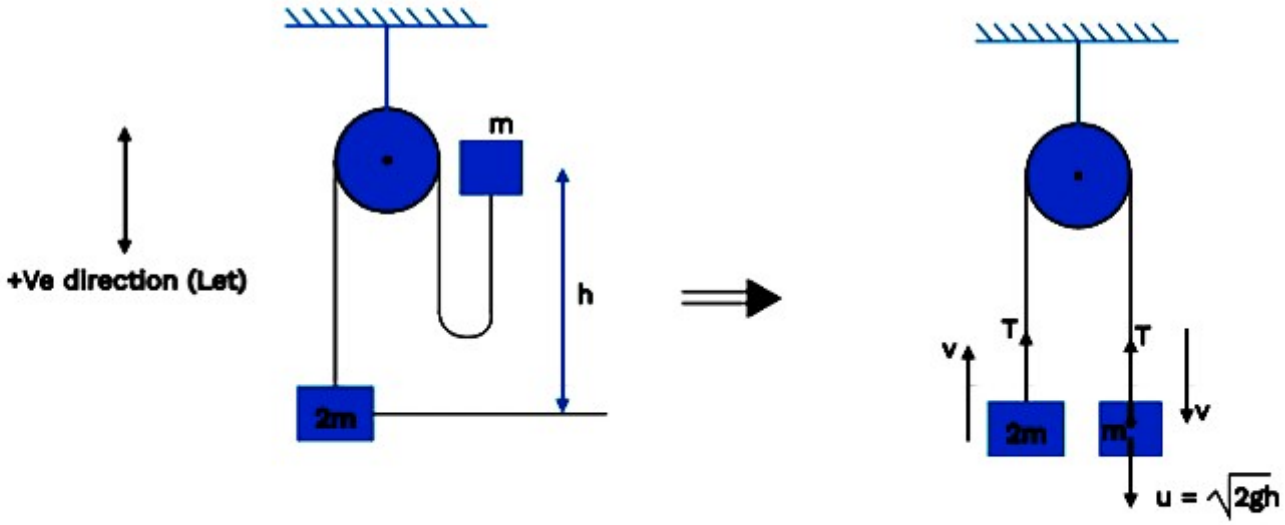
$$3mV - mv = 0$$

$$V = \frac{v}{3}. \quad \textbf{(Ans.)}$$

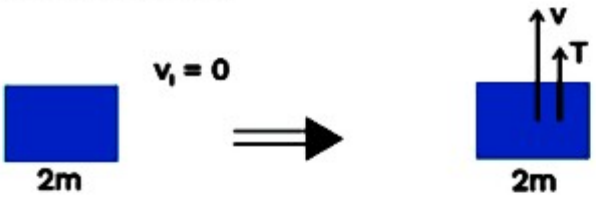
Q. **A mass 2m rests on a horizontal table. It is attached to a light inextensible string which passes over a smooth pulley and carries a mass m at the other end. If the mass m is raised vertically through a distance h and is then dropped, what is the speed with which the mass 2m begins to rise?**



Sol.



1. For Block '2m':

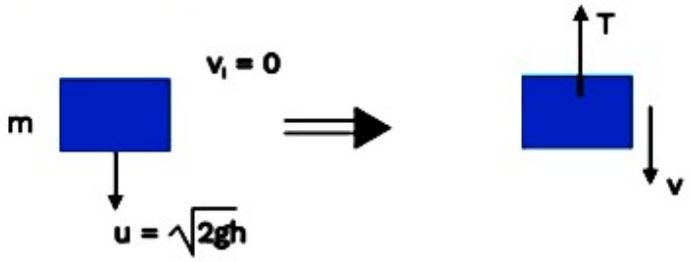


$$\vec{P}_f - \vec{P}_i = \int \vec{F}.dt$$

$$-2m \times v - (2m \times 0) = -\int T.dt \quad \text{..... (1)}$$

$$\Rightarrow 2mv = \int T.dt$$

2. **For Block 'm'**



$\vec{P}_f - \vec{P}_i = \int \vec{F}.dt$

$mv - (m \times \sqrt{2gh}) = -\int T.dt$ ..... (2)

Now add equation—1 and equation—2

$2mv - 0 + mv - m\sqrt{2gh} = 0$

$3mv = m\sqrt{2gh} \Rightarrow v = \frac{\sqrt{2gh}}{3}$ **(Ans.)**

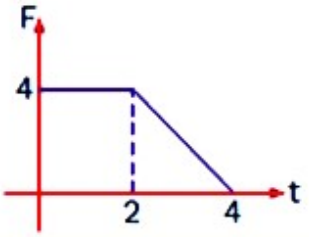
Impulse Momentum Theorem

When a force is acted upon a body, then momentum of the body changes. Impulse of a force is defined as the change in momentum of the body.

Mathematically,

Impulse, $(\vec{I}) = \int_{t_1}^{t_2} \vec{F}.dt$ = Change in momentum during time interval $t_1 \rightarrow t_2$

Q. **A force F is acted on a body as shown in the figure. Find the impulse of the force in first 4 seconds.**



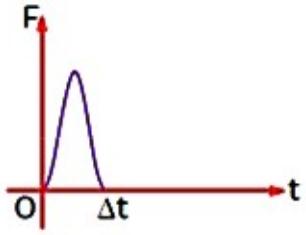
Sol. Impulse can be calculated as area under force time graph. Therefore,

$\text{Impulse} = (4 \times 2) + \left(\frac{1}{2} \times 2 \times 4\right) = 8 + 4 = 12 \text{ Ns}$

IMPULSE AND COLLISIONS

In collisions, large amount of force acts for a very small interval of time.

Variation of force can be shown as follows:



Δt is generally very small, say fractions of a second.

$$(\vec{I}) = \vec{F}_{avg}.\Delta T$$

$$\vec{F}_{avg} = \frac{\vec{I}}{\Delta t}$$

$\vec{F}_{avg}$ is the average force if applied for the same small time

interval Δt will produce the same impulse I or same change in momentum.

Q. **A golf player hits a golf ball of mass 100 g and imparts it a speed of 50 m/s. If golf stick remains in contact with the ball for 0.01 s, then the average force exerted by the golf stick on the ball.**

Sol. Initially, ball was at rest, then ball gains a momentum = $\frac{100}{1000} \times 50 = 5$ N-s

The momentum is gained in 0.01 s.

Hence, the average force = $\frac{5}{0.01} = 500$ N

Q. **Two particles of a mass 2m and m are tied with an inextensible string. The particle of mass m is given a speed v as shown in the figure. Find the speed with which the particle of mass 2m starts moving.**



Sol. A string will apply equal force at its both ends and hence equal impulses.



Thus, for particle of mass 2m,

$I = 2mv_1$ ...(i)

For particle of mass m,

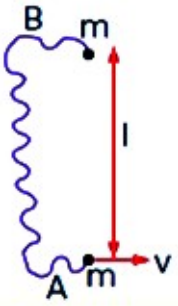
$I = mv - mv_1$ ...(ii)

Solving equation (i) and (ii),

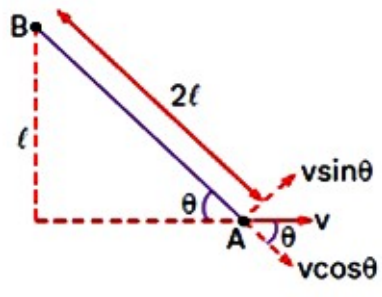
$mv - mv_1 = 2mv_1$

$$v_1 = \frac{mv}{3m} = \frac{v}{3}$$

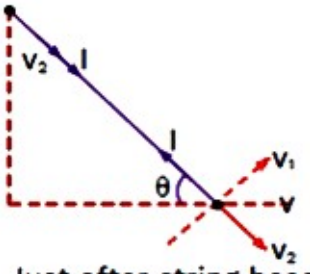
Q. **Two particles A and B of equal masses m are tied with an inextensible string of length 2l. The initial distance between A and B is l. Particle A is given speed v as shown. Find the speed of particle A and B just after the string becomes taut.**



Sol. In this situation, a jerk will be there. Now, the velocities along the length of the string will become same.



Just before string becomes taut



Just after string becomes taut

Now, string will apply impulse only along the string. The component of the velocity of A perpendicular to the string remains constant.

Thus, $v_1 = v \sin\theta$

Along the string,

$$I = mv_2$$

$$I = m(v\cos\theta - v_2)$$

$$v_2 = \frac{v\cos\theta}{2}$$

thus, speed of B is $\frac{v\cos\theta}{2} = \frac{\sqrt{3}v}{4}$.

Speed of A is $\sqrt{(v\sin\theta)^2 + \frac{(v\cos\theta)^2}{4}} = \frac{v}{2}\sqrt{1+3\sin^2\theta} = \frac{\sqrt{7}v}{4}$

$\sin\theta = \frac{\ell}{\ell} = \frac{1}{2} \Rightarrow \theta = 30°$

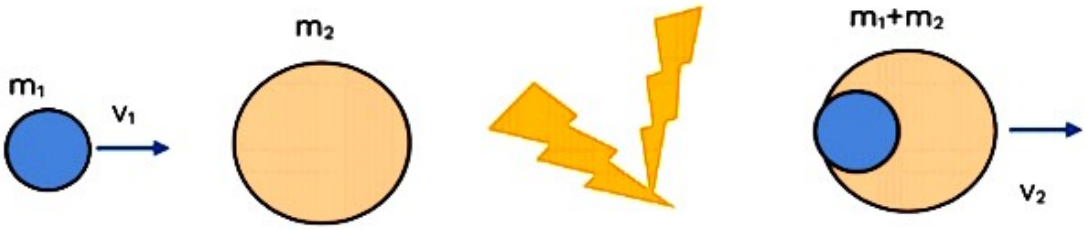
Collisions

COLLISION OR IMPACT

Collision is very common word that we encounter daily.

A cricket ball collides with the bat by the batsman. A car moving with a fast speed on a straight road collides with a truck. The above two examples broadly represent two different types of collisions. In first case, there is negligible deformation produced in the ball and it can be assumed that kinetic energy of the system is not lost due to collision, whereas in the second case, a large amount of kinetic energy is lost.

In this section, the two different types of collisions are being discussed.

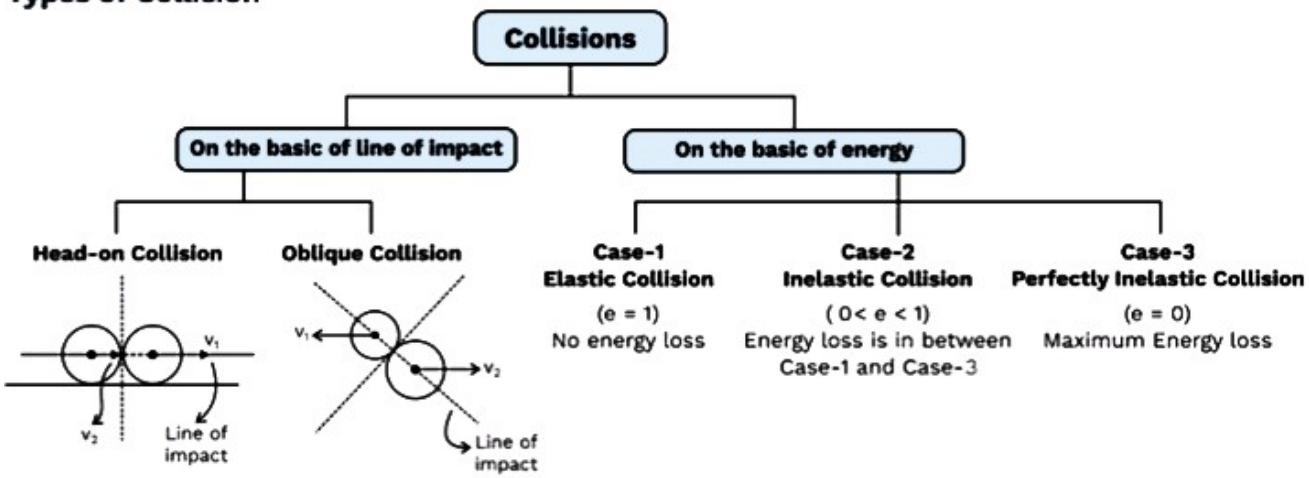


Note :-

1. It is not necessary that a physical contact takes place in a collision.

   Example : Collision between alpha particle $\left(2H_e^4\right)$ and the nucleus of gold ($_{79}Au^{197}$).
2. The duration of collision Δt is negligible as compared to the usual time interval of observation of motion.
3. In a collision, the effects of external non-repulsive forces such as gravity are not taken in account due to the small duration of collision (Δt).
4. Average impulsive force responsible for collision is much larger than external forces acting on the system.

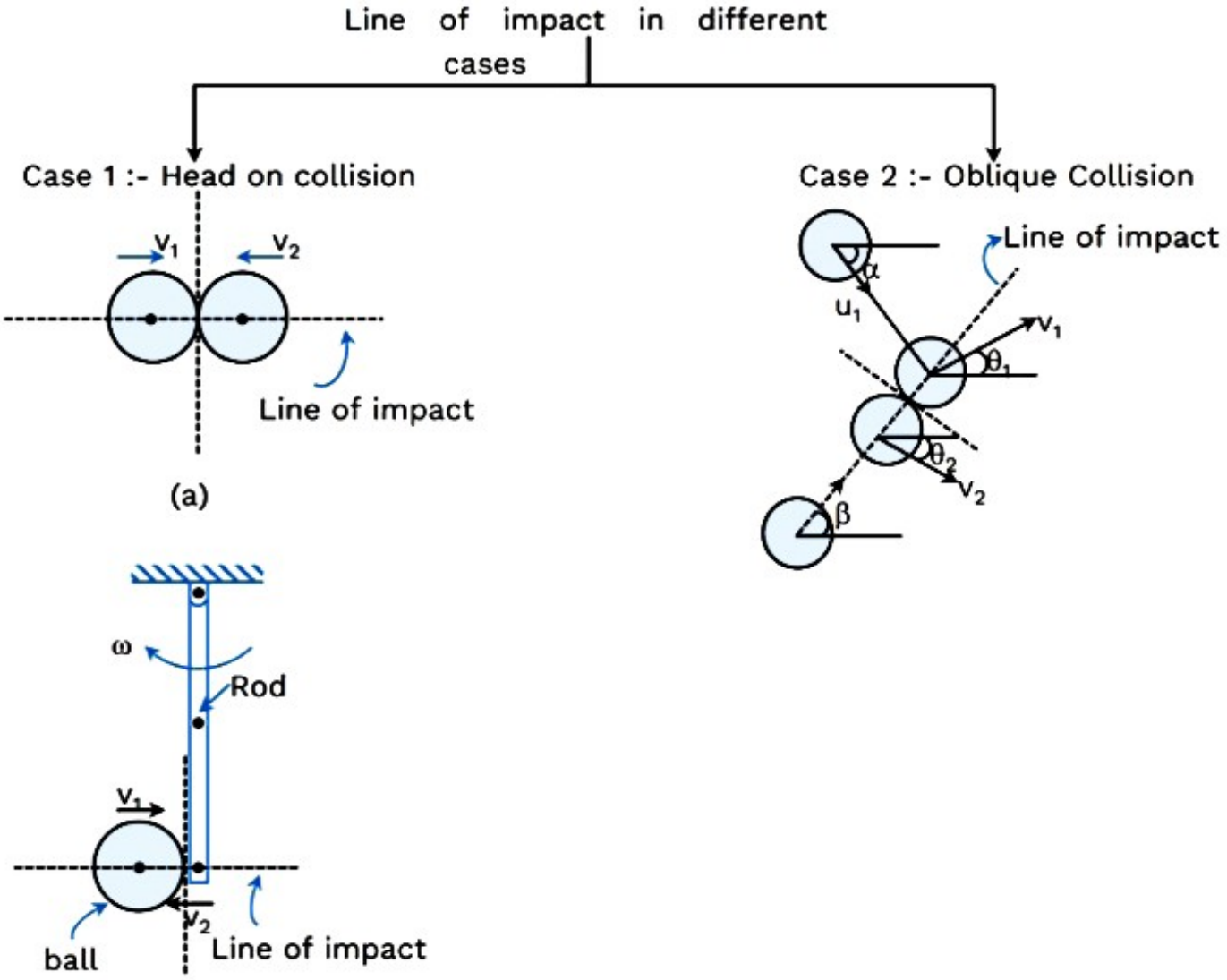
Types of Collision



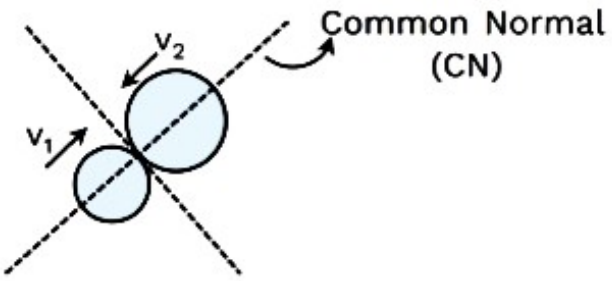
Some important terms related to collision:

1. Line of impact.
2. Common normal
3. Common tangent.
4. Coefficient of Restitution(e).

Line of impact :- 'The line of impact is the line that is co-linear to the common normal of the surfaces that are closest or in contact during impact.



Common Normal (CN)



Force is exerted in CN direction only. Both bodies exert equal and opposite forces (action and reaction) on each other. Due to this, momentum and velocities change accordingly in the CN direction.

Common Tangent

It is tangent line drawn on surfaces of the bodies.

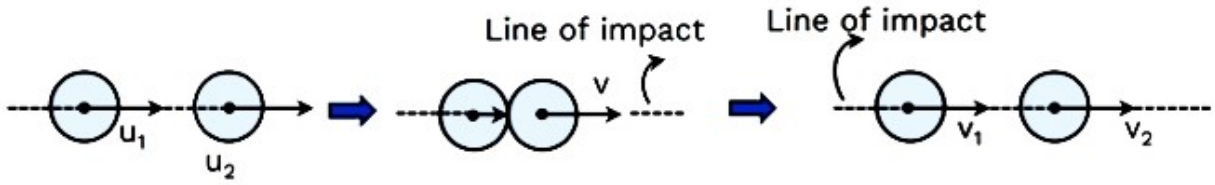
If surface is smooth or bodies are not in contact, then

Force (In tangential direction) = 0

If surface are rough then force (In tangential direction) = μ× (Normal reaction)

Note : Neither momentum nor velocity change in CT direction.

Coefficient of restitution (e)



$$e=\frac{\text{Velocity of separation of point of contact along the line of impact}}{\text{Velocity of approach of points of contact along the line of impact}}=\left(\frac{\vec{v}_2-\vec{v}_1}{\vec{u}_1-\vec{u}_2}\right)$$

For different type of material of bodies, value of e is different.

For elastic collision, e=1.

For partially elastic collision, 0<e<1.

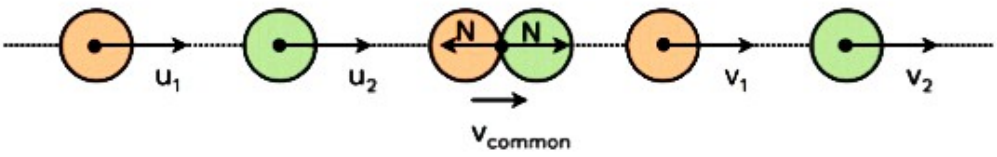
For inelastic collision, e=0.

Classification of Collision Based on Motion Direction :- Collision is divided in two categories-

(1) Head on Collision

(2) Oblique Collision

Head on Collisions :



When the line of motion of two bodies before and after collision is same, then it is termed as head on collision.

As no external force is present, so linear momentum remains conserved.

$$m_1\vec{u}_1+m_2\vec{u}_2=m_1\vec{v}_1+m_2\vec{v}_2$$

(e = 1) for elastic collision.

$$m_1u_1+m_2u_2=m_1v_1+m_2v_2 \qquad ...(1)$$

As we know for elastic collision

$$v_{separation}=(ev_{approach})$$

$$\vec{v}_2-\vec{v}_1=\vec{u}_1-\vec{u}_2 \qquad ...(2)$$

Also for elastic collision, kinetic energy remains conserved.

$\Rightarrow$ $KE_i = KE_f$

$$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \qquad ...(3)$$

Solving any equation of (1), (2), (3), we will get v_1 and v_2 respectively :-

$\Rightarrow$ From eqn (2) : $\vec{v}_2 = (\vec{u}_1 - \vec{u}_2) + \vec{v}_1$ ...(4)

From equation (1) : $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2(u_1 - u_2 + v_1)$$

On solving we get :- $\left[\frac{(m_1 - m_2)}{(m_1 + m_2)}\right]u_1 + \frac{2m_2}{(m_1 + m_2)}u_2 = v_1$

$$\vec{v}_1 = \left[\frac{(m_1 - m_2)}{(m_1 + m_2)}\right]\vec{u}_1 + \left(\frac{2m_2}{(m_1 + m_2)}\right)\vec{u}_2 \quad \text{---(A)}$$

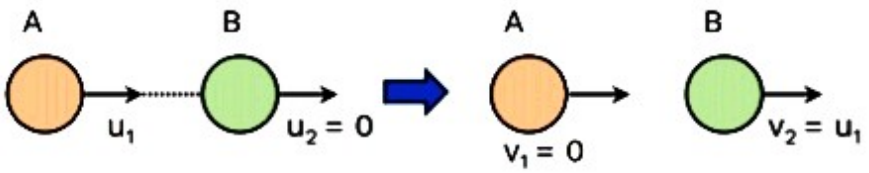
$$\vec{v}_2 = \left[\frac{m_2 - m_1}{(m_1 + m_2)}\right)\vec{u}_2 + \left(\frac{2m_1}{(m_1 + m_2)}\right)\vec{u}_1\Bigg] \quad \text{---(B)}$$

Special cases of Head on Elastic Collision :-

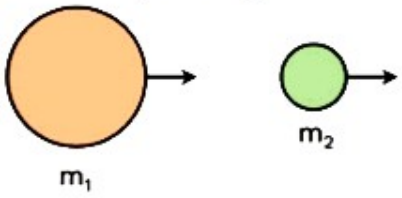
$$\left[\begin{aligned}\vec{v}_1 &= \frac{(m_1 - m_2)}{(m_1 + m_2)}\vec{u}_1 + \frac{2m_2}{(m_1 + m_2)}\vec{u}_2 \\ \vec{v}_2 &= \frac{(m_2 - m_1)}{(m_1 + m_2)}\vec{u}_2 + \frac{2m_1}{(m_1 + m_2)}\vec{u}_1\end{aligned}\right]$$

Case 1 :- If $m_1 = m_2$,

$$\left\{\begin{aligned}\vec{v}_1 &= \vec{u}_2 \\ \vec{v}_2 &= \vec{u}_1\end{aligned}\right\}$$ "After collision, velocities get swapped".



Case 2:- If $m_1 > > m_2$



$$\vec{v}_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)\vec{u}_1 + \left(\frac{2m_2}{(m_1 + m_2)}\right)\vec{u}_2$$

$\{\vec{v}_1 \cong \vec{u}_1\}$ (Think about collision of a bike with train)

$$\vec{v}_2 = \left[\frac{(m_2 - m_1)}{(m_1 + m_2)}\right]\vec{u}_2 + \left(\frac{2m_1}{(m_1 + m_2)}\right)\vec{u}_1$$

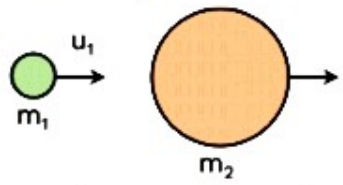
$$\vec{v}_2 = \left[\frac{m_1\left(\frac{m_2}{m_1} - 1\right)}{m_1\left(1 + \frac{m_2}{m_1}\right)}\right]\vec{u}_2 + \left(\frac{2m_1}{m_1\left(1 + \frac{m_2}{m_1}\right)}\right)\vec{u}_1$$

As $m_1 >> m_2 \Rightarrow 1 >> \left(\frac{m_2}{m_1}\right)$, then,

$$\vec{v}_2 = -\vec{u}_2 + \frac{2\vec{u}_1}{1}$$

$$\boxed{\vec{v}_2 = 2\vec{u}_1 - \vec{u}_2}$$

Case 3:- If $m_2 >> m_1$,



$$\Rightarrow \vec{v}_1 = \left(\frac{(m_1 - m_2)}{m_1 + m_2}\right)\vec{u}_1 + \left(\frac{2m_2}{(m_1 + m_2)}\right)\vec{u}_2$$

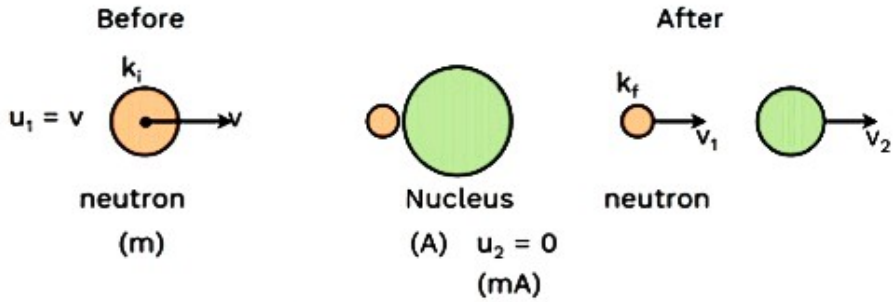
$$\vec{v}_1 = -\vec{u}_1 + 2\vec{u}_2 \quad \Rightarrow \quad \boxed{\vec{v}_1 = 2\vec{u}_2 - \vec{u}_1}$$

$$\vec{v}_2 = \frac{m_2 - m_1}{(m_1 + m_2)}\vec{u}_2 + \left(\frac{2m_1}{m_1 + m_2}\right)\vec{u}_1$$

$$\boxed{\vec{v}_2 = \vec{u}_2}$$

Q. **A neutron moving at a speed v undergoes a head on elastic collision with a nucleus of mass number A at rest. Find the ratio of kinetic energies of neutron after and before collision.**

Sol.



After Head on Collision (Elastic), the speed of Neutron is,

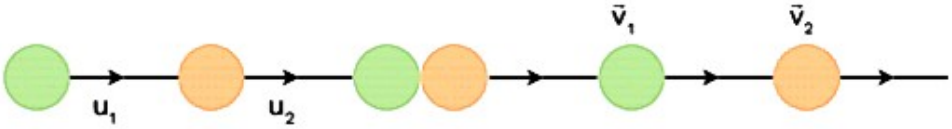
$$\vec{v}_1 = \left[\frac{(m_1 - m_2)}{(m_1 + m_2)}\right]\vec{u}_1 + \frac{2m_2}{(m_1 + m_2)}\vec{u}_2$$

$$\vec{v}_1 = \left(\frac{m - mA}{m + mA}\right)v + 0 = \frac{(1 - A)}{1 + A}v \quad \Rightarrow \quad v_1 = \left(\frac{1 - A}{1 + A}\right)v$$

$$\frac{k_f}{k_i} = \frac{\frac{1}{2}mv_1^2}{\frac{1}{2}mv^2} = \frac{\left(\frac{1 - A}{1 + A}\right)^2 v^2}{v^2} = \left(\frac{1 - A}{1 + A}\right)^2$$

$$\boxed{\frac{k_f}{k_i} = \left(\frac{1 - A}{1 + A}\right)^2}$$

Partially Elastic Head on Collision (0 < e < 1) :



$\boxed{v_{Approach} = u_1 - u_2}$ $\qquad$ $v_{separation} = (\vec{v}_2 - \vec{v}_1)$

By Conservation of Momentum,

$m_1u_1 + m_2u^2 = m_1v_1 + m_2v_2$ ...(1)

Using Coefficient of Restitution,

$v_2 - v_1 = e(u_1 - u_2)$ ...(2)

From equation (1)

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2(v_1 + e(u_1 - u_2))$$

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_1 + e\,m_2u_1 - e\,m_2u_2$$

$$(m_1 - em_2)\,u_1 + m_2(u_2 + eu_2) = (m_1 + m_2)v_1$$

$$v_1 = \frac{(m_1 - em_2)u_1}{m_1 + m_2} + \frac{(m_2 + em_2)u_2}{m_1 + m_2}$$

$$\boxed{\vec{v}_1 = \frac{(m_1 - em_2)}{m_1 + m_2}\vec{u}_1 + \frac{m_2(1+e)}{(m_1 + m_2)}\vec{u}_2} \quad \text{_____(A)}$$

$$\boxed{\vec{v}_2 = \frac{(m_2 - em_1)}{m_1 + m_2}\vec{u}_2 + \frac{m_1(1+e)}{(m_1 + m_2)}\vec{u}_1} \quad \text{_____(B)}$$

Loss of Kinetic Energy

In the collision between the two objects, there may be a loss of kinetic energy. This can be calculated by subtracting final kinetic energy from initial kinetic energy. Thus, we obtain

$$\Delta K = \left(\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2\right) - \left(\frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2\right)$$

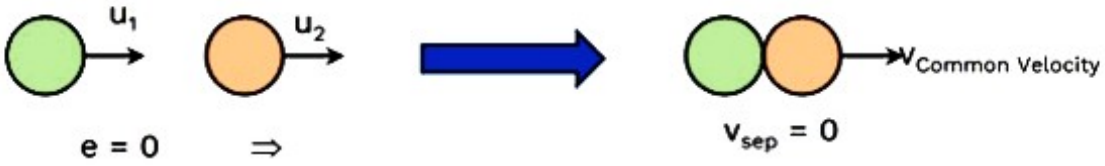
$$\Delta K = \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 - \frac{1}{2}m_1\left[\left(\frac{m_1 - em_2}{m_1 + m_2}\right)\vec{u}_1 + \left(\frac{m_2(1+e)}{m_1 + m_2}\right)\vec{u}_2\right]^2 - \frac{1}{2}m_2\left[\left(\frac{m_2 - em_1}{m_2 + m_1}\right)\vec{u}_2 + \left(\frac{m_1(1+e)}{m_1 + m_2}\right)\vec{u}_1\right]^2$$

or $$\Delta K = \frac{1}{2}\frac{m_1m_2}{m_1 + m_2}(\vec{u}_1 - \vec{u}_2)^2(1 - e^2) \qquad \text{.....(1)}$$

The loss of kinetic energy occurs because of the deformation produced in two objects, during collision.

During collision, first the objects are deformed and then recover their shape. The internal restoring forces developed during deformation and recovery may or may not be conservative. Accordingly, dissipation of energy may or may not be there.

Inelastic Head on Collision (e = 0) :-

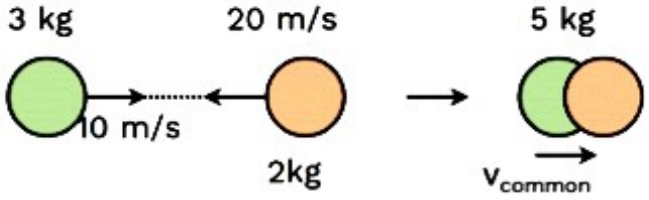


By Conservation of linear momentum,

$$\bar{v}_{Common} = \frac{m_1\bar{u}_1 + m_2\bar{u}_2}{(m_2 + m_2)}$$

Q. **A 3 kg ball moving at 10 m/s collides head on inelastically with another ball of mass 2 kg, moving towards the first ball at 20 m/s. After collision balls will stick to each other, find the amount of total energy loss in this collision.**

Sol.



$$v_{common} = \frac{m_1\bar{v}_1 + m_2\bar{u}_2}{m_1 + m_2} = \frac{3 \times 10\hat{i} + 2 \times (20)(-\hat{i})}{5}$$

$$= \frac{30\hat{i} - 40\hat{i}}{5} = \frac{-10\hat{i}}{5} = -2\hat{i}$$

v_{common} = 2 m/s in negative direction.

k_i = 550 J

k_f = 10 J

Loss of kinetic energy, $\Delta k = k_i - k_f$

= 540 J

Q. **A ball drops from a ceiling of a room and after rebouncing twice from the floor reaches a height equal to half that of the ceiling. Show that the coefficient of restitution is $\sqrt[4]{1/2}$.**

Sol. Let h = height of ceiling

$\Rightarrow$ Speed before the first impact $= \sqrt{2gh}$

$\Rightarrow$ Speed after the first impact $= e\sqrt{2gh}$

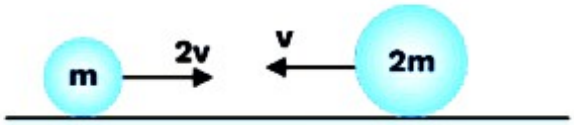
The ball comes back for the second impact.

Before the second impact, speed $= e\sqrt{2gh}$

After the second impact, speed $= e^2\sqrt{2gh}$

$$\text{Height attained} = \frac{u^2}{2g} \Rightarrow \frac{h}{2} = \frac{[e^2\sqrt{2gh}]^2}{2g} \Rightarrow \frac{h}{2} = \frac{e^4(2gh)}{2g} \Rightarrow e = \frac{1}{\sqrt[4]{2}}$$

Q. **Two particles of masses m and 2m moving in opposite directions collide elastically with velocity 2v and v, respectively. Find their velocities after collision.**



Sol. Let the final velocities of m and 2m and v_1 and v_2, respectively, as shown in figure.



By conservation of momentum, we get

$$m(2v) + 2m(-v) = m(v_1) + 2m(v_2)$$

or $0 = mv_1 + 2mv_2$

or $v_1 + 2v_2 = 0$ ...(1)

and since the collision is elastic,

$$v_2 - v_1 = 2v - (-v)$$

or $$v_2 - v_1 = 3v \qquad ...(2)$$

Solving the above two equations, we get

$$v_2 = v \text{ and } v_1 = -2v$$

That is, mass 2m returns with velocity v while mass m returns with velocity 2v in the direction shown in figure.



Q. **A ball of mass m moving at a speed v makes a head-on collision with an identical ball at rest. The kinetic energy of the balls after the collision is 3/4th of the original. Find the coefficient of restitution.**

Sol. As we have seen in the above discussion that under the given conditions :



Before collision **After collision**

$$v_1 = \left(\frac{1+e}{2}\right)v \text{ and } v_2 = \left(\frac{1-e}{2}\right)v$$

Given that $K_f = \frac{3}{4}K_i$

or $$\frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 = \frac{3}{4}\left(\frac{1}{2}mv^2\right)$$

Substituting the values, we get

$$\left(\frac{1+e}{2}\right)^2 + \left(\frac{1-e}{2}\right) = \frac{3}{4}$$

or $$(1+e)^2 + (1-e)^2 = 3 \text{ or } 2 + 2e^2 = 3$$

or $$e^2 = \frac{1}{2} \text{ or } e = \frac{1}{\sqrt{2}}$$

Q. **Two balls of masses 2 kg and 4 kg are moved towards each other with velocities 4 m/s and 2 m/s, respectively, on a frictionless surface. After colliding, the 2 kg ball returns back with velocity 2 m/s. Then find**

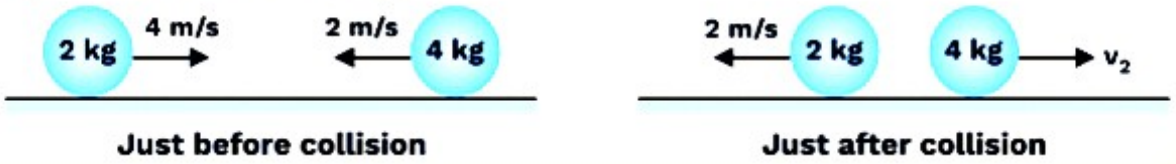
a) velocity of the 4 kg ball after collision.

b) coefficient of restitution e.

c) impulse of deformation J_D .

d) maximum potential energy of deformation.

e) impulse of reformation J_R.



Just before collision **Just after collision**

Sol. a) By momentum conservation,

$$2(4) - 4(2) = 2(-2) + 4(v_2)$$

$$\Rightarrow v_2 = 1 \text{ m/s}$$

b) $$e = \frac{\text{Velocity of separation}}{\text{Velocity of approach}} = \frac{1-(-2)}{4-(-2)} = \frac{3}{6} = 0.5$$

c) At the maximum deformed state, by conservation of momentum, common velocity is v = 0.

$$J_D = m_2(v - u_2) = 4[(0 - (-2)] = 8 \text{ N s}$$

d) Potential energy at the maximum deformed state,

U = loss in kinetic energy during deformation

$$\text{or } U = \left(\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2\right) - \frac{1}{2}(m_1 + m_2)v^2$$

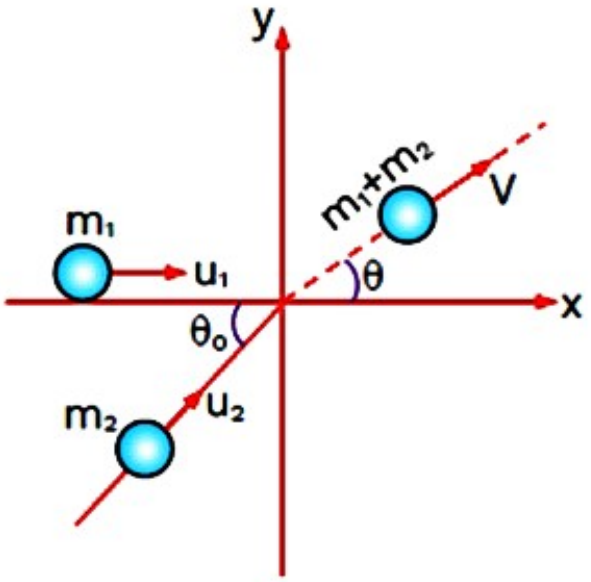
$$= \left(\frac{1}{2}2(4)^2 + \frac{1}{2}4(2)^2\right) - \frac{1}{2}(2+4)(0)^2 = 24 \text{ J}$$

e) $$J_R = m_2(v_2 - v) = 4(1-0) = 4 \text{ N s}$$

also $J_R = eJ_D = 0.5 \times 8 = 4 \text{ N s}$.

TWO DIMENSIONAL COLLISION

The figure shows a 2-dimensional totally inelastic collision.



In given case, first object of mass m_1, initially moves along the x-axis with speed u_1. On the other hand, the second object of mass m_2, initially moves at an angle θ_0 to the x-axis with speed u_2.

After the collision, the two objects stick together and move at angle θ to the x-axis with speed v.

By conservation of linear momentum,

$$m_1\vec{u}_1 + m_2\vec{u}_2 = (m_1 + m_2)\vec{v}$$

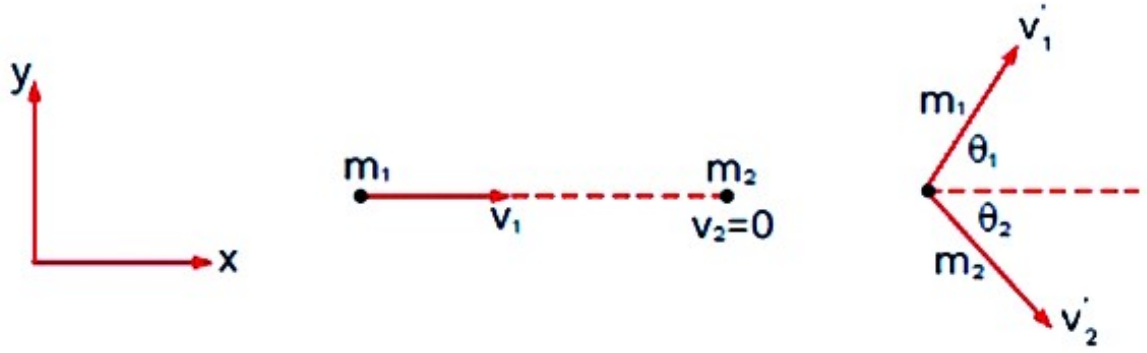
or $$m_1(u_1\hat{i}) + m_2(u_2\cos\theta_0\hat{i} + u_2\sin\theta_0\hat{j}) = (m_1 + m_2)(V\cos\theta\hat{i} + V\sin\theta\hat{j})$$

or $$m_1u_1 + m_2u_2\cos\theta_0 = (m_1 + m_2)\,V\cos\theta \qquad ...(i)$$

and $$m_2u_2\sin\theta_0 = (m_1 + m_2)\,V\sin\theta \qquad ...(ii)$$

This is a system of two equations and two unknows (i.e., V and θ) They can be solved to obtain the values of V and θ.

Q. **A particle of mass m_1 and speed v_1 (in the +x-direction) collides with another particle of mass m_2. m_2 is at rest before the collision occurs, thus $v_2 = 0$. After the collision, the particles have velocities $\vec{v}_1'$ and $\vec{v}_2'$ in the x-y plane in the directions of θ_1 and θ_2 with the x-axis. There are no external forces.**



(A) What is the total momentum before the collision (direction and magnitude)?

(B) What is the total momentum after the collision (direction and magnitude)?

(C) What is the total kinetic energy before the collision?

(D) What is the ratio of the speed $\frac{v_2'}{v_1'}$?

(E) What is the magnitude (speed) of $\vec{v}_1'$. If $m_1 = m_2$?

Sol. (A) $\vec{p} = m_1 v_1 \hat{i}$

(B) $\vec{p} = m_1 v_1 \hat{i}$ (by conservation of momentum)

(C) $\frac{1}{2} m_1 v_1^2$

(D) By conservation of linear momentum along y-axis,

$$m_1 v_1' \sin\theta_1 = m_2 v_2' \sin\theta_2 \Rightarrow \frac{v_2'}{v_1'} = \frac{m_1 \sin\theta_1}{m_2 \sin\theta_2}$$

(E) By conservation of momentum along x-axis,

$$m_1 v_1' \cos\theta_1 + m_2 v_2' \cos\theta_2 = m_1 v_1$$

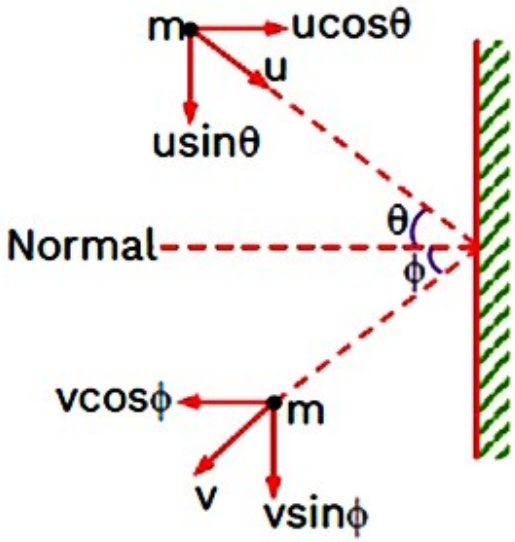
$$m_1 v_1' \cos\theta_1 + \left(m_2 v_1' \frac{\sin\theta_1}{\sin\theta_2}\right) \cos\theta_2 = m_1 v_1$$

$$v_1'\left(\cos\theta_1 + \frac{\sin\theta_1}{\sin\theta_2}\cos\theta_2\right) = v_1 \qquad (\because m_1 = m_2)$$

$$\Rightarrow \quad v_1' = v_1\frac{\sin\theta_2}{\cos\theta_1\sin\theta_2 + \sin\theta_1\cos\theta_2} = \frac{v_1\sin\theta_2}{\sin(\theta_2+\theta_1)}$$

Q. **A particle of mass m is projected towards a wall such that the angle of incidence is θ and the speed just before collision is u. Assuming that the wall is smooth and the collision is elastic, show that the ball rebounds at same angle.**

Sol. The given situation is shown in figure. As the collision is elastic, there is no loss of energy and hence speed of rebound is same.



The contact force between the wall and ball is along the normal, as friction is absent.

So, the magnitude of momentum of ball along the wall must not change.

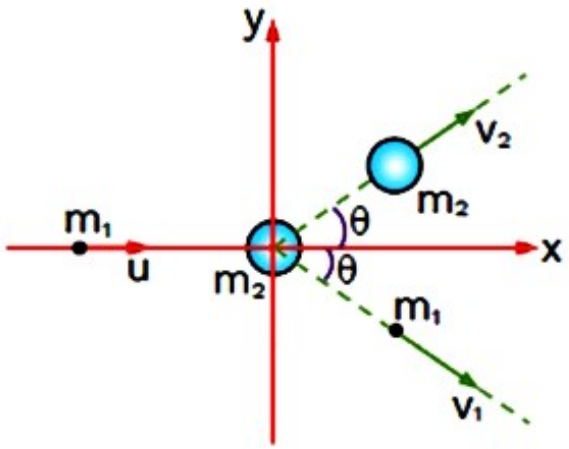
Now, v = u (No loss of energy)

Also, $v\sin\phi = u\sin\theta$ (No force along the wall)

$\Rightarrow \quad \theta = \phi$

Q. **Two objects slide over a frictionless horizontal surface. The first object, mass m_1 = 5 kg, is propelled with a speed u = 4.5 m/s towards the second object, mass m_2 = 2.5 kg, which is initially at rest.**
After the collision, both objects have velocities which are directed at $\theta = 30°$ on either side of the original line of motion of the first object. What are the final speeds of the two objects? Is the collision elastic or inelastic?

Sol.



By conservation of momentum along x-axis,

$$m_1u = m_1v_1\cos\theta + m_2v_2\cos\theta \quad \text{...(i)}$$

By conservation of momentum along the y-axis,

$$0 = -m_1v_1\sin\theta + m_2v_2\sin\theta$$

or $$m_1v_1\sin\theta = m_2v_2\sin\theta \quad \text{...(ii)}$$

From (i) & (ii), we get,

$$v_1 = \frac{u}{2\cos\theta} = \frac{4.5}{2\cos 30^\circ} = 2.598 \text{ m/s}$$

and $$v_2 = \frac{m_1v_1}{m_2} = \frac{5 \times 2.6}{2.5} = 5.2 \text{ m/s}$$

The initial kinetic energy of the system is, $K_1 = \frac{1}{2}m_1u^2$

And, final kinetic energy of the system is,

$$K_2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$$

$$= \frac{1}{2}m_1\left(\frac{u}{2\cos 30^\circ}\right)^2 + \frac{1}{2}m_2\left(\frac{m_1v_1}{m_2}\right)^2$$

$$= \frac{1}{2}m_1\frac{u^2}{4\cos^2 30^\circ} + \frac{1}{2}\frac{m_2 m_1^2}{m_2^2} \times \frac{u^2}{4\cos^2 30^\circ}$$

$$= \frac{1}{2}m_1u^2\left[\frac{1}{4\times\frac{3}{4}} + \frac{5}{2.5\times 4\times\frac{3}{4}}\right]$$

$$= \frac{1}{2}m_1u^2\left[\frac{1}{3}+\frac{2}{3}\right]$$

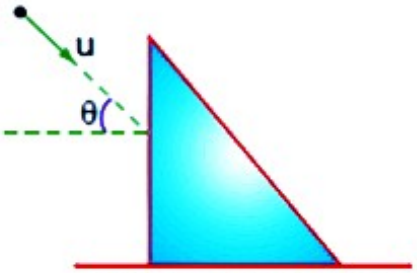
$$= \frac{1}{2}m_1u^2$$

Since, $K_1 = K_2$, so the collision is elastic.

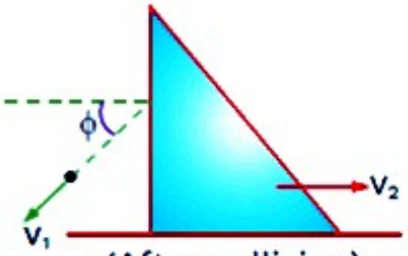
COEFFICIENT OF RESTITUTION IN OBLIQUE COLLISIONS

Equation of coefficient of restitution in oblique collisions can only be applied in the direction normal to the plane of colliding surface.

Q. **If a particle of mass m collides with a wedge a shown, then find coefficient of restitution.**



(Before collision)



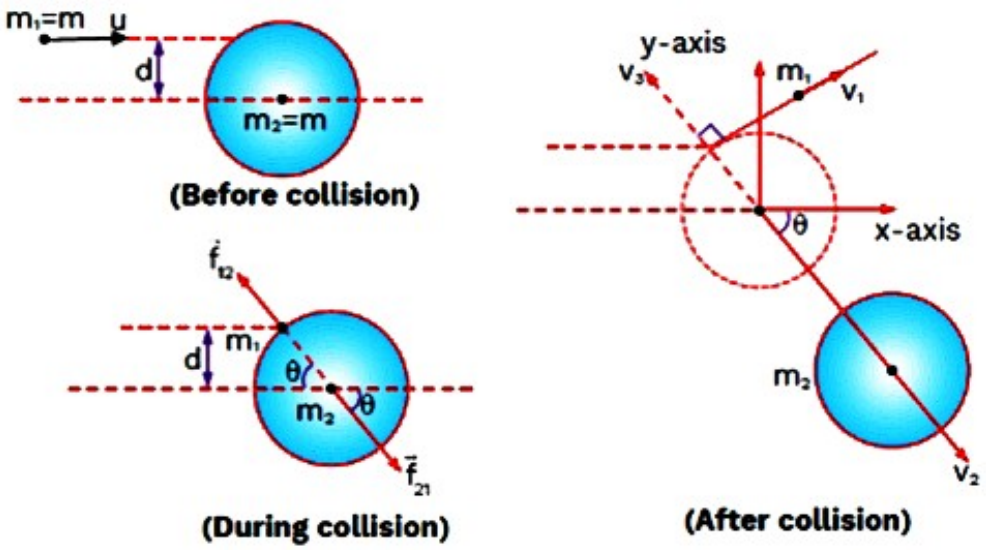
(After collision)

Sol. Along the normal to the surface, $e = \dfrac{v_2 - (-v_1\cos\phi)}{u\cos\theta}$

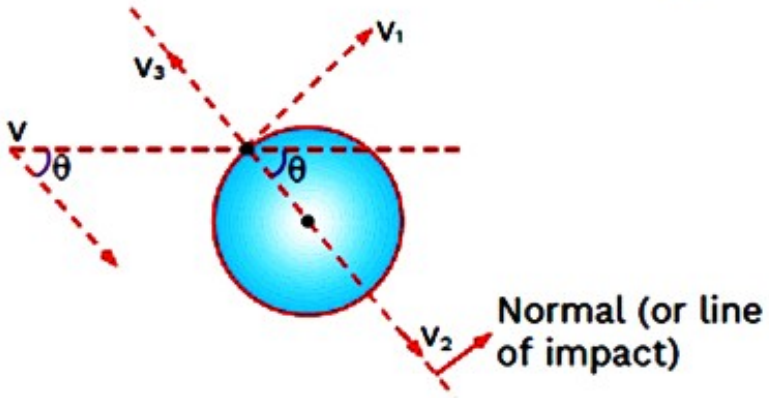
Here, initial and final velocities of particle and wedge are broken in components along the normal, then the equation of coefficient of restitution is applied.

Q. **A particle of mass m moving with speed u collides elastically with a sphere of radius R and same mass, at rest, at an impact parameter d. Find**
(A) Angle between their final velocities, and
(B) Magnitude of their final velocities

Sol.



We first locate the line of impact or normal to the colliding surface. Along this line, equation of coefficient of restitution can be applied.



As the collision is perfectly elastic and masses of both the particles are same, thus their velocity vectors should be exchanged along the line of impact.
Thus, $v_2 = u\cos\theta$ and $v_3 = 0$
Now, if we focus on the particle alone, there is no force on it perpendicular to the line of impact. Thus, its velocity does not change in this direction and hence,

$$v_1 = u\sin\theta$$

Now, from geometry,

$$\sin\theta = \frac{d}{R}$$

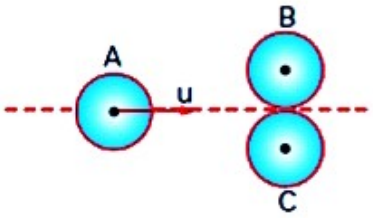
$$\cos\theta = \sqrt{1-\frac{d^2}{R^2}}$$

$$\Rightarrow \quad v_2 = \frac{u}{R}\sqrt{R^2 - d^2}$$

$$v_1 = \frac{ud}{R}$$

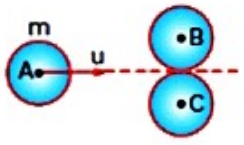
$$v_3 = 0$$

Q. **B and C are two identical spheres kept in contact with each other on a horizontal frictionless floor. Another identical sphere A moving with velocity u along the common tangent to B and C strikes B and C simultaneously.**

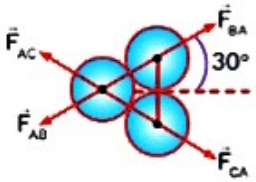


Find their velocities after collision if
(A) Collision is elastic
(B) Collision is inelastic with coefficient of restitution e.

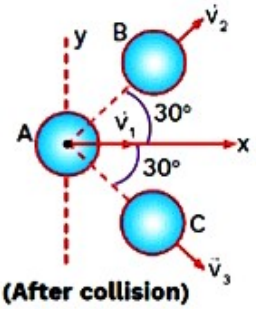
Sol.



(Before collision)



(During collision)



(After collision)

(A) For elastic collision

By conservation of linear momentum along y-axis,

$$mv_2\sin30° - mv_3\sin30° = 0$$

$\Rightarrow$ $v_2 = v_3 = v_0$(say) ...(i)

By conservation of linear momentum along x-axis,

$mu = mv_1 + mv_2\cos30° + mv_3\cos30°$

$u = v_1 + 2v_0\cos30°$

$u = v_1 + \sqrt{3}\, v_0$ ...(ii)

Coefficient of restitution,

$$e = \frac{v_{separation}}{v_{apprcach}}$$

$$1 = \frac{v_0 - v_1\cos 30°}{u\cos 30°} \quad \text{(for elastic collision)}$$

$\Rightarrow$ $$\frac{\sqrt{3}u}{2} = v_0 - \frac{v_1\sqrt{3}}{2}$$

$$u = \frac{2v_0}{\sqrt{3}} - v_1 \quad ...(iii)$$

Adding equation (i) and (ii),

$$2u = \sqrt{3}\, v_0 + \frac{2v_0}{\sqrt{3}} = \frac{3v_0 + 2v_0}{\sqrt{3}}$$

$$u = \frac{5v_0}{2\sqrt{3}} \Rightarrow v_0 = \frac{2\sqrt{3}u}{5} \quad ...(iv)$$

Putting value of v_0 in equation (iii),

$$v_1 = \frac{2v_0}{\sqrt{3}} - u = \frac{2}{\sqrt{3}}\frac{2\sqrt{3}u}{5} - u$$

$$= \frac{4u}{5} - u$$

$$= -\frac{u}{5} \quad ...(v)$$

(B) For inelastic collision

Total momentum of the system is conserved.

We obtain the same equations as in part (A)

$\Rightarrow$ equation (i), $v_1 = v_2 = v_0$

Equation (ii), $u = v_1 + \sqrt{3}\, v_0$

Coefficient of restitution,

$$e = \frac{v_0 - v_1\cos 30°}{u\cos 30°}$$

$$\Rightarrow \quad eu\frac{\sqrt{3}}{2} = v_0 - v_1\frac{\sqrt{3}}{2}$$

$$\Rightarrow \quad u = \frac{2v_0}{e\sqrt{3}} - \frac{v_1}{e} \qquad \text{...(vi)}$$

Dividing equation (ii) by e,

$$\frac{u}{e} = \frac{v_1}{e} + \frac{\sqrt{3}v_0}{e}$$

$$\frac{u}{e} = \frac{v_1}{e} + \frac{3v_0}{e\sqrt{3}}$$

Equation (vi), $u = \frac{2v_0}{e\sqrt{3}} - \frac{v_1}{e}$

Adding, $u\left(\frac{1}{e}+1\right) = \frac{1}{e}\left[\frac{5v_0}{\sqrt{3}}\right]$

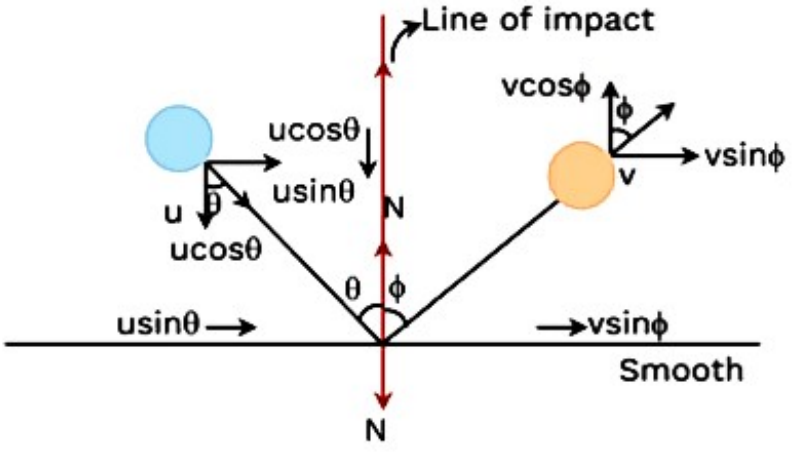
$$\Rightarrow \quad v_0 = \frac{\sqrt{3}u}{5}(1+e) \qquad \text{...(vii)}$$

Putting value of v_0 in equation (vi),

$$v_1 = \left(\frac{2v_0}{\sqrt{3}} - eu\right)$$

$$= \frac{u}{5}(2-3e) \qquad \text{...(viii)}$$

Oblique Collision of a ball on a flat surface :-



Case 1 - Elastic Collision (e = 1) :-

$v_{App} = v_{sep}$ along the line of impact.

$u \cos\theta = v \cos\phi$ ...(1)

Also, no external force is acting along the surface so, momentum will be conserved, then,

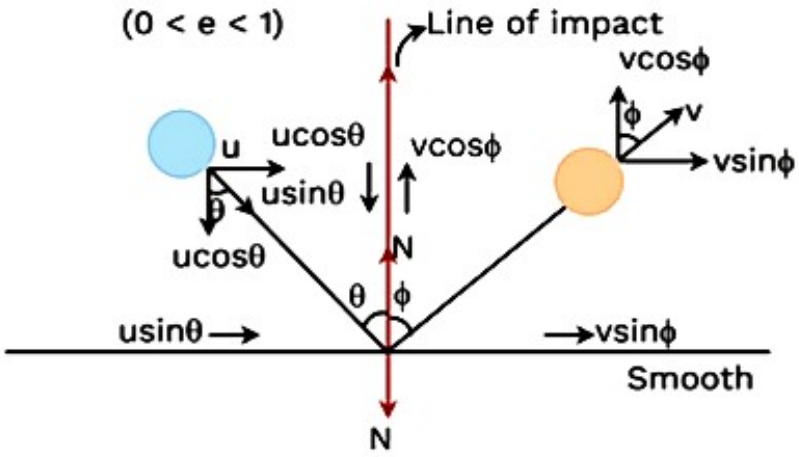
$u\sin\theta = v\sin\phi$ ...(2)

Dividing (1) by (2),

$$\frac{u\cos\theta}{u\sin\theta} = \frac{v\cos\phi}{v\sin\phi} \Rightarrow \cot\theta = \cot\phi \Rightarrow \theta = \phi \quad ...(3)$$

$\Rightarrow$ $\boxed{v = u \ ; \ \phi = \theta}$

Case 2 - Partial Elastic Collision :-



Along the Line of Impact,

$v_{sep} = e(v_{App})$

$v\cos\phi = e(u\cos\theta)$ ...(1)

As along the surface, no external force is acting, so momentum will be conserved,

$v\sin\phi = u\sin\theta$ ...(2)

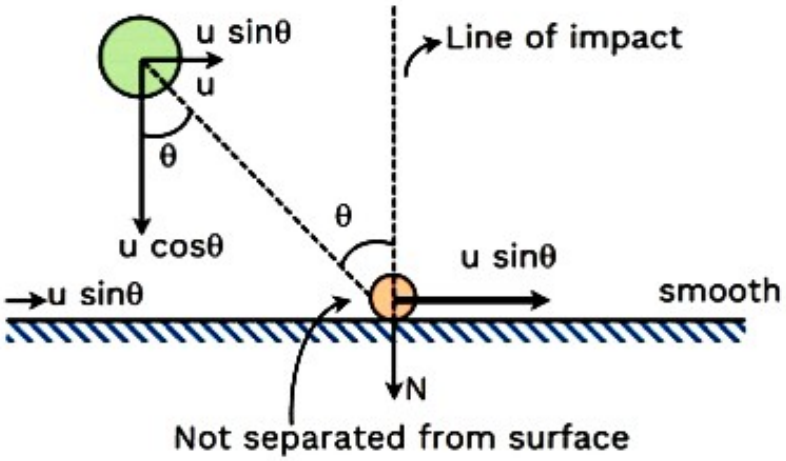
Squaring and adding (1) and (2),

$v^2\cos\phi + v^2\sin^2\phi = e^2u^2\cos^2\theta + u^2\sin^2\theta$

$v^2 = (e^2\cos^2\theta + \sin^2\theta)u^2$

$$v = u\sqrt{\sin^2\theta + e^2\cos^2\theta} \qquad \tan\phi = \left(\frac{1}{e}\right)(\tan\theta).$$

Case -3 : Inelastic Collision (e = 0) :-

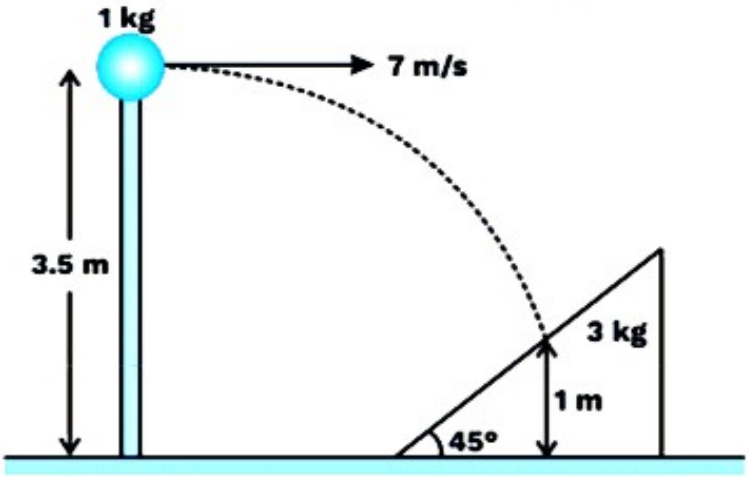


There is no component of velocity perpendicular to the surface after the collision, but as the surface is smooth, so there is no external force along the surface and the ball's momentum will be conserved along the surface.

$mu\sin\theta = m\vec{v}$

$\boxed{\vec{v} = u\sin\theta}$ ...along surface.

Q. **A smooth ball of mass 1 kg is projected with velocity 7 m/s horizontal from a tower of height 3.5 m. It collides elastically with a wedge of mass 3 kg and inclination of 45° kept on ground (in figure). The ball collides with the wedge at a height of 1 m above the ground. Find the velocity of the wedge and the ball after collision. (Neglect friction at any contact.)**

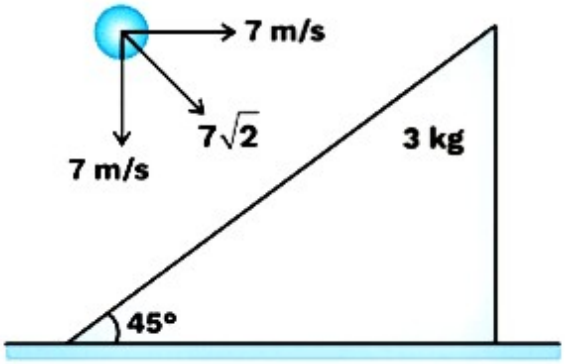


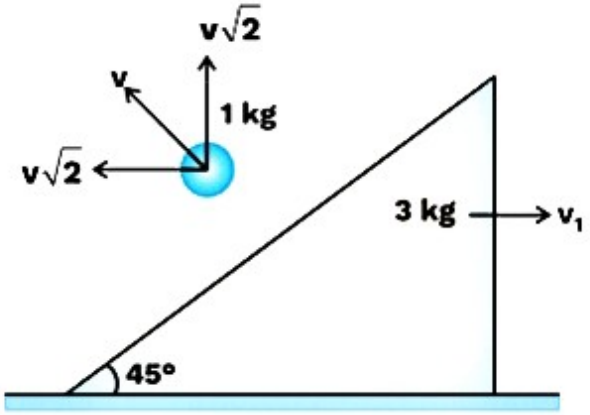
Sol. v_x = 7 m/s, velocity along y-axis of the ball just before collision.

$$v_y = \sqrt{2 \times 9.8 \times 2.5} = 7 \text{ m/s}$$

As $v_x = v_y$, so it strikes the plane of incline perpendicularly.

Let the ball rebound with velocity V and let v_1 be the velocity of the wedge.





Applying the principle conservation of momentum in horizontal direction, we get

$$1 \times 7 = -1 \times \frac{v}{\sqrt{2}} + 3v_1$$

$$7\sqrt{2} = 3\sqrt{2}\, v_1 - v \qquad \ldots(1)$$

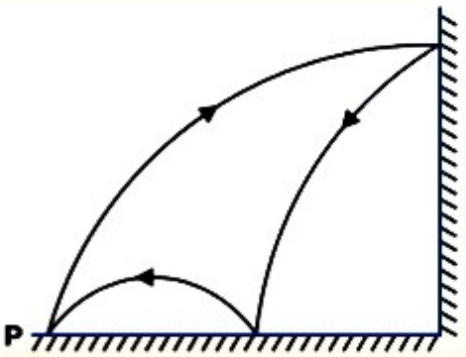
Applying the equation for coefficient of restitution, we get

$$e = 1 = \frac{v_1/\sqrt{2} + v}{7\sqrt{2}}, \quad 7\sqrt{2} = \frac{v_1}{\sqrt{2}} + v \qquad \ldots(2)$$

Solving Eqs. (1) and (2)

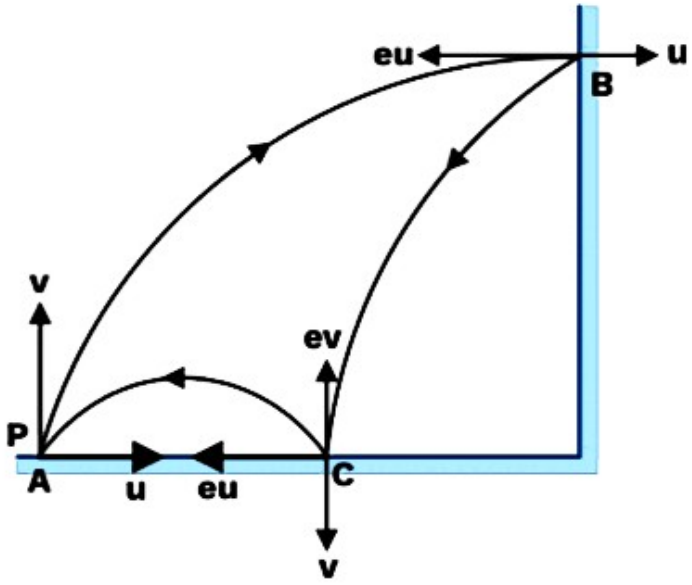
$$v_1 = 4 \text{ m/s and } v = 5\sqrt{2} \text{ m/s}$$

Q. **A small ball is projected from point P towards a vertical wall as shown in figure. It hits the wall when its velocity is horizontal. Ball reaches point P after one bounce on the floor. The coefficient of restitution assuming it to be same for two collisions is n/2. All surfaces are smooth. Find the value of n.**

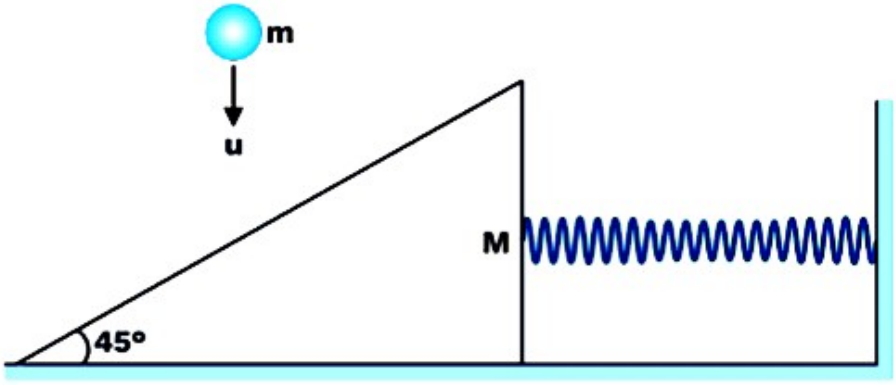


Sol. u time (AB) = eu (time BC) + eu (time CA)

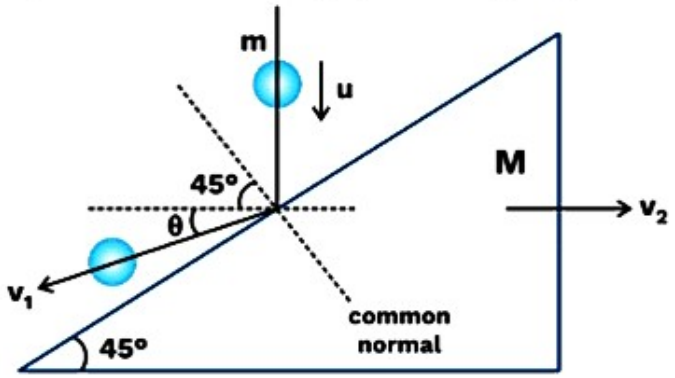
$$u\frac{v}{g} = eu\left(\frac{v}{g}\right) + eu \times 2\frac{ev}{g} \quad \Rightarrow e = \frac{1}{2}$$



Q. **A ball of mass m hits a wedge of mass M vertically with speed u, which is placed, on a smooth horizontal surface. Find the maximum compression in the spring, if the collision is perfectly elastic and no friction anywhere. Spring constant of spring is K.**



Sol. Let after collision, the ball has velocity v_1 and wedge v_2 as shown.



Apply conservation of momentum in horizontal direction :

$$mv_1 \cos\theta = Mv_2 \qquad ...(1)$$

From Newton's experimental law :

$$e = 1 = \frac{v_2 \sin 45° - [-v_1 \cos(45 + \theta)]}{u \cos 45°}$$

$$\Rightarrow \quad u = v_2 + v_1 \cos\theta - v_1 \sin\theta \qquad ...(2)$$

Velocity component of the ball along the surface of wedge will remain the same as during collision there is no force on the ball in this direction. So,

$$u \sin 45° = v_1 \cos(45 - \theta)$$

$$\Rightarrow \quad u = v_1 \cos\theta + v_1 \sin\theta \qquad ...(3)$$

From (1), (2) and (3)

$$v_2 = \frac{2mu}{m + 2M}$$

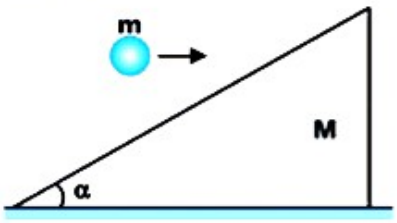
To find maximum compression in the spring, after collision apply conservation of energy for the system of wedge and spring :

$$\frac{1}{2} Mv_2^2 = \frac{1}{2} Kx^2$$

$$\Rightarrow \quad x = \sqrt{\frac{M}{K}}\, v_2$$

$$\Rightarrow \quad x = \left(\frac{2mu}{m + 2M}\right)\sqrt{\frac{M}{K}}$$

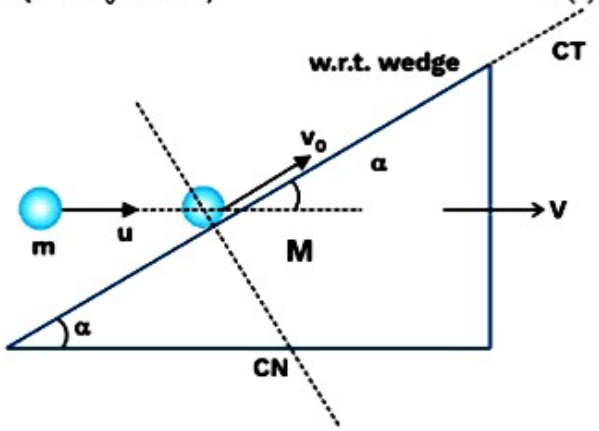
Q. **A ball of mass m moving with constant horizontal velocity u strikes a stationary wedge of mass M on its inclined surface as shown in the figure. After collision, the ball starts moving up the inclined plane. The wedge is placed on frictionless horizontal surface.**



a) Calculate the velocity of wedge immediately after collision.
b) Calculate the maximum height the ball can ascend on the wedge.

Sol. a) Let after collision velocity of wedge be V and that of ball be v_0 as shown. v_0 is w.r.t. wedge. Apply conservation of linear momentum along x-axis, we get

$$mu = Mv + m(V + v_0 \cos\alpha) \quad \ldots(1)$$



Velocity along CT remains, constant

$$u\cos\alpha = v_0 + V\cos\alpha \quad \ldots(2)$$

Net velocity of ball just after collision is

$$\vec{v}_b = (v_0\cos\alpha + V)\hat{i} + (v_0\sin\alpha)\hat{j}$$

From (i) and (ii),

$$mu = MV + mV + m\cos\alpha(u\cos\alpha - V\cos\alpha)$$

$$= (M+m)V + mu\cos^2\alpha - mV\cos^2\alpha$$

$$\Rightarrow \quad V = \frac{mu\sin^2\alpha}{M + m\sin^2\alpha}$$

b) When the ball ascends maximum height, then ball will be at rest w.r.t wedge and their velocity will be same. Let their common velocity be v'.

By conservation of momentum, we can find v' as,

$v' = mu/(M + m)$.

From (ii), $v_0 = (u - V)\cos\alpha$, put the value of V in this to get

$$v_0 = \frac{Mu\cos\alpha}{M + m\sin^2\alpha}$$

Also, $v_0\cos\alpha + V = \left[\frac{M\cos^2\alpha + m\sin^2\alpha}{M + m\sin^2\alpha}\right]u$

To find maximum height:

From energy conservation :

$$mgh + \frac{1}{2}(m+M)v'^2$$

$$= \frac{1}{2}MV^2 + \frac{1}{2}m(v_0\cos\alpha + V)^2 + \frac{1}{2}m(v_0\sin\alpha)^2$$

$$2mgh + \frac{m^2u^2}{M+m} = MV^2 + m(v_0\cos\alpha + V)^2 + m(v_0\sin\alpha)^2$$

$$2mgh + \frac{m^2u^2}{M+m} = \frac{Mm^2u^2\sin^4\alpha + m(M\cos^2\alpha + m\sin^2\alpha)^2u^2 + mM^2u^2\cos^2\alpha\sin^2\alpha}{(M + m\sin^2\alpha)^2}$$

Solve to get : $h = \frac{u^2}{2g}\left[\frac{M^2\cos^2\alpha}{(M+m)(M + m\sin^2\alpha)}\right]$.

Variable Mass System

Mass Variation Cases :- Whenever mass of a system (or Body) changes with line during motion such cases can be solved by concept of impulse and conservation of momentum.

Two Cases can be there :-

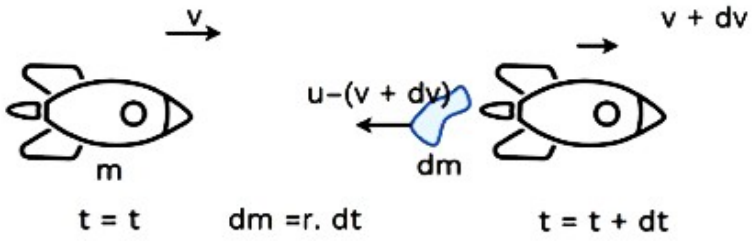
Case 1 :- When external force is absent.

Case 2 :- When external force is present.

Case 1 : - When $\vec{F}_{external}$ is not present:

Hint : Conservation of momentum at intermediate time of motion from 't' do "t + dt"

Example of Rocket Problem :-



u is velocity of exhaust gases relative to the rocket. r is the fuel consumption rate in the rocket.

By conservation of momentum,

$$mv = (m - dm)(v + dv) - dm[u - (v + dv)]$$

$$\Rightarrow \quad m.dv = u.dm$$

Differentiating both sides,

$$\frac{mdv}{dt} = \frac{dm}{dt}(u) \qquad m.a = r.u$$

$ru \rightarrow$ mass of ejected gases

$a \rightarrow$ instantaneous acceleration of the rocket.

$m \rightarrow$ Instantaneous mass of the rocket.

Now, ma = Net force acting on Rocket

But there is no external force acting on rocket,

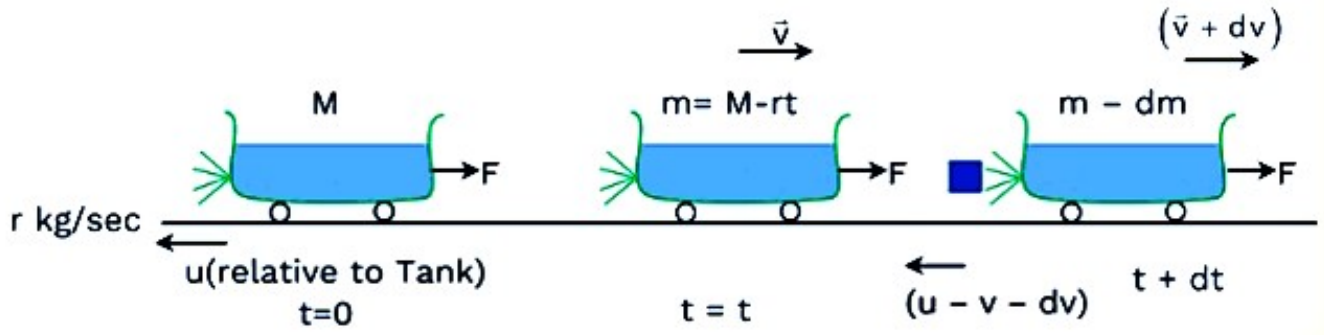
So, the thrust on rocket,

$$\boxed{\vec{F}_{Th} = ru}$$

Case II :- When $\vec{F}_{ext}$ is present :

In such cases we use impulse momentum equation for the motion of body at intermediate time t and t + dt.

Q. **Find the velocity of tank as time function in the given situation**



Sol. Here we write impulse momentum equation for tank,

$mv + Fdt = (m - dm)(v + dv) - dm(u - v - dv)$

which gives,

$F.dt = m.dv - dm.u$

$$\begin{Bmatrix} m = M - rt \\ dm = rdt \end{Bmatrix}$$

$\Rightarrow$ $Fdt = (M - rt)\,dv - (r\,u)dt$

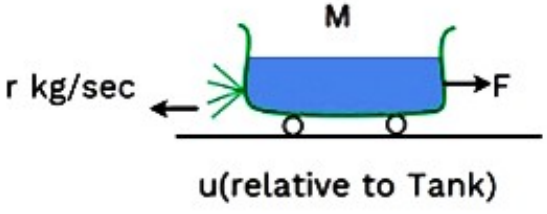
$(F + r\,u)dt = (M - r\,t)dv$

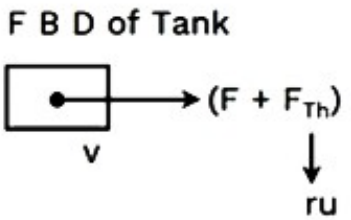
$$\int_0^v dv = \int_0^t \frac{(F + ru)}{(M - rt)} at$$

$$[v]_0^v = \frac{(F + ru)\left|\ell n(m - rt)\right|_0^t}{(-r)}$$

$$v = \frac{(F + ru)}{r}\left\{\ell n\left(\frac{M}{M - rt}\right)\right\}$$

Alternate Method :





$$F + ru = m\left(\frac{dv}{dt}\right) \qquad \text{where, } m = M - rt$$

$$dv = \frac{(F + ru)\,dt}{m}$$

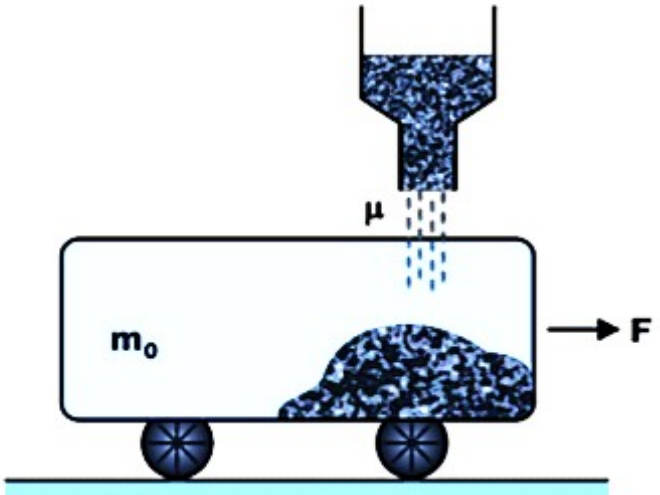
$$\int_0^v dv = \int_0^t \frac{(F + ru)\,dt}{(M - rt)}$$

$$= (F + ru)\left|\ell n(M - rt)\right|_0^t \left(-\frac{1}{r}\right)$$

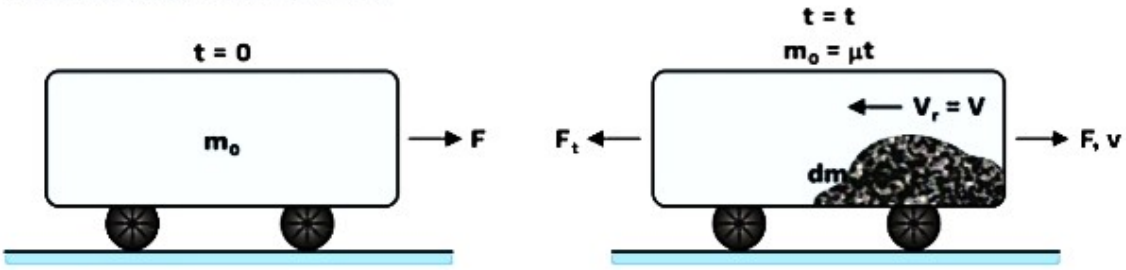
$$= \frac{(F + ru)}{-(r)}\left\{\ell n(M - rt) - M\right\}$$

$$\boxed{v = \frac{F + ru}{r}\,\ell n\left(\frac{M}{M - rt}\right)}$$

Q. **A flat car of mass m0 starts moving to the right due to a constant horizontal force F. Sand spills on the flat car from a stationary hopper. The rate of loading is constant and equal to μ kg/s. Find the time dependence of the velocity and the acceleration of the flat car in the process of loading. The friction is negligibly small.**



Sol. Initial velocity of the flat car is zero. Let v be its velocity at time t and m its mass at that instant. Then



At t = 0, v = 0 and m = m_0 at t = t, v = v and m = $m_0 + \mu t$

Here, $v_r = v$ (backwards)

$$\frac{dm}{dt} = \mu$$

$$\therefore \quad F_t = v_r \frac{dm}{dt} = \mu v \qquad \text{(backwards)}$$

Net force on the flat car at time t is $F_{net} = F - F_t$

$$\text{or} \quad m\frac{dv}{dt} = F - \mu v \qquad ...(1)$$

$$\text{or} \quad (m_0 + \mu t)\frac{dv}{dt} = F - \mu v$$

$$\text{or} \quad \int_0^v \frac{dv0}{F - \mu} = \int_0^t \frac{dt}{m_0 + \mu t}$$

$$\therefore \quad -\frac{1}{\mu}[\ell n(F - \mu v)]_0^v = \frac{1}{\mu}[\ell n(m_0 + \mu t]_0^t$$

$$\Rightarrow \quad \ell n\left(\frac{F}{F - \mu v}\right) = \ell n\left(\frac{m_0 + \mu t}{m_0}\right)$$

$$\therefore \quad \frac{F}{F - \mu v} = \frac{m_0 + \mu t}{m_0}$$

$$\text{or} \quad v = \frac{Ft}{m_0 + \mu t}$$

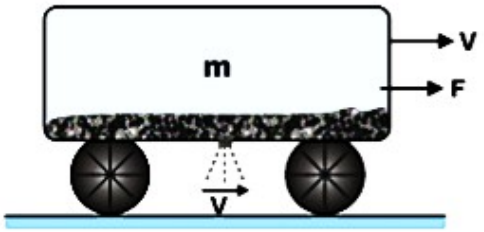
From Eq. (i), $\frac{dt}{dt}$ = acceleration of flat car at time t

$$\text{or} \quad \frac{F - \mu v}{m}$$

$$a = \left(\frac{F - \frac{F\mu t}{m_0 + \mu t}}{m_0 + \mu t}\right) \quad \text{or} \quad a = \frac{Fm_0}{(m_0 + \mu t)^2}$$

Q. **A cart loaded with sand moves along a horizontal floor due to a constant force F coinciding in direction with the cart's velocity vector. In the process sand spills through a hole in the bottom with a constant rate μkg/s. Find the acceleration and velocity of the cart at the moment t, if at the initial moment t = 0 the cart with loaded sand had the mass m_0 and its velocity was equal to zero. Friction is to be neglected.**

Sol. In this problem the sand spills through a hole in the bottom of the cart. Hence, the relative velocity of the sand v_r will be zero because it will acquire the same velocity as that of the cart at the moment.



$v_r = 0$

Thus, $F_t = 0$ $\left(\text{as } F_t = v_r \frac{dm}{dt}\right)$

and the net force will be F only.

$\therefore \quad F_{net} = F$

or $\quad m\left(\frac{dv}{dt}\right) = F \quad \ldots(1)$

But here $m = m_0 - \mu t$

$\therefore \quad (m_0 - \mu t)\frac{dv}{dt} = F$

or $\quad \int_0^v dv = \int_0^t \frac{F dt}{m_0 - \mu t}$

$\therefore \quad v = \frac{F}{-\mu}[\ell n(m_0 - \mu t)]_0^t$

or $\quad v = \frac{F}{\mu} \ln\left(\frac{m_0}{m_0 - \mu t}\right)$

From eq. (1), acceleration of the cart

$$a = \frac{dv}{dt} = \frac{F}{m}$$

or $\quad a = \frac{F}{m_0 - \mu t}$

Q. **A rocket, with an initial mass of 1000 kg, is launched vertically upwards from rest under gravity. The rocket burns fuel at the rate of 10 kg per second. The burnt matter is ejected vertically downwards with a speed of 2000 ms^{-1} relative to the rocket. If burning stops after one minute. Find the maximum velocity of the rocket. (Take g as at 10 ms^{-2})?**

Sol. Using the velocity equation

$$v = u - gt + v_r \ln\left(\frac{m_0}{m}\right)$$

Here u = 0, t = 60s, g = 10 m/s^2, v_r = 2000 m/s, m_0 = 1000 kg

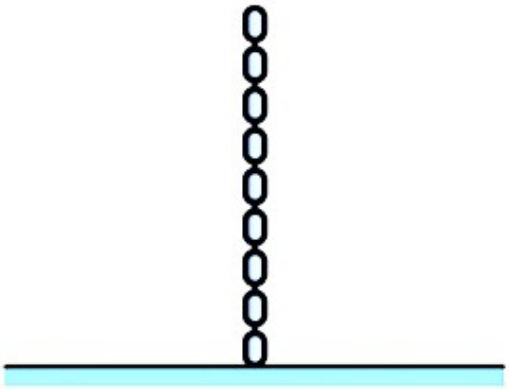
and $m = 1000 - 10 \times 60 = 400$ kg

We get $v = 0 - 600 + 2000 \ln\left(\frac{1000}{400}\right)$

or $v = 2000 \ln 2.5 - 600$

The maximum velocity of the rocket is $200(10 \ln 2.5 - 3) = 1232.6\ ms^{-1}$

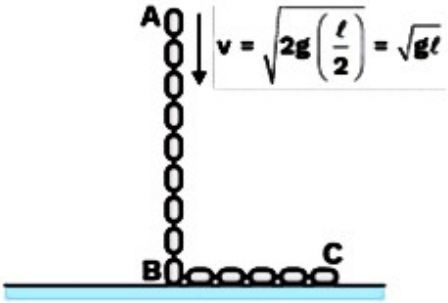
Q. **A uniform chain of mass m and length ℓ hangs on a thread and touches the surface of a table by its lower end. Find the force exerted by the table on the chain when half of its length has fallen on the table. The fallen part does not form heap.**



Sol. 1) Weight of the portion BC of the chain lying on the table,

$$W = \frac{mg}{2} \text{ (downwards)}$$

Using $v = \sqrt{2gh}$



2) Thrust force $F_t = v_r\left(\frac{dm}{dt}\right)$

$v_r = v$

$$\frac{dm}{dt} = \lambda v$$

$F_t = \lambda v^2$ (where, $\lambda = \frac{m}{\ell}$, is mass per unit length of chain)

$v^2 = (\sqrt{g\ell})^2 = g\ell$

$\therefore \quad F_t = \left(\frac{m}{\ell}\right)(g\ell) = mg$ (downwards)

$\therefore$ Net force exerted by the cain on the table is

$$F = W + F_t = \frac{mg}{2} + mg = \frac{3}{2}mg$$

So, from Newton's third law the force exerted by the table on the chain will be $\frac{3}{2}$ mg (vertically upwards).